

Economic evaluation of a pharmacist-led IT-based intervention with simple feedback in reducing rates of clinically important errors in medicines management in general practices (PINCER)



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Abstract

Title

Economic evaluation of a pharmacist-led IT-based intervention with simple feedback in reducing rates of clinically important errors in medicines management in general practices, based on a cluster randomised trial (PINCER).

Authors

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Background

Medication errors in general practice are considered an important source of potentially preventable morbidity and mortality. There is also a usually implicit assumption that improving safety is a “good thing” even though most errors documented are minor and unlikely to affect patient outcome and associated cost. Initiatives to reduce medication errors are usually costly. In an increasingly financially constrained healthcare environment, it is essential to be clearer about the true economic impact of medication error reduction.

Objectives

The overall aim of this study was to determine the cost-effectiveness associated with a pharmacist - led IT-based intervention to reduce rates of potentially harmful prescribing and monitoring errors in general practices (PINCER).

Methods

The economic analysis compared the costs and health benefits of a pharmacist-led IT-based intervention (PINCER) with simple feedback in reducing rates of six clinically important errors in medicines management in general practices. An economic evaluation was carried out to determine the cost per extra quality-adjusted life-year (QALY) generated, from the perspective of the National Health Service (NHS). This analysis combined the results from the PINCER trial with error-specific projected harm and NHS cost to allow generation of estimates of overall patient benefit and NHS costs. Six error-specific treatment pathway Markov models were constructed to quantify the economic impact of the medication errors included in the PINCER intervention. Incremental cost effectiveness ratios, cost effectiveness acceptability curves and net benefit, whereby a monetary value to QALYs was assigned, were generated. Results from the base case analysis were tested using sensitivity and scenario analysis.

Results

In the probabilistic analysis, PINCER was cost-saving (mean ICER was -£2519 per QALY gained (SD 97,460; median -£159; 2.5th percentile: -£23,939; 97.5th percentile £21,767). At a ceiling willingness to pay of £20,000, the PINCER pharmacist intervention reaches 59% probability of being cost effective. The probability of PINCER being cost effective does not increase beyond 59%. The net benefit statistic generated suggests a mean of £16 net benefit (SD £121; median £22; 2.5th percentile: -£218; 97.5th percentile £242), at a ceiling willingness to pay for a QALY of £20000. The mean cost per QALY generated suggested that PINCER increased health gain at a cost per QALY well below most accepted thresholds for implementation. However, the range around this ICER is extremely wide, reflecting the large degree of uncertainty around effect in some of the individual outcome models. If the PINCER intervention targeted one of the errors only, the mean (SE) costs per QALY generated were: NSAIDs prescribing: cost-saving (-£21731, £94); Betablockers prescribing: cost-saving (-£2381, £3906); ACEI monitoring: £19140 (£18008); Methotrexate monitoring: £2060 (£4654); Lithium monitoring: cost-saving (-£523544, £453550); Amiodarone monitoring: £475 (£15). Targeting NSAID prescribing and amiodarone monitoring errors were the most cost effective activities within the PINCER intervention. These were also the models with the most data to support them. Varying the cost of the intervention or the practice size had a negligible effect on results.

Conclusions

This study estimated the economic impact of a safety-focused intervention in health care, which is known to be effective in reducing rates of key prescribing and monitoring errors in general practice. The intervention was more effective and less costly than the alternative but the huge levels of uncertainty present in the analysis meant that the PINCER intervention could not be considered cost effective with a large degree of certainty under current decision rules. However, correction of some errors has a larger clinical and economic effect, such that the PINCER intervention could be cost effective if the “right” errors are targeted. Conclusions from this economic analysis are hampered by the paucity of data around the real clinical and economic impact of medication errors. Better evidence on the impact of errors is required. Further work is required to address the economic impact of including other errors not included in the PINCER intervention. More importantly, given that reducing medication errors may produce non-health benefits such as trust and increased engagement with the health service, the role of cost effectiveness in allocating resources to safety-focused interventions in health care needs to be examined and explored.

List of Abbreviations

ACE:	Angiotensin converting enzyme (inhibitor)
ADE:	Adverse drug event
CEAC:	Cost effectiveness acceptability curve
CHD:	Coronary heart disease
CPOE:	Computerised physician order entry
CTU:	Clinical Trials Unit
DMEC:	Data Monitoring and Ethics Committee
EMIS:	Egton Medical Information Systems (the name of a GP computer system)
GP:	General practitioner (or family practitioner)
ICC:	Intraclass correlation coefficient
ICER:	Incremental cost effectiveness ratio
IMD:	Index of Multiple Deprivation
INR:	International normalised ratio
IT:	Information technology
Li:	Lithium
MRC:	Medical Research Council
NHS:	The UK National Health Service
NPSA:	National Patient Safety Agency
NSAIDs:	non-steroidal anti-inflammatory drugs
ONS:	Office for National Statistics
OR:	odds ratio
PCT:	Primary Care Trust
PPI:	proton pump inhibitor
TFT:	Thyroid Function test
TPP:	The Phoenix Partnership (the name of a GP computer system)
TSC:	Trial Steering Committee
U&E:	Urea and electrolytes

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1 Background

This report presents analysis determining the cost effectiveness of a pharmacist-led IT-based intervention with simple feedback in reducing rates of clinically important errors in medicines management in general practices, based on a cluster randomised trial (PINCER).¹ Detail on the trial hypothesis and methods, main clinical results, within-trial economic analysis and associated costs of the intervention are available elsewhere.^{2,3}

1.1 Estimating the true economic impact of medication error reduction

Most healthcare systems around the world are attempting to improve safety in healthcare, with associated policy development and a range of national or regional initiatives, mostly in secondary care.⁴ In England, the new National Health Service (NHS) outcomes framework, as proposed in the recent NHS White Paper, Equity and Excellence: liberating the NHS refers specifically to the formation of strategies to improve patient safety.⁵ Despite a general agreement that improving patient safety in primary care is a priority, there appears to be little agreement about how best to do it. There is also a usually implicit assumption that improving safety is a “good thing” even though most errors documented are minor and unlikely to affect patient outcome and associated cost.

1.2 What is the economic impact of medication errors?

Researchers have been measuring the rates of medication errors for over forty years,⁶⁻⁹ but there has been little work beyond these epidemiological studies to assess the true impact on patient outcomes and cost. A report to the National Patient Safety Agency (NPSA) on the economic perspective of adverse events in the NHS¹⁰ stated that there had been few attempts to examine medical errors and adverse events from an economic perspective, far less to examine the cost effectiveness of policies and initiatives to reduce error rates.

There have been many attempts to estimate costs associated with medical errors and adverse events.¹⁰ Not all medical errors lead to an adverse event and associated costs, hence most costing studies concentrate on adverse events and not on medical errors. Not all adverse events are caused by medical error, however, so it is necessary for the researchers to separate preventable from non-preventable adverse events to identify the costs associated with errors. The type of costs included in the study (medical costs, lost household production), and over what timescale they have been assessed will also affect the final figures

derived. Furthermore, serious adverse events may be associated with malpractice claims, which can be highly expensive.¹¹

Most research has centred on costs incurred by errors occurring in secondary care or costs incurred by secondary care in managing errors occurring in primary care. The earliest studies in preventable drug-related morbidity in 1969 reported increased resource use in terms of increased length of hospital stays,⁷ and more recent studies have confirmed the presence of increased secondary care resource use caused by medication errors.¹²⁻²¹ The use of variable measurement methods, parameters collected and study quality has limited the usefulness of some of these studies.²² In a key US study in 1997, Bates *et al* reported that estimated post-event costs attributable to an adverse drug event (ADE) were US\$2595 for all ADEs and US\$4685 for preventable ADEs.¹⁸ From the same patient cohort, the relative risk of death among patients experiencing an ADE was 1.88 (95% confidence interval, 1.54-2.22; P<.001).²³

In 2000, the Department of Health estimated that adverse events in England were associated with 850,000 inpatient episodes, costing £2 billion in additional bed-days.²⁴ It is likely that only the most severe, and thus most rare, consequences of medication errors occurring in primary care result in a secondary care stay. There is much less research around the costs associated with errors originating in primary care, and costs occurring in primary care. This is despite the fact that this is where most prescribing occurs, and thus where most morbidity associated with errors is likely to occur. In a recent literature review looking at routinely recorded patient safety events in primary care, 50 studies were reviewed.²⁵ Approximately 6.5% of adult emergency admissions were due to drug-related events. Between 0.7% and 2.3% of deaths following adverse events were attributed to treatment in primary care. Field *et al* reported that older adults experiencing adverse events in primary care incurred US\$65631 per patient per year (2005 costs), and they attributed US\$27365 of this to preventable events.¹⁹

There are likely to be costs outside the perspective of the healthcare provider. A Dutch study quantified hospital costs due to preventable hospital admissions related to medication, but also attempted to quantify production loss costs.²⁰ These authors estimated that the average production losses for one admission in a person <65 years was €1712 (2011 prices). In 2002, Rothschild *et al* estimated that costs incurred from malpractice claims associated with preventable and non-preventable inpatient and outpatient medication errors ranged from US\$64700 to US\$376500 per individual case.¹¹

In summary, the results of these studies appear to suggest that there is sufficient economic impact from errors to support efforts to reduce medication error rates and associated preventable adverse events in primary care.

1.3 What is the economic impact of interventions to reduce medication error rates?

Interventions to reduce medication errors are not new. In 1972, an educational intervention in digoxin prescribing reduced “digitalis intoxication”.²⁶ There have been many reviews of these studies.^{10 27-29} Most studies about reducing medication errors have been undertaken in secondary care and tend to be focused on computerised tools, educational strategies or professional roles.²⁸ Strategies and initiatives that aim to change prescribing behaviour are generally costly, with little evidence presented around their cost effectiveness.^{28 30} For example, reducing prescribing errors may reduce costs, but the true economic implications of implementing this intervention is uncertain, given that prescribing behaviour may not change as anticipated, or that the clinical and economic effects of most errors may be minor.³¹

There is very little evidence that describes the clinical and economic impact of medication error reduction.⁴ Studies reporting interventions to reduce error reduction may provide information around costs of the intervention, or even the effects of the intervention on prescribing budgets³² but generally do not report evidence around the effect of the intervention on patient outcome or costs.

Kaushal attempted to quantify the return on investment of a computerised physician order entry (CPOE) system.³³ Between 1993 and 2002, the Boston Women’s Hospital (BWH) spent US\$11.8 million (2002 prices) to develop, implement, and operate CPOE. Over ten years, the system saved BWH \$28.5 million. Costs were saved through reductions in unnecessary medications, investigations, and staff time utilisation. The authors also determined cost savings from drug adverse event alerts by multiplying the number of averted ADEs by the average cost of an ADE derived from Bates et al (US\$4,685 in 1997 dollars).¹⁸

One modelling study was found that aimed to detect the economic impact of a pharmacy-based intervention to reduce medication errors.³⁴ No cost per error was reported. However, using a range of assumptions, this UK study estimated the potential to cause harm based on the error rates. Probability of harm from undetected errors was divided into harm associated with errors of omission and errors of commission.³⁴ Probability of harm was divided into

significant (resulted in temporary harm to the patient and required intervention without (increase in) hospital stay); serious (resulted in temporary harm and required hospitalisation) and severe, life-threatening or fatal (resulted in permanent patient harm, required intervention to sustain life, or contributed to a patient's death). Utility weights were attached to harm from undetected errors divided into significant, serious, severe, life-threatening or fatal. These were hypothetical estimates as there are no relevant data available to describe the utility effects of the broadly defined severity categories. This approach has not been used in our study, as knowledge of the types of errors affected by the PINCER intervention means that we can use a more data-driven approach.

Medication errors in primary and secondary care are considered an important cause of morbidity and mortality, and a number of reports from the UK, USA and other countries have highlighted the need to reduce error rates to prevent patients suffering from avoidable harm.^{24 35} In England, publication by the Government of *An organisation with a memory*²⁴ and *Building a safer NHS for patients*³⁶ illustrates a strong commitment to reducing errors; the establishment of the NPSA was a clear further example of this commitment.

Recent UK Government reports have suggested that while there may still be a need to understand more about medication errors and the reasons for their occurrence,^{36 37} the priority now must be to find effective, cost-effective, acceptable and sustainable ways of preventing patients from being harmed as a result of such errors.

Given the large quantities of money generally consumed in these initiatives, in an increasingly financially constrained healthcare environment, it is essential to be clearer about the true economic impact of medication error reduction.

2 Work already completed by this research team

2.1 Summary of PINCER trial methods²

We have conducted a pragmatic cluster randomised trial investigating the effectiveness of a pharmacist-led information technology-enabled (PINCER) intervention in reducing risk of hazardous prescribing and medicines management errors.

2.1.1 Study sites and patient participants

We wrote to 240 general practices in PCTs in Nottinghamshire, Staffordshire and Central and Eastern Cheshire, England informing them of the study. of which 72 (30%) were recruited between July 2006 and August 2007. Participating and non-participating practices had comparable number of GPs and socioeconomic profiles; participating practices were however larger, more likely to be training practices and had slightly higher Quality and Outcomes Framework scores (<http://www.qof.ic.nhs.uk>). The main reason practices gave for not taking part was that they were too busy.

Baseline characteristics of practices are reported in Table 1. Overall, treatment arms were well balanced in terms of participant and practice characteristics at baseline. Seventy-two general practices with a combined list size of 480,942 patients were recruited to the study and randomised to one of the two study interventions.

Table 1 Characteristics of practices and patients at baseline by treatment arm²

Practice characteristics	Simple feedback arm (%)	Pharmacist intervention arm (%)
Number of practices	36 (50.0)	36 (50.0)
Study centre		
Nottingham	22 (61.1)	21 (58.3)
Manchester	14 (38.9)	15 (41.7)
Median list size (IQR)	6438 (3834, 9707)	6295 (2911, 9390)
Age of practice population		
0-14	38804 (16.3)	39818 (17.4)
15-64	159277 (67.1)	152156 (66.5)
65-74	20683 (8.7)	19151 (8.4)
>=75	18648 (7.9)	17623 (7.7)
Total	237412 (100.0)	228748 (100.0)
Sex of practice population		
Male	118469 (49.9)	113284 (49.5)
Female	118943 (50.1)	115464 (50.5)
Median Index of Multiple Deprivation 2004 score (IQR)	26.3 (18.8, 36.5)	30.3 (18.2, 39.6)
GP training practices (%)	10 (27.8)	13 (36.1)
Median Quality and Outcomes	42 (38,42)	42 (38,42)

Framework medicines management points (IQR)		
Median total Quality and Outcomes Framework points (IQR)	1041 (1004, 1049)	1036 (993, 1048)

2.1.2 Study interventions

General practices were centrally randomised to computer-generated simple feedback on at-risk patients (control arm) or the PINCER intervention comprising feedback, educational outreach and dedicated support (intervention arm). We did not feel it would be appropriate to randomise practices to a no intervention control arm because it would have meant identifying patients at risk from medication errors with there being no prospect of these being rectified.

2.1.3 Simple feedback

Those practices randomly allocated to this arm received computerised feedback on patients identified to be at risk from potentially hazardous prescribing and medicines management from the practice computer system, along with brief written educational materials explaining the importance of each type of error in terms of the evidence-base and risks associated with each error. Practices in the simple feedback arm were asked to try to make any changes to patients' medications within a 12 week (intervention) period following the baseline data collection.

2.1.4 Pharmacist intervention

Those practices randomly allocated to this arm received simple feedback and in addition, had a complex pharmacist-led IT-based intervention.

First, the trial pharmacists arranged to meet with members of the practice team to discuss the computer-generated feedback on patients with medication errors. All doctors were encouraged to attend this meeting along with at least one member of the nursing staff, the practice manager and at least one member of the reception staff.

Before the meeting, wherever possible, all relevant members of staff were provided with a brief summary of the objectives of the pharmacist-led intervention and a summary of the findings from the computer search.

At the meeting the pharmacists were asked to use the following approach derived from the principles of educational outreach³⁸ while also taking account of human error theory³⁹:

- Establish professional credibility by explaining their own background in clinical pharmacy and their affiliation with either the University of Manchester or University of Nottingham (depending on the site they are working from).
- Take a non-judgemental approach in all discussions with members of the practice team.
- Outline the findings from the computer search.
- Explore the views of team members about the findings.
- Investigate the baseline knowledge of team members regarding the importance of each of the errors.
- Provide clear, concise, evidence-based materials on each of the errors, encouraging active participation by team members.
- Explore the views of team members on the underlying causes of the medication errors (using root-cause analysis techniques where appropriate).⁴⁰
- Explain their availability to work part-time with the practice over the following 12 weeks to:
 - Help take corrective action in individual patients with medication errors.
 - Help improve the systems operating in the practice in order to prevent future errors.
- Encourage the team to agree on an action plan with clear objectives.
- Ask for a member of the practice team to volunteer to liaise with the pharmacist over arrangements for making changes to individual patients' medication and introducing changes to systems within the practice.
- Ask the practice to agree to a follow-up meeting within four to six weeks of the initial meeting.

Following this initial meeting, the pharmacists used a range of techniques to help correct the medication errors that had been identified and prevent future medication errors. They were asked to work closely with the practice team member assigned to provide liaison with other members of the practice.

2.1.5 Study outcomes

Our primary outcomes were the proportions of patients at six months post-intervention who experienced any of the following three clinically important errors: i). non-selective non-steroidal anti-inflammatory drugs (NSAIDs) prescribed to those with a history of peptic ulcer without co-prescription of a proton pump inhibitor; ii). beta-blockers prescribed to those with

a history of asthma; and iii). long-term prescriptions of angiotensin converting enzyme (ACE) inhibitor or loop diuretics to those aged ≥ 75 years without assessment of urea and electrolytes in the preceding 15 months. Secondary outcomes included were: i). Patients prescribed methotrexate for ≥ 3 months without a full blood count or liver function test in last three months; ii). Patients prescribed lithium for ≥ 3 months without a lithium level in last three months; and iii). Patients prescribed amiodarone for ≥ 6 months without a thyroid function test in the last six months.

The cost per error avoided (from the perspective of the English NHS) was estimated using incremental cost-effectiveness analysis.

2.2 Summary of PINCER findings²

At six months follow up, patients in the PINCER arm were significantly less likely to have experienced one of the six errors. The results are summarised in Table 2.

Table 2 Prevalence of prescribing and monitoring problems at six months follow-up by treatment arm

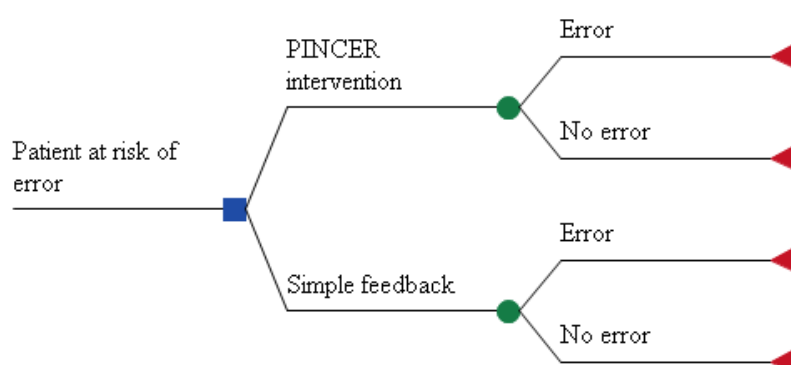
Outcome/population at risk*	Simple feedback arm (%)	Pharmacist intervention arm (%)	Relative risk reduction
NSAID (OM1): Patients with a history of peptic ulcer prescribed an NSAID without a PPI / Patients with a history of peptic ulcer without a PPI	86/2014 (4.3)	51/1852 (2.8)	0.35 p=0.01
BETA-BLOCKER (OM2): Patients with asthma prescribed a beta-blocker / Patients with asthma	658/22224 (3.0)	499/20312 (2.5)	0.17 p=0.006
ACEI (OM3): Patients aged ≥ 75 on long term ACE inhibitors or diuretics without urea and electrolyte monitoring in the previous 15 months / Patients aged ≥ 75 on long term ACE inhibitors or diuretics	436/5329 (8.2)	255/4851 (5.3)	0.36 p=0.003
METHOTREXATE (OM5): Patients prescribed methotrexate for ≥ 3 months without a full blood count or liver function test in last 3 months / Patients prescribed methotrexate for ≥ 3 months	162/518 (31.3)	122/494 (24.7)	0.19 p=0.45
LITHIUM (OM7): Patients prescribed lithium for ≥ 3 months without a lithium level in last 3 months / Patients prescribed lithium for ≥ 3 months	84/211 (39.8)	67/190 (35.3)	0.11 p=0.12
AMIODARONE (OM8): Patients prescribed amiodarone for ≥ 6 months without a thyroid function test in the last 6-months / Patients prescribed amiodarone for ≥ 6 months	106/235 (45.1)	81/242 (33.5)	0.25 p=0.02

*The nomenclature for each error in the PINCER report (Outcome Measure (OM) 1-8) has been converted to the principal drug featured in each error.

2.3 Within-trial PINCER economic analysis

We have undertaken a two-stage economic analysis from the perspective of a payer within the English NHS: a within-trial analysis of cost per error avoided and a modelling analysis of economic impact of error reduction. The principal objective of the within-trial analysis was to identify and value the resource use associated with the interventions used in the trial, in relation to changes in error rates between intervention and control practices. This analysis did not attempt to estimate the changes in cost results from changes in error rates. The evaluation compared the pharmacist-led intervention with simple feedback. Figure 1 illustrates the comparators and the probabilistic events that are associated with each strategy in the within-trial analysis.

Figure 1 A decision analytic model of pharmacist intervention versus simple feedback in patients at risk of error



The key results of the within-trial analysis are summarised in Table 3.

Table 3 Simple feedback and PINCER intervention arm costs and error rates and incremental economic analysis²

Mean cost per practice (range)/£	Simple feedback	PINCER intervention
Report generation	93 (n/a)	934 (n/a)
Pharmacist training costs	0	276 (80 – 591)
Quarterly facilitated strategic meetings	0	195 (56 – 418)
Monthly operational meetings	0	57 (16 – 122)

Practice feedback	0	22 (6 – 47)
Management of errors*	0	407 (57 – 1 319)
Total cost	93 (n/a)	1 050 (329 – 2 087)
Mean incremental cost (95% CI)/£	872 (766 – 978)	
Mean incremental errors (95% CI)	-13.90 (-13.42 – -12.39)	
Mean ICER (2.5-97.5th percentile)/£ per error avoided	66 (58 – 73)	

*time spent by PINCER pharmacist following up errors identified in initial practice error report

The PINCER intervention had a 95% probability of being cost-effective if the decision-maker's ceiling willingness to pay reached £75 per error avoided at six months.

3 Methods

3.1 Overall rationale

The economic analysis outlined in the sections above has allowed the generation of cost per error avoided by the use of the PINCER intervention. This statistic, while informative, does not provide us with information about the overall impact of the intervention on patient health and NHS budgets. The design of this trial did not allow primary observation of patient outcomes and NHS costs resulting from errors, or reduction of errors. Therefore, we have used a modelling approach to simulate the effect of the observed error reductions on overall patient outcomes and NHS costs. Modelling within economic analysis allows research questions to be answered beyond the data obtained from primary research. It provides a simplified version of reality to allow analysis, links diverse sources of information into a coherent whole rather than using one trial and allows more questions to be asked (and answered) than just within one setting. There are limitations associated with the use of modelling, particularly as the researcher defines the model structure and the data used to populate that model. Misspecification of the model will produce erroneous and misleading results. Furthermore, data used to populate models come from diverse sources and there is an assumption that it is appropriate to combine these data and use them as though they come from a homogenous source.

The model required for this study should allow us to piece together the process of care such that the model is realistic in terms of alternatives under investigation and the sequence of events. This section describes and justifies the specification of the models and data sources used, whereby all assumptions and limitations are made explicit.

3.2 Aims and objectives

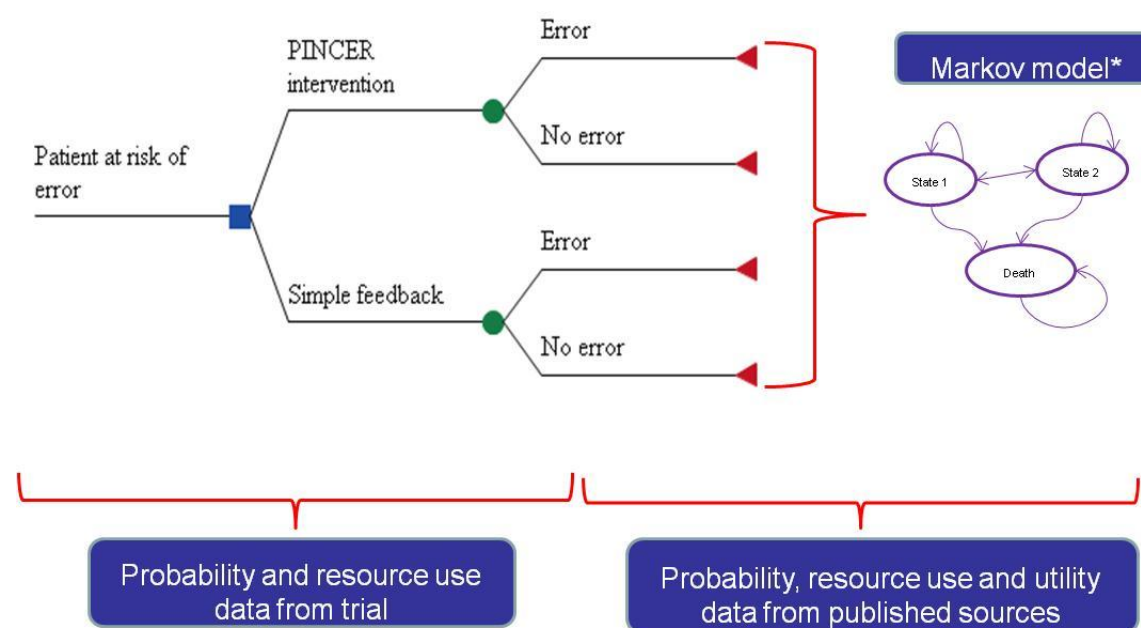
The overall aim was to determine the cost effectiveness of an IT-based intervention to reduce potentially harmful prescribing and monitoring errors in primary care. This aim was achieved by meeting the following objectives:

- Development of models that represent the treatment pathways associated with the consequences of errors targeted in the PINCER intervention;
- Completion of the treatment pathway models with UK relevant probabilities, utilities and resource use data;
- Combination of the treatment pathway models with the within-trial PINCER analysis to generate probabilistic cost per QALY and net benefit statistics.

3.3 Methodology

The “within-trial” economic analysis was expanded by incorporating error-specific projected harm and NHS cost to allow generation of estimates of overall patient benefit and NHS costs incurred in the compared strategies as described in the PINCER study (see Figure 2).

Figure 2 Overview of economic model developed to combine PINCER trial results with estimates of harm caused by errors



*number and type of health states will depend on the prescribing or monitoring error

The six outcomes described in Table 1 were included in this analysis requiring development of six treatment pathway models. A Markov model was developed for each treatment pathway, using a five year time horizon. Each treatment pathway described the consequences of being prescribed or monitored appropriately, compared with being prescribed or monitored inappropriately. These models were combined with the “within-trial” economic analysis to generate cost per QALY and net benefit. The methods describe the generic and treatment pathway specific aspects of developing the models. The development of each model is also described in more detail in Appendices 1 to 6. The approach used to combine these models with the “within-trial” economic analysis to generate cost per QALY and net benefit is outlined.

3.4 Model specification

We undertook the economic analysis from the perspective of the funder of the PINCER intervention or simple feedback intervention (the English NHS) in terms of the direct costs of providing an intervention to reduce prescribing errors in general practice and the costs of managing the consequences of errors.

The models developed for this study are stochastic probabilistic models where events occur with specified probabilities. The stochastic nature of the data used to populate the model provides a measure of uncertainty around the data and thus provides more useful cost effectiveness information to decision-makers. A Markov model was developed for each treatment pathway, using a three month cycle length with half cycle correction, five year time horizon and the UK Treasury recommended 3.5% discount rate for both costs and outcomes. Each treatment pathway described the consequences of being prescribed or monitored appropriately, compared with being prescribed or monitored inappropriately. Age-related mortality was included in each model.

3.5 Sources of clinical outcome, health status and resource use data

A literature search was conducted through the electronic databases Medline, Embase and Web of Science using the treatment pathway specific search terms. References in English and limited to humans were included. Databases were searched to the end of 2010. After excluding duplicate records, references that remained for further evaluation were selected on title and/or abstract. Studies were included if they examined issues on the incidence and/or prevalence, treatment or resource use of the consequences of the error. Subsequently, full text of the retrieved references of the previous selection was evaluated. Finally, reference lists of the retrieved references of the first search were hand-searched.

Transition probability and health status data were taken preferentially from up-to-date UK sources that reflected the characteristics of the populations seen within the PINCER trial. When this was not possible, other data sources had to be used. Some models were more difficult to populate with evidence than others. The limitations of, and uncertainties around, specific data sources are presented in the Appendices.

The resource use data were obtained preferentially from up-to-date UK sources of observation of normal clinical practice, where units of resource use have been reported in a disaggregated manner, to allow attachment of current unit prices for drugs, patient stays and

so on. If possible, individual patient data were used, with associated measures of mean and variation. If these were not available, point estimates were used, with carefully specified deterministic ranges, and standard methods for allocating distributions to these data were used. Where no detailed patient-level resource use data were collected in clinical trials, or those data were not considered to reflect normal clinical practice, published estimates of resource use reflecting normal clinical practice in the UK were used. Cost year was 2010.

Each model was discussed with clinicians on the Pincer team and clinical experts in the area to ensure face validity. The lithium model and the amiodarone model were also discussed with Richard Morriss (Professor of Psychiatry & Community Mental Health, Faculty of Medicine & Health Sciences, University of Nottingham) and Jayne Franklyn (Professor of Medicine and Head of School of Clinical and Experimental Medicine, University of Birmingham), respectively.

3.6 Incremental economic analysis

Each error-specific model was populated with probability, cost and health status data. This allowed the generation of the outcomes and costs in a cohort of patients with the error present, and in a cohort with the error absent. The probability, cost and utility data were assigned beta, gamma and beta distributions respectively.

The incremental cost per extra QALY generated in the absence of an error was calculated using the following equation:

$$(\text{Cost}_{\text{error absent}} - \text{Cost}_{\text{error present}}) / (\text{QALY}_{\text{error absent}} - \text{QALY}_{\text{error present}})$$

Statistical analysis is not appropriate to test the robustness of ICERs. It is not possible to generate 95% confidence intervals around ICERs because the ratio of two distributions does not necessarily have a finite mean, or therefore, a finite variance.⁴¹ Therefore, generation of a bootstrap estimate of the ICER sampling distribution to identify the magnitude of uncertainty around the ICERs is required. Bootstrapping with replacement was employed, utilising Microsoft Excel, using a minimum of 5000 iterations to obtain 2.5% and 97.5% percentiles of the ICER distribution. These ICERs were plotted on cost effectiveness planes.

The error rates generated in the Pincer trial in the Pincer and simple feedback arms are reported at practice level. Pincer intervention costs were also generated at a practice level. Therefore, our model needed to reflect this when incorporating incremental costs and outcomes from errors.

The probability of each error occurring in the PINCER and simple feedback practices was combined with the error-specific models described above. This allowed us to generate the incremental effect of the PINCER intervention on the costs and outcomes of each error. Probabilistic estimates of costs and outcomes were derived, the analysis generating 5000 iterations for each error. The incremental costs and outcomes associated with each error were incorporated additively into the economic model.

Both deterministic and probabilistic incremental economic analyses were carried out using the adjusted cost and outcome data outlined above, in combination with the PINCER intervention costs reported in the first part of this report. This generated deterministic and probabilistic estimates of the overall costs and outcomes of the PINCER versus simple feedback arms. The model assumes that no new patients enter the practice during the five year period.

The incremental cost per extra QALY generated by the PINCER intervention over simple feedback was calculated using the following equation:

$$(\text{Cost}_{\text{PINCER}} - \text{Cost}_{\text{Simple feedback}}) / (\text{QALY}_{\text{PINCER}} - \text{QALY}_{\text{Simple feedback}})$$

Bootstrapping with replacement was employed, utilising Microsoft Excel, using a minimum of 5000 iterations to obtain 2.5% and 97.5% percentiles of the ICER distribution. These ICERs were plotted on cost effectiveness planes for base case, sensitivity and scenario analyses. Points in the NW quadrant are never considered cost-effective (the intervention is more costly and less effective, so dominated by the alternative). Points in the SE are always considered cost-effective (the intervention is less costly and more effective, so dominates the alternative). Points in the NE and SW quadrants may or may not be considered cost-effective depending upon the maximum monetary values that a decision-maker might be willing to pay for a particular unit change in outcome.

Cost effectiveness acceptability curves (CEACs) are a method for summarising information on uncertainty in cost-effectiveness. A CEAC shows the probability that an intervention is cost-effective compared with the alternative, given the observed data, for a range of maximum monetary values that a decision-maker might be willing to pay for a particular unit change in outcome.⁴² CEACs were constructed to express the probability that the cost per QALY gained (y-axis) is cost-effective as a function of the decision-maker's ceiling cost effectiveness ratio (λ) (x-axis) for base case, sensitivity and scenario analyses.⁴³ The CEAC

is constructed by plotting the proportion of the costs and effects pairs that are cost-effective for a range of values of λ . This proportion is easily identifiable from the incremental cost-effectiveness plane as the proportion of points falling to the south and east of a ray through the origin with slope equal to λ . The process of constructing a CEAC begins by calculating this proportion with a ray of slope zero (equivalent to the x-axis). The process is repeated numerous times for rays of larger and larger slopes, up to a maximum value of infinity (equivalent to the y-axis).⁴² A CEAC simply presents the probability that an intervention is cost-effective compared with the alternative for a range of values of λ , that is, the probability that the ICER falls below the maximum acceptable ratio. Statements concerning CEACs should be restricted to the uncertainty of the estimate of cost-effectiveness.

The incremental net monetary benefit (INB) was estimated from the incremental costs and QALYS for PINCER compared with simple feedback using the formula:

$$INB(\lambda) = \lambda(QALY_{PINCER} - QALY_{Simple\ feedback}) - (Cost_{PINCER} - Cost_{Simple\ feedback})$$

The incremental net benefit approach was used due to well-known problems associated with incremental cost-effectiveness ratios (ICERs) when bootstrap replicates cover all four-quadrants of the cost-effectiveness plane.^{44 45}

The incremental net-monetary benefit statistic is positive when, for a given threshold (λ), the PINCER intervention represents good value relative to simple feedback. The threshold is typically interpreted as society's willingness-to-pay for an additional unit of health, the QALY. However, when the intervention results in fewer QALYs, the threshold (λ) can be interpreted as the lower bound on the savings an intervention must create for an observed reduction in QALYs.⁴⁵ Incremental net monetary benefit was calculated for a threshold range from £0 to £160,000 using increments of £10,000.

3.7 Sensitivity and scenario analysis

The errors included in the intervention were varied to see if this affected the overall cost effectiveness of the PINCER intervention in the following four scenario analyses:

- Each error separately
- PINCER primary outcome errors only
- Prescribing errors only
- Monitoring errors only.

The costs associated with the PINCER intervention were varied, to reflect possible variations in how the intervention might be delivered in practice. The practice size affects the intervention cost, so this was also varied to examine the effect this might have.

4 Results

4.1 Results from outcome measure-specific models

A brief summary for each of the outcome measure-specific models is presented here. The models are provided in detail in Appendices 1 to 6.

4.1.1 Patients with a past medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI

The model structure is presented in Figure 3, transition probabilities, health states and costs are summarised in Table 4 and Table 5.

Figure 3 Markov model for patients with a past medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI

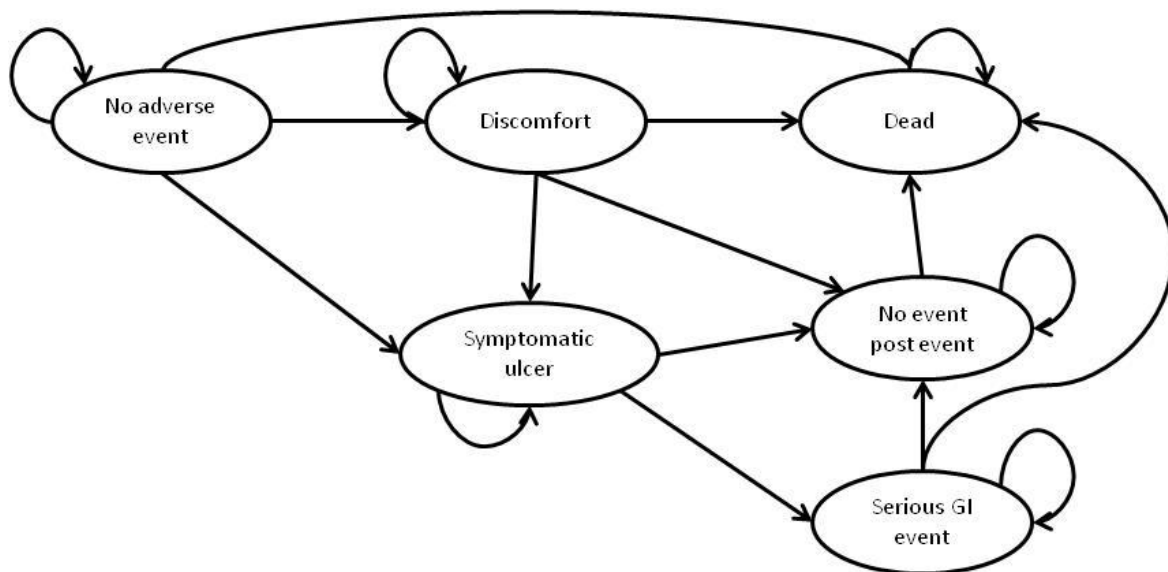


Table 4 Probabilities for the 3-month cycle Markov model in the error and non-error groups (NSAIDs)

Transition probability	No error	Error
No GI adverse event → No GI adverse event	0.894*	0.829*
No GI adverse event → GI discomfort	0.099 ⁴⁶	0.154 ^{46 47}

No GI adverse event → Symptomatic ulcer	0.0047 ⁴⁶	0.0142 ^{46 47}
No GI adverse event → Serious GI event	0.0001 ⁴⁶	0.0002 ^{46 47}
No GI adverse event → Death	0.0003 ⁴⁸	
Discomfort → Discomfort	0.188 ⁴⁶	
Discomfort → Symptomatic ulcer	0.0069 ⁴⁶	
Discomfort → Serious GI event	0.00015 ⁴⁶	
Discomfort → No further GI event	0.802*	
Discomfort → Death	0.0003 ⁴⁸	
Symptomatic ulcer → Discomfort	0.148 ⁴⁶	
Symptomatic ulcer → Symptomatic ulcer	0.0183 ⁴⁶	
Symptomatic ulcer → Serious GI event	0.00039 ⁴⁶	
Symptomatic ulcer → No further GI event	0.824*	
Symptomatic ulcer → Death	0.001 ⁴⁹	
Serious GI event → Discomfort	0.148 ⁴⁶	
Serious GI event → Symptomatic ulcer	0.0183 ⁴⁶	
Serious GI event → Serious GI event	0.00039 ⁴⁶	
Serious GI event → No further GI event	0.725*	
Serious GI event → Death	0.1083 ⁵⁰	

* Net of other probabilities at this node

Table 5 Summary of utility weights and cost per health state for NSAID model

Health state	Utility weights ⁵¹	Mean (SE) cost per patient /£
No GI adverse events	0.85 ⁵²	7 (0.4)
Discomfort	0.760 ⁵³	165 (8)
Symptomatic ulcer	0.720 ⁵³	714 (36)
Serious GI event	0.674 ⁵³	9596 (480)
No further GI event following initial GI event	0.85 ⁵²	135 (7)
Death	0	1590 (79)

4.1.2 Patients with a history of asthma who have been prescribed a beta-blocker

The model structure is presented in Figure 4, transition probabilities, health states and costs are summarised in Table 6 and Table 7.

Figure 4 Markov model for patients with asthma and a β -blocker prescription

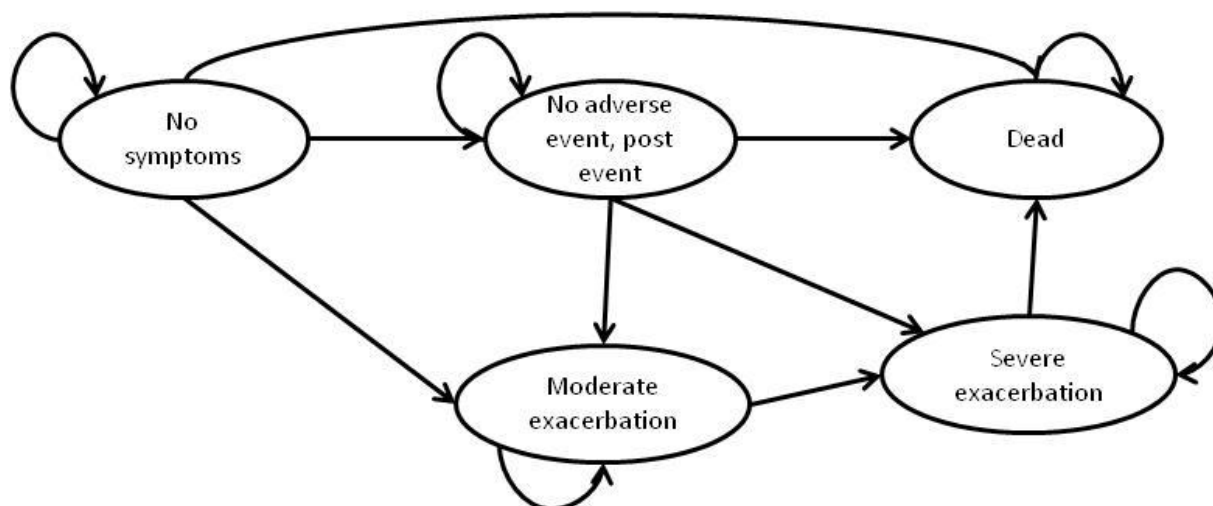


Table 6 Probabilities for the 3-month cycle Markov model in the error and non-error groups (beta-blockers)

Transition probability	No error	Error
No symptoms→No symptoms	0.982*	0.955*
No symptoms→Moderate exacerbation	0.008 ⁵⁴	0.031 ⁵⁵
No symptoms→Severe exacerbation	0.002 ⁵⁴	0.007 ⁵⁶
No symptoms→Death	0.008 ⁵⁴	
Severe exacerbation→No symptoms post event	0.664*	
Moderate exacerbation→Moderate exacerbation	0.111 ⁵⁴	
Moderate exacerbation→Severe Exacerbation	0.111 ⁵⁴	
Moderate exacerbation→Death	0.114 ⁵⁴	
Severe exacerbation→No symptoms post event	0.797*	

Severe exacerbation→Death	0.203 ⁵⁴
No symptoms post exacerbation→Death	0.008 ⁵⁴

* Net of other probabilities at this node

Table 7 Summary of utility weights and cost per health state for beta-blocker model

Health state	Mean (SE) Utility	Mean (SE)cost per patient /£*
No symptoms	0.73 (0.03)	7 (1.5)
No symptoms post event	0.73 (0.03)	5 (1)
Moderate exacerbation	0.67 (0.02)	77 (15)
Severe exacerbation	0.66 (0.04)	4147 (829)
Death	0	0

*derivation provided in Appendix 2

4.1.3 Patients aged 75 years and older who have been prescribed an Angiotensin-Converting Enzyme Inhibitor (ACEI) long-term who have not had a recorded check of their renal function and electrolytes in the previous 15 months

The model structure is presented in Figure 5, transition probabilities, health states and costs are summarised in Table 8 and Table 9Table 8 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups for ACEI

Transition probability	Not monitored	Monitored
No symptoms →No symptoms	0.979 (1-0.016-0.001-0.004)*	0.988 (1-0.008-0.0005-0.004)*
No symptoms→ hyperkalaemia	0.016**	0.008 ⁵⁷
No symptoms→ARF	0.0010 ^{58 59}	0.0005 ⁶⁰
Hyperkalaemia→ Post Hyperkalaemia	0.996 (1-0.004)*	
ARF→Post ARF	0.974 (1-0.026)*	
No symptoms→Dead	0.004 ^{61 62}	
Hyperkalaemia →Dead	0.004**	
Post hyperkalaemia →Dead	0.004**	
ARF→Dead	0.026 ⁶³	
Post ARF→Dead	0.026 ⁶³	

* Net of other probabilities at this node

**derivation provided in Appendix 3

Table 9 Summary of utility weights and cost per health state for ACEI model.

Figure 5 Markov model for patients aged 75 years and older who have been prescribed an ACEI long-term who have not had a recorded check of their renal function and electrolytes in the previous 15 months

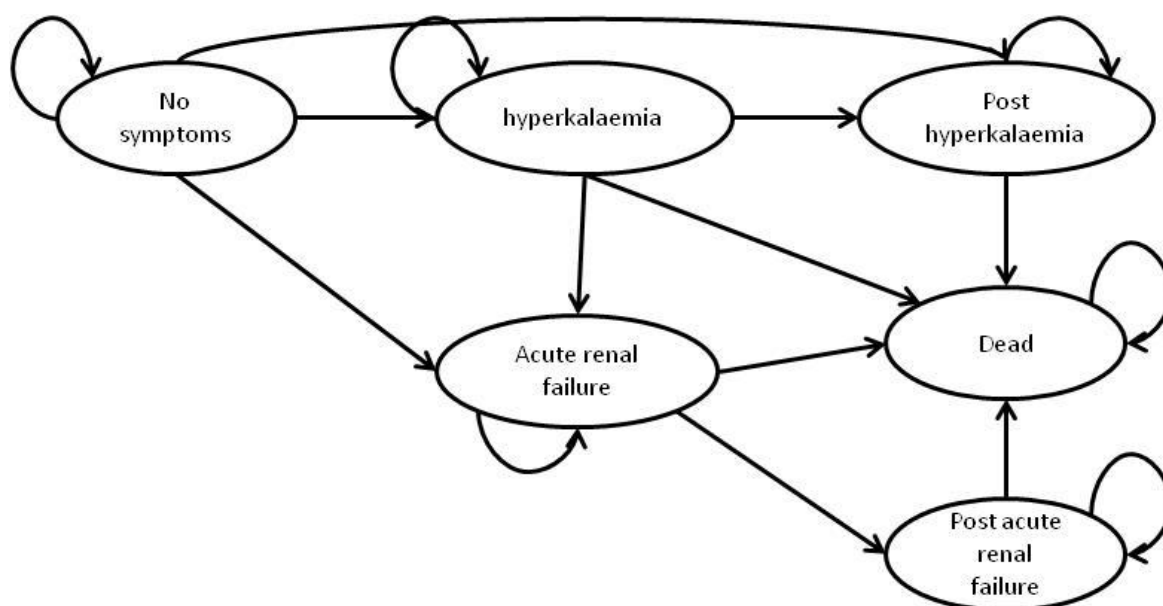


Table 8 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups for ACEI

Transition probability	Not monitored	Monitored
No symptoms → No symptoms	0.979 (1-0.016-0.001-0.004)*	0.988 (1-0.008-0.0005-0.004)*
No symptoms → hyperkalaemia	0.016**	0.008 ⁵⁷
No symptoms → ARF	0.0010 ^{58 59}	0.0005 ⁶⁰
Hyperkalaemia → Post Hyperkalaemia	0.996 (1-0.004)*	
ARF → Post ARF	0.974 (1-0.026)*	
No symptoms → Dead	0.004 ^{61 62}	
Hyperkalaemia → Dead	0.004**	
Post hyperkalaemia → Dead	0.004**	
ARF → Dead	0.026 ⁶³	
Post ARF → Dead	0.026 ⁶³	

* Net of other probabilities at this node

**derivation provided in Appendix 3

Table 9 Summary of utility weights and cost per health state for ACEI model

Health state	Mean (SE) utility weights	Mean (SE) cost per patient /£
No symptoms	0.78 (0.013) ⁶⁴	39 (2) ^{65 66}
Hyperkalaemia	0.60 (no range reported) ⁶⁷	1480 (74) ⁶⁸
Acute renal failure	0.44 (0.32) ⁶⁹	3043 (152) ⁶⁸
No hyperkalaemia post hyperkalaemia	0.73 (0.19) ⁷⁰	117(6) ^{65 66}
No ARF post ARF	0.60 (no range reported) ⁷¹	117(6) ^{65 66}
Death	0	0

4.1.4 Patients receiving methotrexate for at least three months who have not had a recorded full blood count and/or liver function test within the previous three months

The model structure is presented in Figure 6, transition probabilities, health states and costs are summarised in Table 10 and Table 11.

Figure 6 Markov model for patients receiving methotrexate for at least three months who have not had a recorded full blood count and/or liver function test within the previous three months

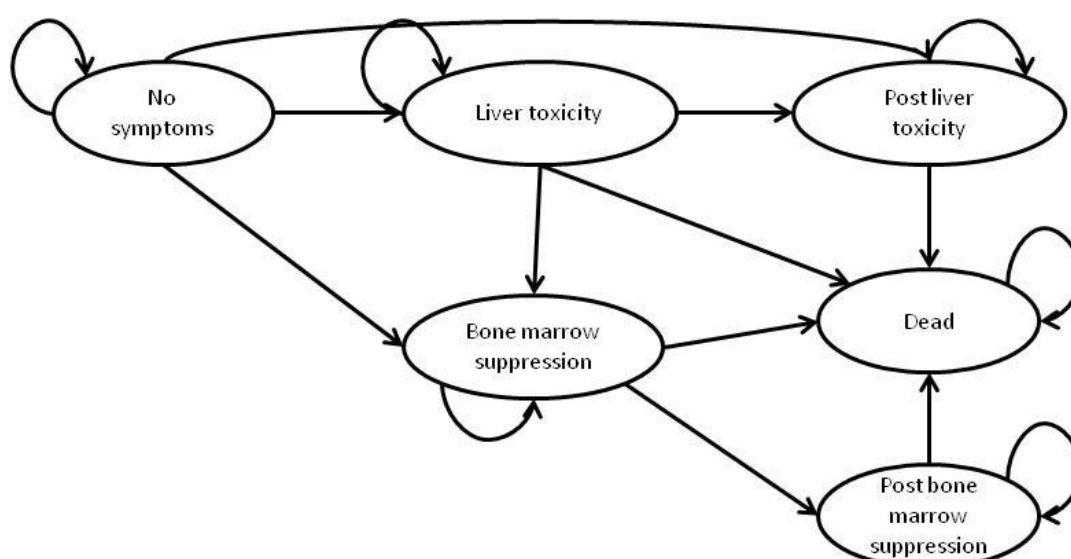


Table 10 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups (methotrexate)

Transition probability	Not monitored	Monitored
No symptoms → No symptoms	0.9474 (1-0.0434-0.0038-0.0054)*	0.9686 (1-0.0228-0.0032-0.0054)*
No symptoms → Liver Toxicity	0.0434 ⁷²	0.0228 ⁷³
No symptoms → BMS	0.0038 ⁷⁴	0.0032 ⁷³
BMS → Post BMS	0.9270 (1-0.0730)*	
No symptoms → Dead	0.0054 ⁷⁵	
Liver Toxicity → Post Liver Toxicity	0.9946 (1-0.0054)*	
Liver Toxicity → Dead	0.098 ⁷⁶	
BMS → Dead	0.0730 ⁷⁷	
Post-BMS → Dead	0.0054 ⁷⁵	

BMS: bone marrow suppression

* Net of other probabilities at this node

**derivation provided in Appendix 4

Table 11 Summary of utility weights and cost per health state for methotrexate model

Health state	Mean (SE) utility weights	Mean (SE) cost per patient /£
No symptoms	0.90(0.20) ⁷⁸	40 (2) ^{65 79}
Liver toxicity	0.76 (0.02) ⁸⁰	2472 (124) ⁶⁸
Bone marrow suppression	0.75*	2776 (6) ⁶⁸
No liver toxicity post liver toxicity	0.84*	118 (139) ^{65 79}
No BMS post PMS	0.80*	118 (6) ^{65 79}
Death	0	0

**derivation provided in Appendix 4

4.1.5 Patients receiving lithium for at least three months who have not had a recorded check of their lithium levels within the previous three months

The model structure is presented in Figure 7, transition probabilities, health states and costs are summarised in Table 12 and Table 13.

Figure 7 Markov model for patients receiving lithium for at least three months who have not had a recorded check of their lithium levels within the previous three months

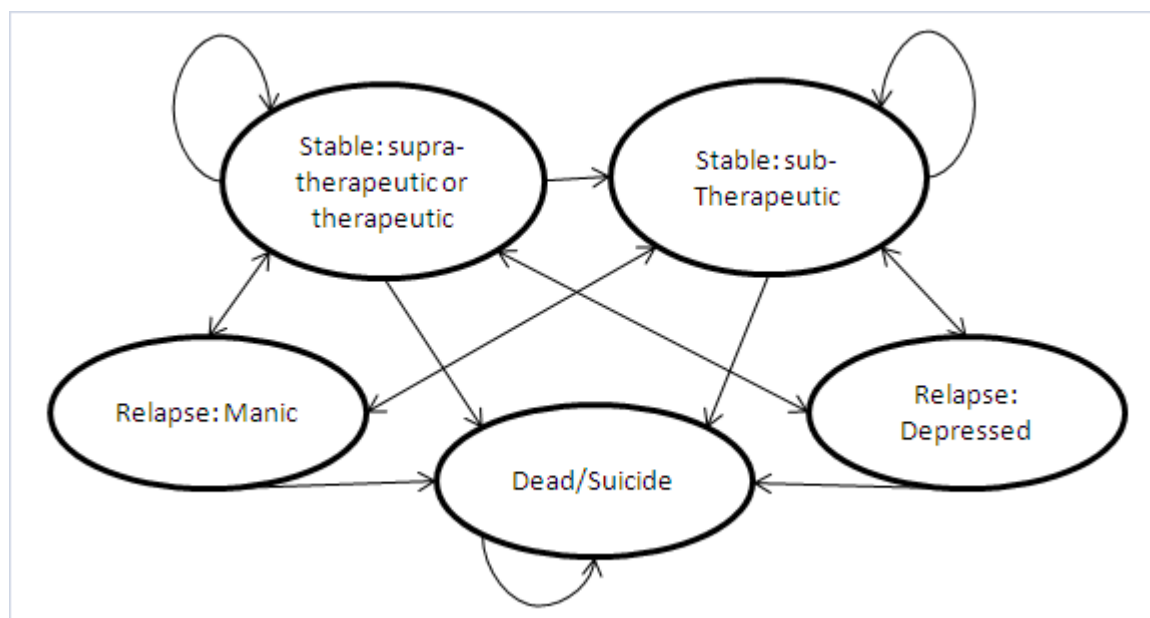


Table 12 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups (lithium)

Transition probability	Not monitored	Monitored
Supra or therapeutic → Supra or therapeutic	0.6089* (1-(0.2463+0.0427+0.0987+0.0034))	0.7113* (1-(0.1440+0.0427+0.0987+0.0034))
Supra or therapeutic → Sub-therapeutic	0.2463 ⁸¹	0.1440 ⁸²
Subtherapeutic → subtherapeutic	0.8147* (1-(0.0725+0.0037+0.1091))	
Subtherapeutic → relapse: manic	0.0725 ⁸³	
Subtherapeutic → relapse: depressed	0.1091 ⁸³	
Subtherapeutic → suicide/dead	0.0037 ^{48 84}	
Supra or therapeutic → relapse: manic	0.0427 ⁸³	
Supra or therapeutic → relapse: depressed	0.0987 ⁸³	
Supra or therapeutic → suicide/dead	0.0034 ^{48 84}	
Relapse: manic → sub-therapeutic	0.1440 ⁸²	

Relapse: manic → supra or therapeutic	0.8530 ⁸²
Relapse: manic → suicide/dead	0.0030 ⁴⁸
Relapse: depressed → sub-therapeutic	0.1440 ⁸²
Relapse: depressed → supra or therapeutic	0.8530 ⁸²
Relapse: depressed → suicide/dead	0.0030 ⁴⁸

* Net of other probabilities at this node

Table 13 Summary of utility weights and cost per health state for lithium model

Health state	Mean (SD) utility weights	Mean (SE) cost per patient /£*
Stable: sub-therapeutic	0.74 (0.23) ⁸⁵	0 (Error) 16 (2) (No error)
Stable: supra/therapeutic	0.71 (0.22) ⁸⁵	192 (13) (Error) 208 (13) (No error)
Mania (OP) mild relapse**	0.56 (0.27) ⁸⁵	5862 (569)
Mania (IP) mild relapse**	0.26 (0.29) ⁸⁵	7822 (952)
Mania (OP) moderate relapse**	0.54 (0.26) ⁸⁵	5862 (568)
Mania (IP) moderate relapse**	0.23 (0.29) ⁸⁵	7822 (952)
Depression (OP) mild relapse~	0.70 (0.20) ⁸⁶	7248(707)
Depression (IP) mild relapse~	0.33 (0.36) ⁸⁶	9697 (952)
Depression (OP) moderate relapse~	0.63 (0.19) ⁸⁶	7248(707)
Depression (IP) moderate relapse~	0.27 (0.34) ⁸⁶	9697 (952)
Death	0	268 (27)

IP inpatient; OP outpatient

*derivation provided in Appendix 5

**20% patients treated as outpatients, 80% patients treated as inpatients

~90% patients treated as outpatients, 10% patients treated as inpatients

4.1.6 Patients receiving amiodarone for at least six months who have not had a thyroid function test within the previous six months

The model structure is presented in Figure 8, transition probabilities, health states and costs are summarised in Table 14 and Table 15.

Figure 8 Markov model for patients receiving amiodarone for at least six months who have not had a thyroid function test within the previous six months

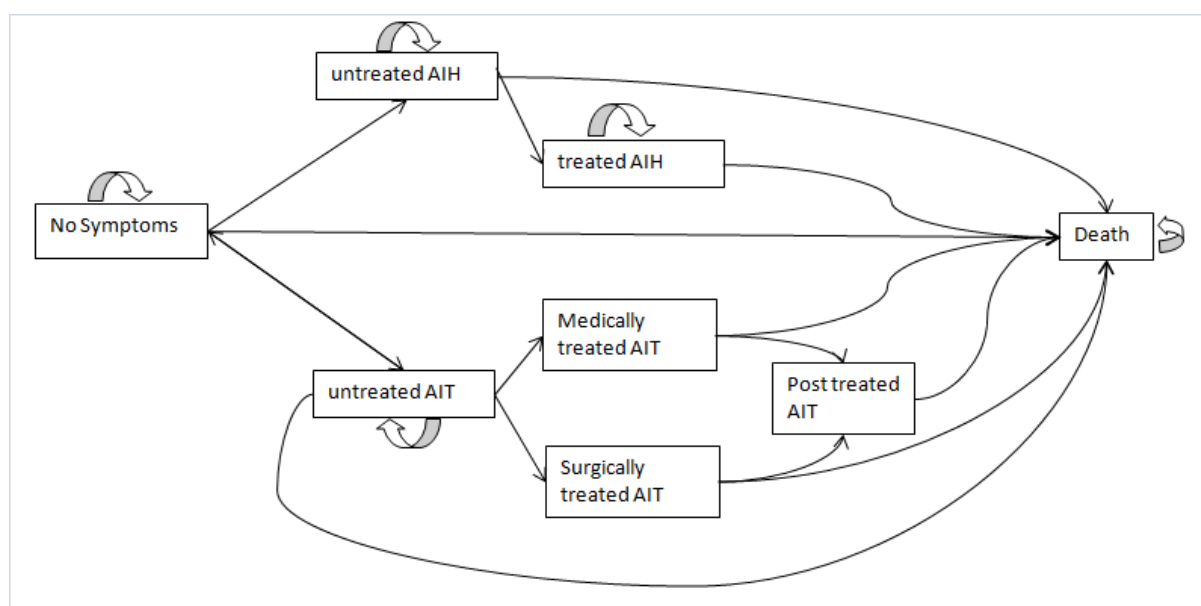


Table 14 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups (amiodarone)

Transition probability	Not monitored	Monitored
AIT untreated → AIT surgical management	0.0081*	0.081 ⁸⁷
AIT untreated → AIT medical management	0.0988*	0.9879**
AIT untreated → AIT untreated	0.8964**	0*
AIH untreated → AIH medical management	0.0995*	0.9945**
AIH untreated → AIH untreated	0.8950**	0*
No Symptoms → No Symptoms	0.9622**	

No Symptoms → AIT untreated	0.0233 ⁸⁸
No Symptoms → AIH untreated	0.0115 ⁸⁸
No Symptoms → Death	0.0035 ^{48 89}
AIT untreated → Death	0.0400 ^{48 89 90}
AIT surgical management → Post treated AIT	0.9083**
AIT surgical management → Death	0.0917 ^{48 89 91}
AIT medical management → Post treated AIT	0.9965**
AIT medical management → Death	0.0035 ^{48 89}
Post treated AIT → Post treated AIT	0.9965**
Post treated AIT → Death	0.0035 ^{48 89}
AIH untreated → Death	0.0055 ^{48 89 92}
AIH treated → AIH treated	0.9965**
AIH treated → Death	0.0035 ^{48 89}

*derivation provided in Appendix 6

**Net of other probabilities at this node

Table 15 Summary of utility weights and cost per health state for amiodarone model

Health state	Mean (SE) utility weights**	Mean (SE) cost per patient /£**
No Symptoms	0.78 (0.21) ⁹³	Error 91 (5) No error 93 (5)
Untreated AIH	0.60 (0.21) ⁹⁴	Error 91 (5) No error 93 (5)
Treated AIH	0.65 (0.21) ⁹⁵	151 (8)
Untreated AIT	0.58 (0.21) ⁹⁴	Error 143 (7) No error 145 (7)
Medically treated AIT	0.76 (0.21) ⁹⁵	339 (17)
Surgically treated AIT	0.73 (0.21) ⁹⁶	3028 (151)
Post-treated AIT	0.76 (0.21) ⁹⁵	95 (5)
Death	0	0

**derivation provided in Appendix 6

4.1.7 Summary of outputs for outcome-measure specific models

Table 16 provides a summary of key cost and outcome parameters derived from each outcome measure-specific model, for inclusion in the overall composite-error model. The ICER planes (cost per QALY gained) for each outcome measure-specific model are presented in Figures 9 to 14.

Table 16 Summary of key cost and outcome parameters derived from each outcome measure-specific model

Error	QALYs (SE) generated per patient		Cost/£ (SE) per patient	
	Error	Non-error	Error	Non-error
NSAID	3.89 (0.002)	3.89 (0.002)	2031 (200)	1663 (308)
Beta-blocker	2.90 (0.23)	3.00 (0.26)	728 (410)	309 (255)
ACEI	3.40 (0.28)	3.43 (0.29)	1688 (501)	1456 (504)
Methotrexate	3.83 (0.24)	3.92 (0.26)	2354 (571)	1757 (544)
Lithium	3.05 (0.60)	3.05 (0.60)	12270 (1585)	11723 (1716)
Amiodarone	3.43 (0.001)	3.51 (0.001)	1843 (252)	1876 (260)

Figure 9 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present (NSAID)

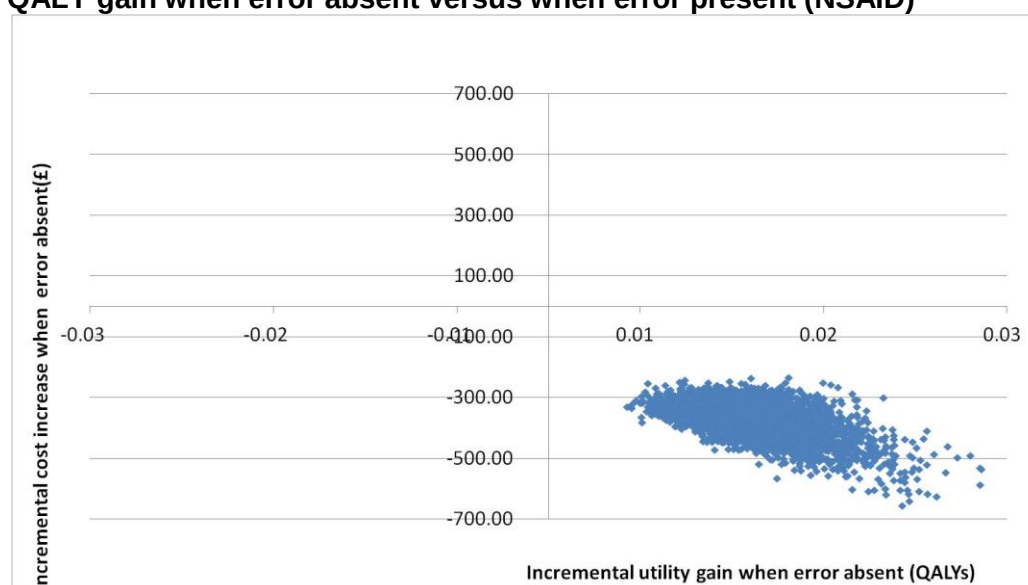


Figure 10 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present (Beta-blocker)

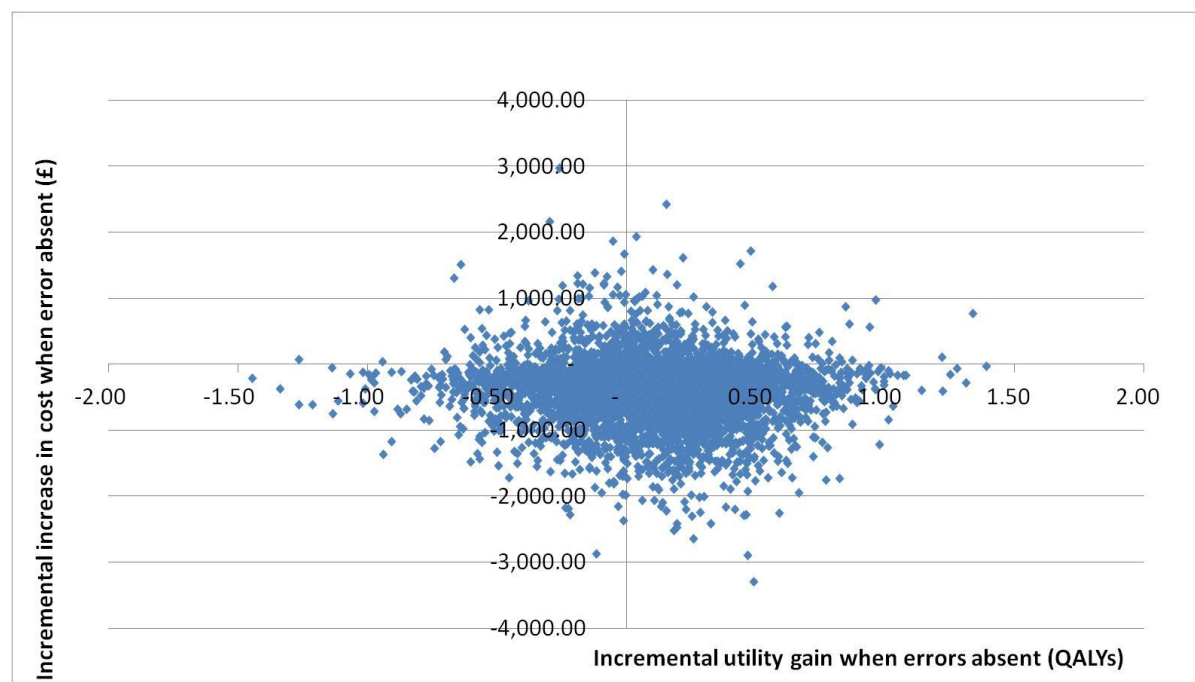


Figure 11 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present (ACEI)

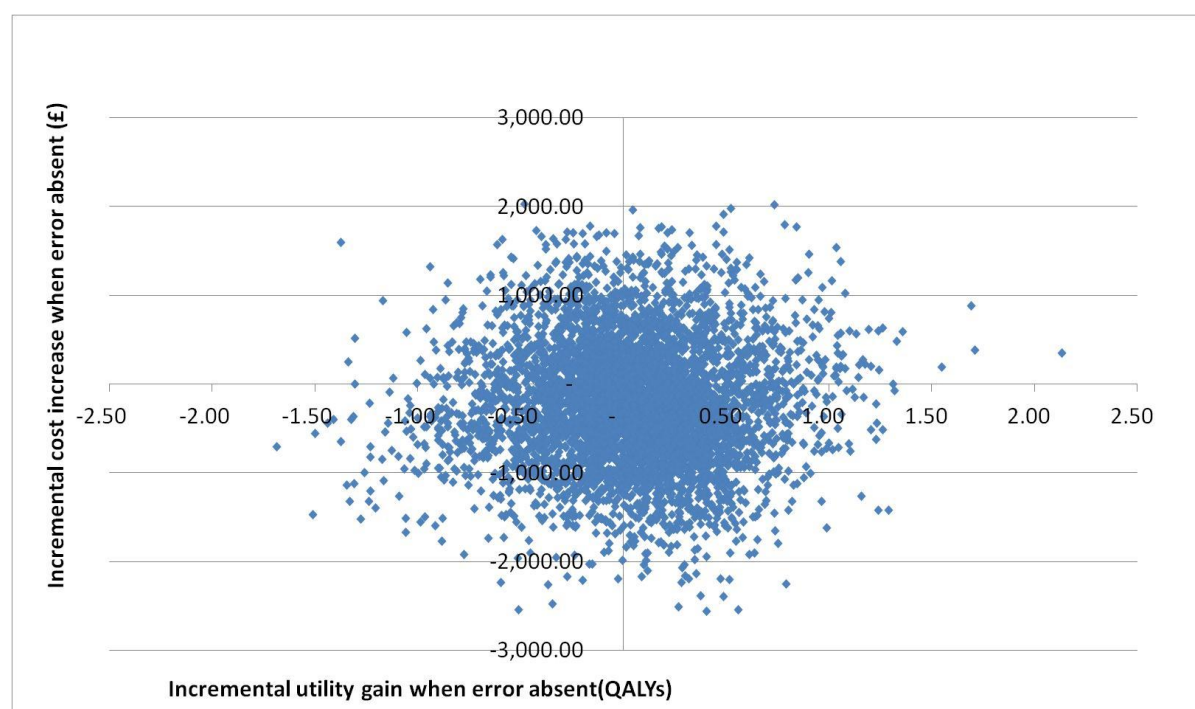


Figure 12 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present (Methotrexate)

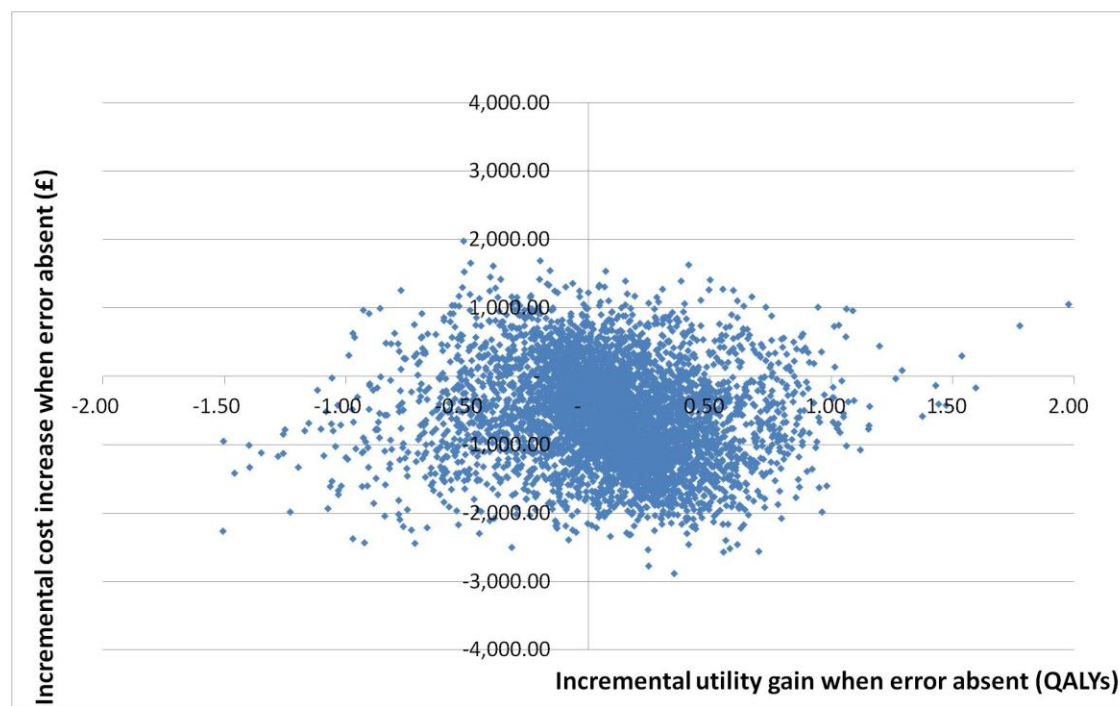


Figure 13 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present (Lithium)

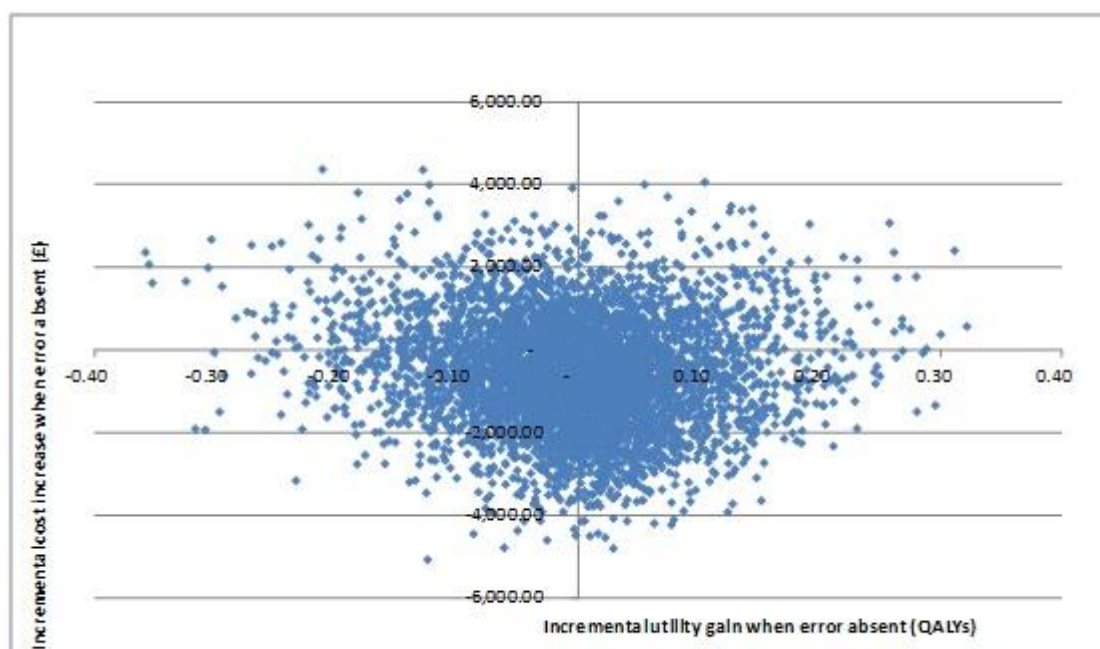


Figure 14 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present (Amiodarone)

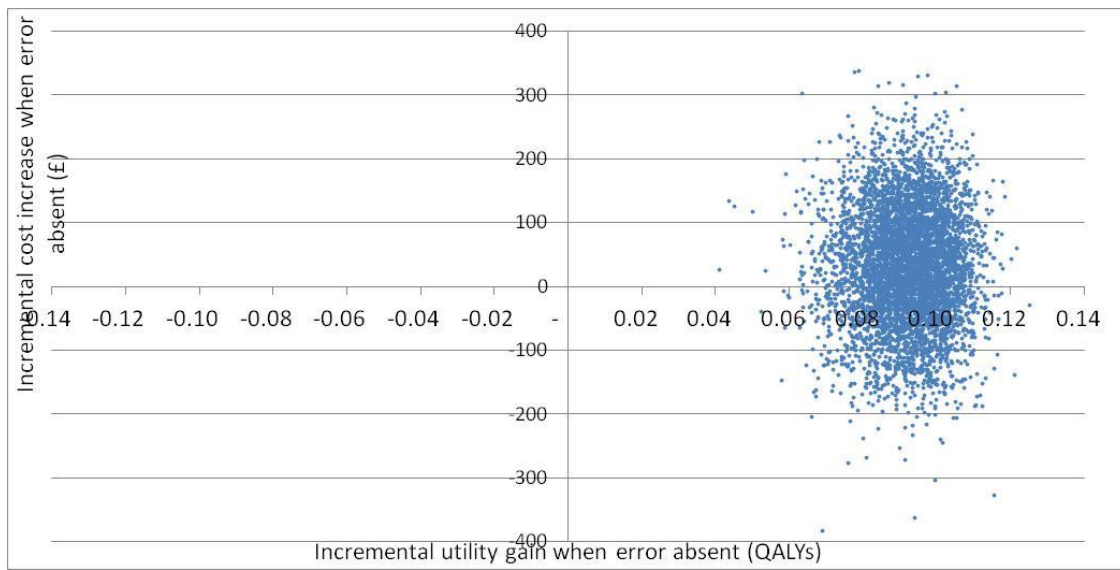
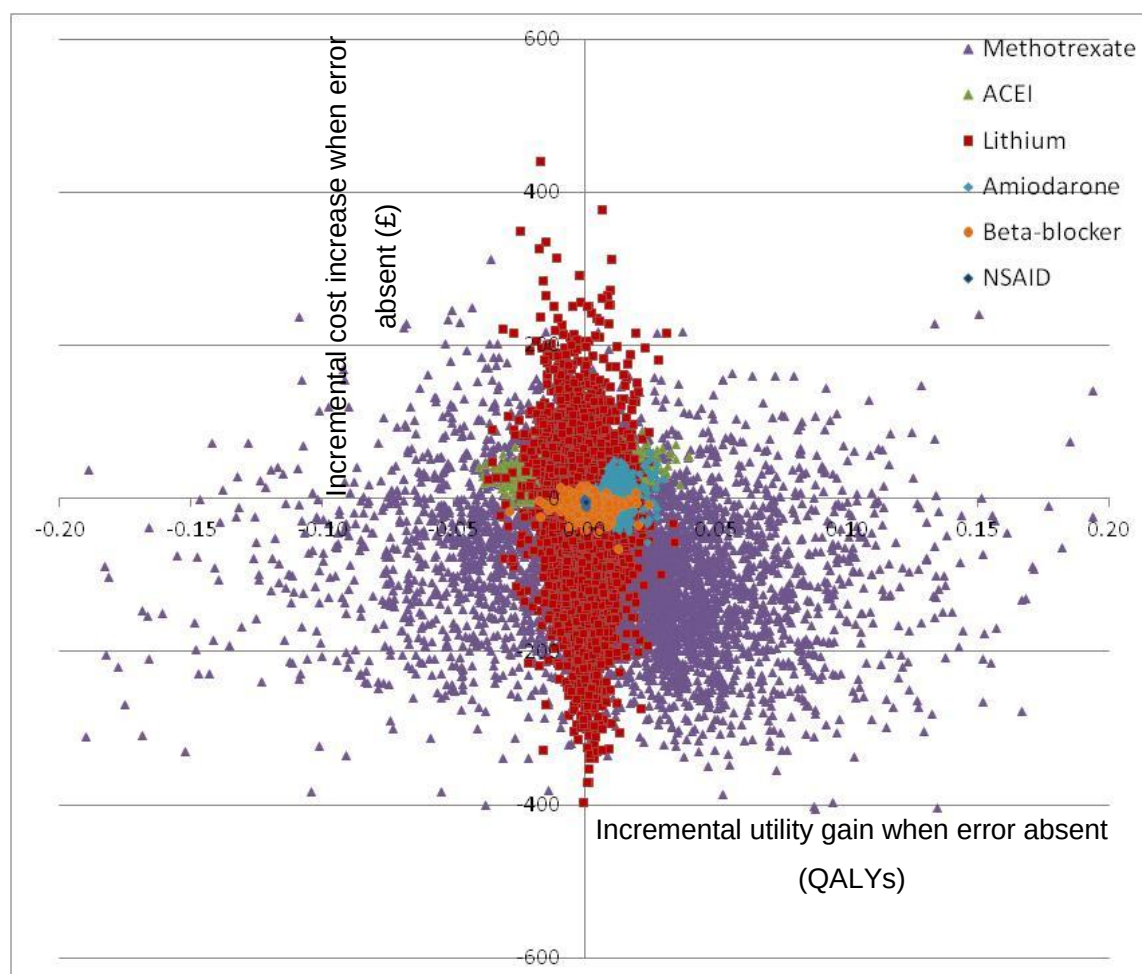


Figure 15 shows the relative magnitude and distribution of incremental costs and effects for each outcome, plotted on a common scale. This combined presentation of the ICER distribution for each error illustrates very clearly the different magnitudes of uncertainty around the point estimates of ICERs for the six errors. We return to the significance of this in Sections 4.2, 4.3 and the discussion.

Figure 15 Cost-effectiveness plane of probabilistic incremental costs and incremental QALY gain when error absent versus when error present for each outcome measure on a common scale



4.2 Incremental analysis of PINCER intervention

The treatment pathways were used to generate incremental cost and QALY, per practice, per error, given the error rates observed in the PINCER trial, summarised in Table 2.

Patients at risk of one of the errors included was 799 in the mean practice size (NSAID: 7%; Beta-blockers: 71%; ACEI: 16%; Methotrexate: 4%; Lithium: 1%; Amiodarone: 1%) and this was used in the base case analysis.

4.2.1 Deterministic incremental analysis

Table 17 presents an overview of the QALY's gained and costs for the intervention and simple-feedback practices (deterministic model). The PINCER intervention dominated simple feedback as it was £2611 less costly per practice and generated 0.81 extra QALYs per practice.

Table 17 Summary of inputs and ICERs generated for deterministic incremental analysis of PINCER intervention versus simple feedback.

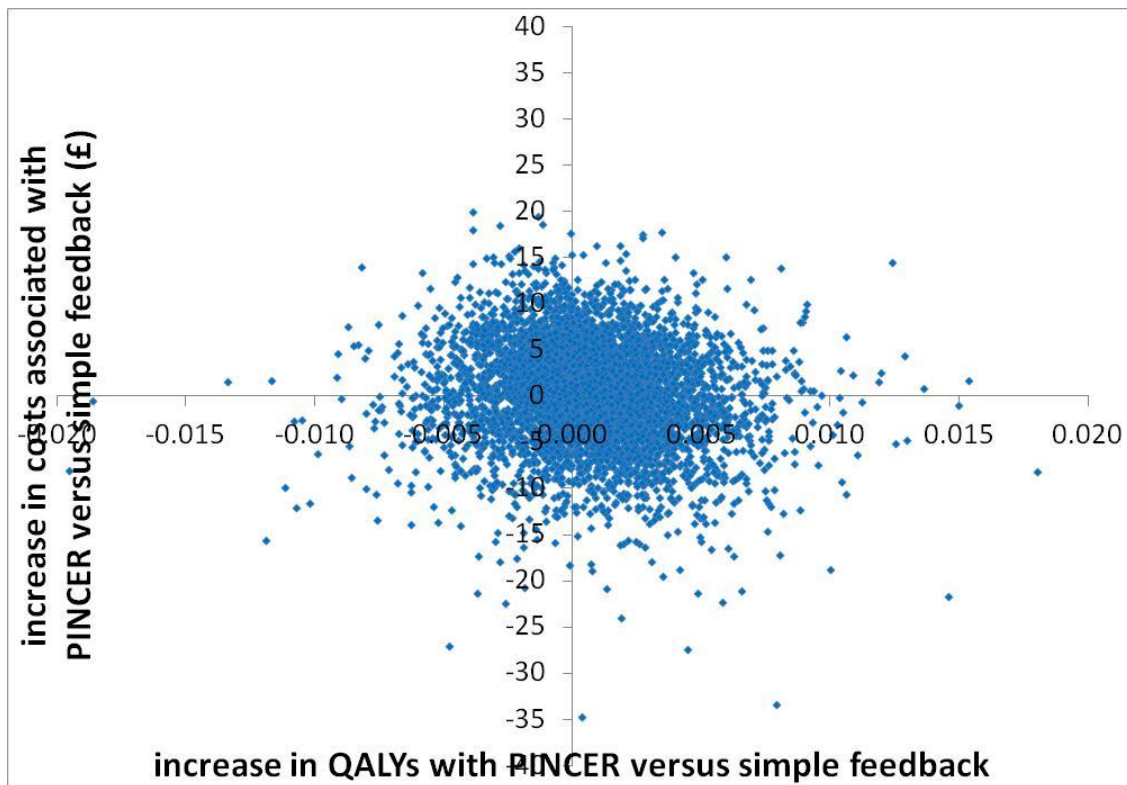
Error	Prevalence of patient group in practice	Simple feedback event rate per practice	RRR PINCER	QALYs generated per practice*		Cost/£ per practice		QALY difference per practice	Cost difference per practice (£)
				Simple feedback	PINCER	Simple feedback	PINCER		
NSAID	7%	0.04	0.35	256.6	256.6	95253	94939	0.01	-314
Beta-blocker	71%	0.03	0.17	1530.3	1530.5	241723	240759	0.26	-964
ACEI	16%	0.08	0.36	407.6	407.8	112326	111077	0.16	-1249
Methotrexate	4%	0.31	0.19	124.6	124.8	53792	52822	0.16	-968
Lithium	1%	0.40	0.11	24.2	24.2	95148	94940	0.00	-209
Amiodarone	1%	0.45	0.25	36.8	37.1	15838	16059	0.21	221
				Difference in intervention cost /practice					872
							Total	0.81	-2611
							ICER	-3,243	

RRR: relative risk reduction; *:QALYs and cost per practices are calculated for a practice with a population at risk of the six errors of 799 patients.

4.2.2 Probabilistic incremental analysis

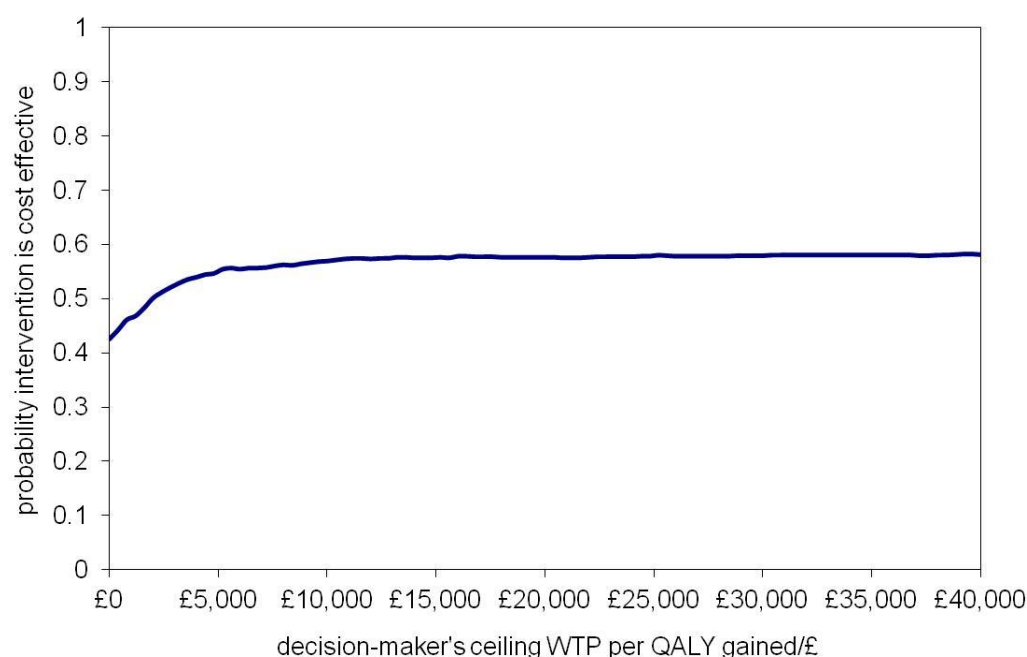
In the probabilistic analysis, the PINCER intervention was cost-saving. The mean ICER was -£2519 per QALY gained (SD 97,460; median -£159; 2.5th percentile: -£23,939; 97.5th percentile £21,767). Figure 16 illustrates the ICER distribution at 6 months. Negative ICERs are difficult to interpret and often arise when some or all of the ICERs fall in the SE or NW quadrant. It is not possible to tell this from the ICER itself. In this analysis, as can be seen in Figure 16 33% of the ICER estimates were in the NE quadrant (PINCER more effective, more costly), 27% were in the SE quadrant (PINCER more effective, less costly: “dominant”), 14% in the SW quadrant (PINCER less effective, less costly) and 26% were in the NW quadrant (PINCER less effective, more costly: “dominated”).

Figure 16 Incremental economic analysis of PINCER intervention versus simple feedback



The NE quadrant, with positive costs and positive effects, and the SW quadrant, with negative costs and negative effects, involve tradeoffs. These two quadrants represent the situation where the intervention may be cost-effective compared with the alternative, depending upon whether the ICER is above or below the decision-maker's maximum willingness to pay for an extra QALY. A CEAC shows the probability that an intervention is cost-effective compared with the alternative, given the observed data, for a range of maximum monetary values that a decision-maker might be willing to pay for a particular unit change in outcome.⁴² Figure 17 illustrates the CEACs at 6 months. There is 42% probability that the Pincer intervention is both more effective and less costly. This analysis suggests that at a ceiling willingness to pay of £20000, the Pincer pharmacist intervention reaches 59% probability of being cost effective. The probability of Pincer being cost effective does not increase beyond 59% irrespective of how high the decision-maker's ceiling willingness to pay becomes.

Figure 17 Cost effectiveness acceptability curve of PINCER intervention versus simple feedback



The net benefit statistic generated suggests a mean of £16 net benefit (SD £121; median £22; 2.5th percentile: -£218; 97.5th percentile £242) at λ of £20000.

4.3 Scenario and sensitivity analysis

The ICERs and CEACs were regenerated for:

- assuming that each of the errors was the only one targeted by the PINCER intervention, to assess the effect each error has on the cost effectiveness of the PINCER intervention (Figure 18),
- only including the primary outcomes of the PINCER intervention (Figure 19),
- prescribing and monitoring errors only (Figure 20),
- different costs of the PINCER intervention (Figure 21)
- different practice sizes (Figure 22).

Mean ICERs, percentage ICERs in each quadrant and probability of cost effectiveness at $\lambda < £20000$ base case, sensitivity and scenario analyses are summarised in Table 18.

Table 18 ICERs, percentage ICERs in each quadrant and probability of cost effectiveness at $\lambda < £20000$ for base case, sensitivity and scenario analyses

	Mean,SE ICER (£/QALY)	% ICERs in each quadrant				Prob. CE at $\lambda < £20000$
		NE	SE	SW	NW	
Base case	CS* (-2519, 1378)	33	27	14	26	59%
NSAIDs only	CS (-21731, 94)	1	99	0	0	99%
Beta-blocker only	CS (-2381, 3906)	28	37	18	17	64%
ACEI only	19140, 18008	41	1	1	58	35%
Methotrexate only	2060, 4654	9	55	26	10	67%
Lithium only	CS (-523544, 453550)	13	38	32	18	63%
Amiodarone only	475, 15	68	32	0	0	100%
Primary errors only	CS (-201, 1737)	46	6	4	44	46%
Monitoring errors only	CS (-143, 907)	26	40	20	15	53%
Prescribing errors only	9609, 9483	36	20	12	32	65%
Reduction in intervention costs						
-10%	CS (-2589, 1364)	33	28	14	25	59%
-20%	CS (-2659, 1352)	32	28	14	25	59%
-30%	CS (-2729, 1342)	32	29	15	25	60%
-40%	CS (-2799, 1334)	31	29	15	25	60%
-50%	CS (-2869, 1328)	31	30	15	24	60%
Number of patients at risk per practice (proxy for practice size), base case n=799.						
600	CS (-2286, 1438)	35	26	13	27	59%
700	CS (-2419, 1401)	34	27	13	26	59%
900	CS (-2597, 1363)	33	28	14	25	59%
1000	CS (-2659, 1352)	32	28	14	25	59%
1500	CS (-2846, 1330)	31	30	15	24	60%
2000	CS (-2939, 1325)	30	30	16	24	60%

*CS: cost saving

Figure 18 Cost effectiveness acceptability curve of PINCER intervention versus simple feedback for all included outcomes, aggregated and disaggregated

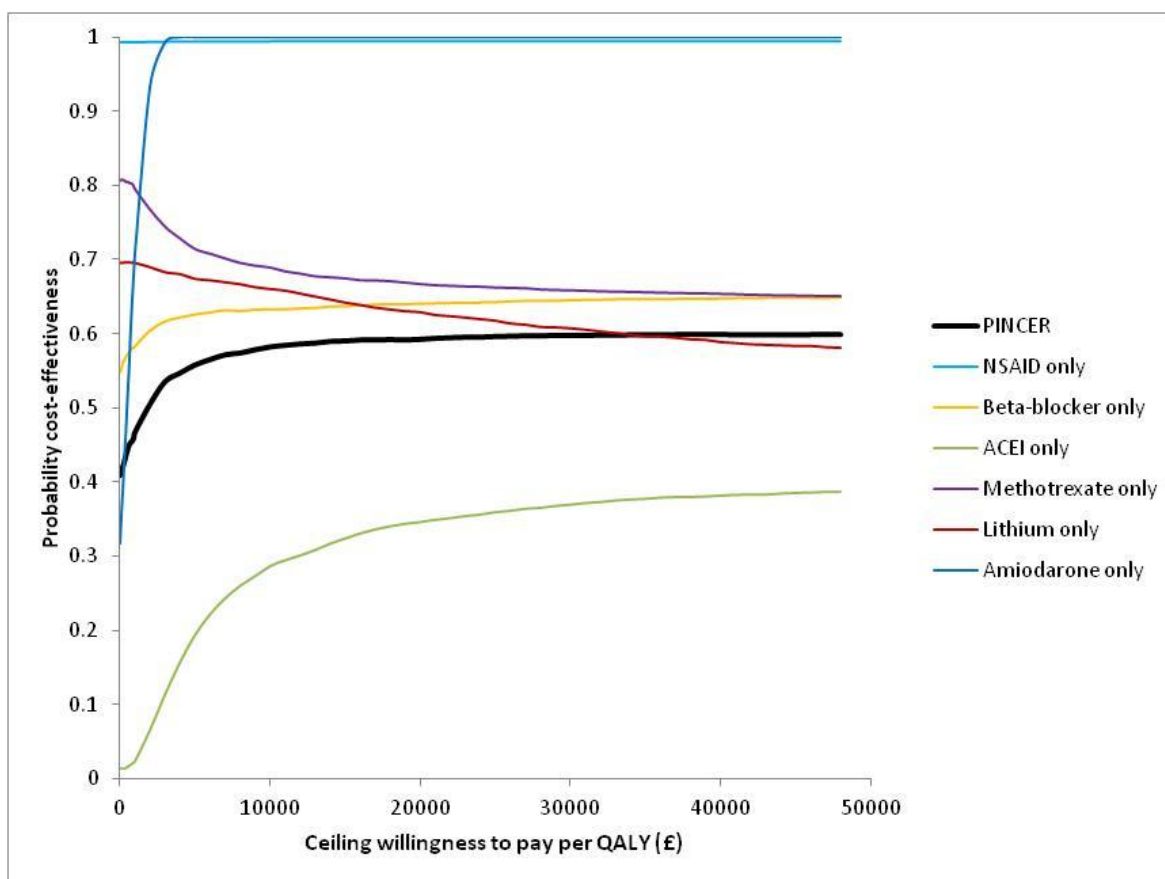


Figure 19 Cost effectiveness acceptability curve of PINCER intervention where only the primary outcomes are included

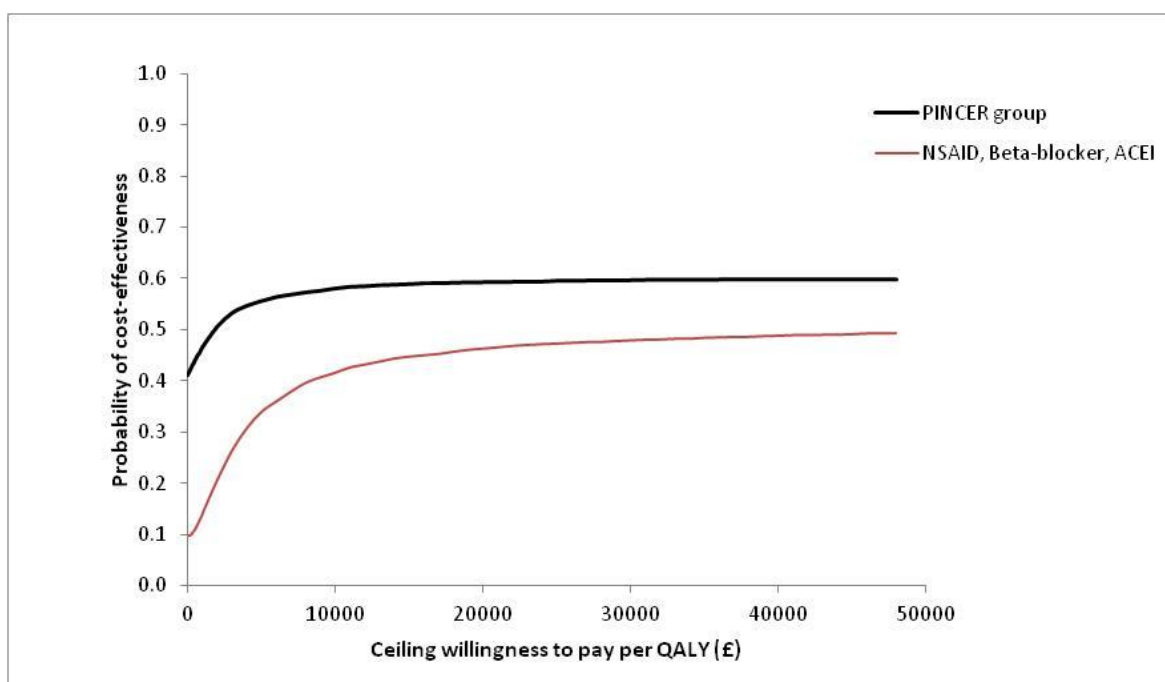


Figure 20 Cost effectiveness acceptability curve of Pincer intervention where the prescribing and monitoring outcomes are considered separately

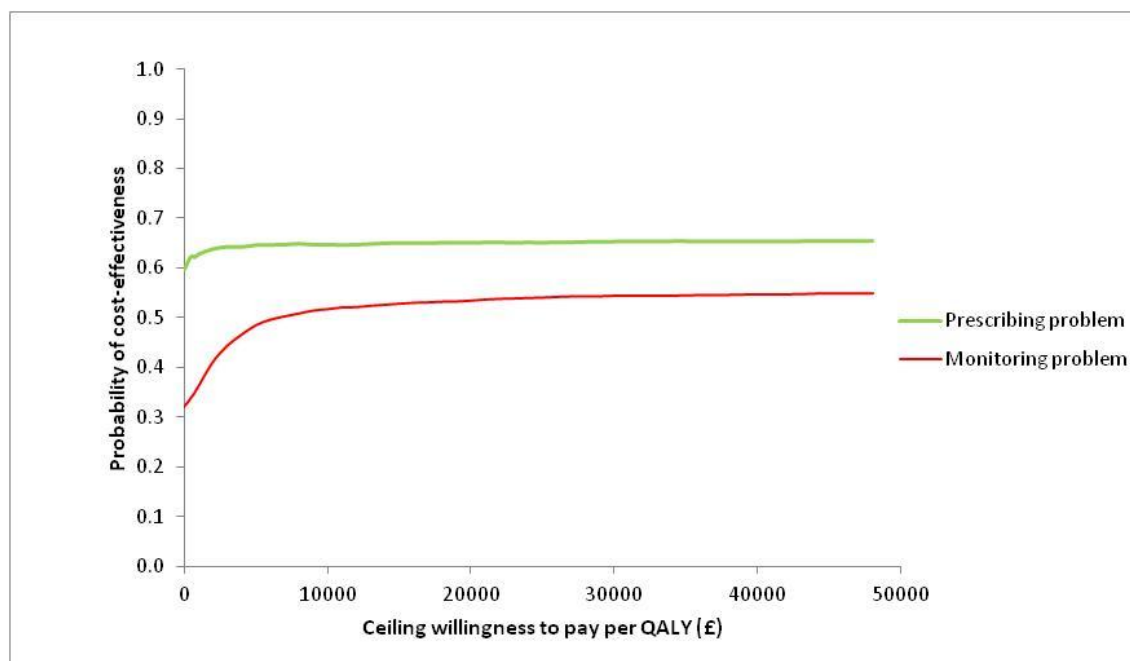


Figure 21 Cost effectiveness acceptability curve of Pincer intervention where cost of the intervention is varied

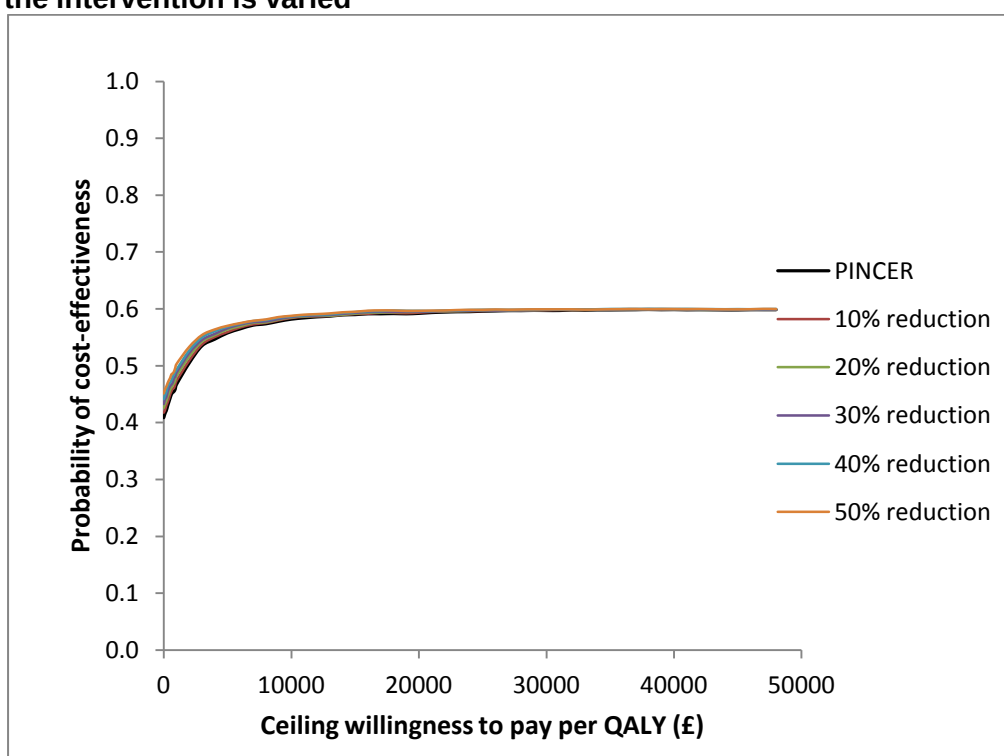
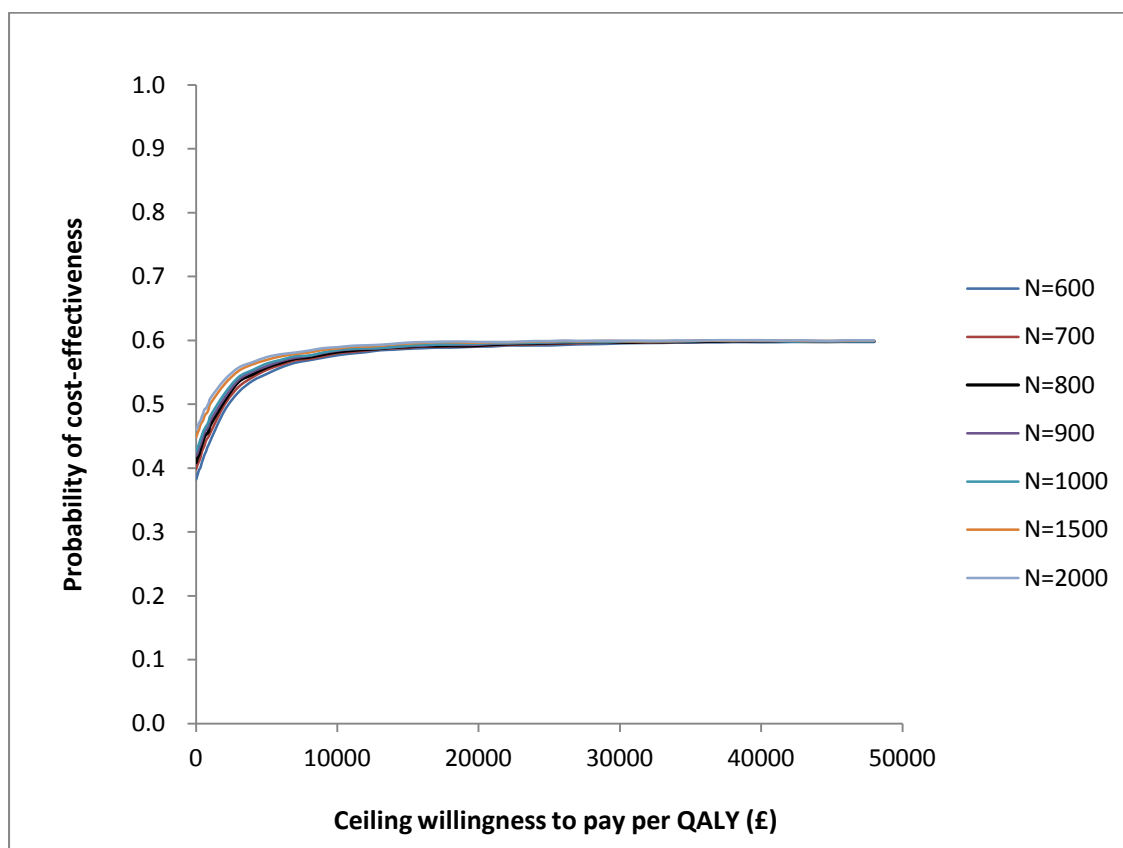


Figure 22 Cost effectiveness acceptability curve of Pincer intervention where practice size is varied



5 Discussion

5.1 Key findings from individual models

The errors associated with primary outcomes for the Pincer trial all lead to an increase in QALY production and a reduction in overall cost. This result confirms the Pincer team's rationale for considering that these were the three key errors to focus upon.

However, the model examining the effect of beta-blocker prescribing in asthma was associated with very large levels of uncertainty. This is because evidence that cardioselective beta-blockers lead to significant levels of asthma exacerbation was not found when we were building the model. A recent Scottish study found that prescribing new oral beta-blockers in people with asthma did not lead to large increases in patients treated with oral steroids acutely in general practice.⁹⁷ Beta-blockers are no longer contraindicated in COPD because of the known cardioprotective effect,⁹⁸⁻¹⁰⁰ but the use of beta-blockers in asthma is still contra-indicated. A recent study suggests that while short term use of beta-blockers in people in asthma causes airways constriction, this effect does not remain during long term use.¹⁰¹ In fact, some studies now suggest that beta-blockers may actually have a therapeutic (anti-inflammatory) effect in asthma and COPD, other than just through cardioprotection in those patients with concomitant cardiovascular disease.^{99 102} This has led to some clinicians starting to rethink their position of not prescribing beta-blockers in people with asthma.¹⁰³ The lack of empirical evidence to support prescribers highlights the need for more research to investigate the extent to which (different types of) beta-blockers cause problems in patients with asthma and (in carefully monitored patients) the potential benefits of beta-blockers in patients with asthma and CHD/heart failure (as has been done for COPD).

There is a mixed picture when the errors associated with the secondary outcomes are considered. The models examining the effect of lithium and methotrexate monitoring were associated with very large levels of uncertainty. Better management with methotrexate does appear to lead to increased QALY production with an overall reduction in cost. Better management with lithium does not appear to have much effect on QALYs, but leads to an overall reduction in cost. This may reflect that regular monitoring of lithium reduces the frequency of hospital admissions for manic or depressive relapse, but does not have much impact on chronic health state, which is generally poor in people with bipolar disorders. Better management with amiodarone does appear to lead to increased QALY production but

with an overall increase in cost. This may reflect that better monitoring of amiodarone leads to patients with thyroid dysfunction being picked up more quickly. As a small proportion of these cases will have thyroidectomy, this will incur extra cost. However, this procedure has a 10% perioperative mortality rate, but the overall utility gain for the cohort is positive if the error is avoided.

5.2 Key findings from composite error PINCER model

In the probabilistic analysis, the PINCER intervention was cost-saving. The mean ICER was negative at -£2519 per QALY gained (SD 97,460; median -£159; 2.5th percentile: -£23,939; 97.5th percentile £21,767), and ICERs were distributed across all four quadrants of the cost effectiveness plane, so it is difficult to interpret. At a ceiling willingness to pay of £20,000, the PINCER pharmacist intervention reaches 59% probability of being cost effective. The probability of PINCER being cost effective does not increase beyond 59%. The net benefit statistic generated suggests a mean of £16 net benefit (SD £121; median £22; 2.5th percentile: -£218; 97.5th percentile £242) at λ of £20000. The mean cost per QALY generated suggested that PINCER increased health gain at a cost per QALY well below most accepted thresholds for implementation. However, the range around this ICER is extremely wide, reflecting the large degree of uncertainty around effect in some of the individual outcome models. This uncertainty translates into the probability of cost effectiveness never reaching 90% and the net benefit statistic, whilst having a positive mean, having a range that incorporates both positive and negative values. Varying the cost of the intervention or the practice size had a negligible effect on results.

Investigation of the effect each outcome has on the cost effectiveness of PINCER demonstrates that correcting errors in NSAID prescribing alone and amiodarone monitoring alone would generate 95% probabilities of PINCER being cost effective at £10,000 and £0 per QALY gained, respectively. However, correcting errors in beta-blocker prescribing, ACEI, diuretic, lithium and methotrexate monitoring does not appear to be cost effective, within current thresholds for cost effectiveness. Because NSAID prescribing and amiodarone monitoring account for only 8% of the overall errors corrected per practice, the effects are swamped by the effects from the other errors. The quality of the evidence for the clinical and economic impact of NSAID prescribing and amiodarone monitoring errors is better than for the other errors. The errors examined that do not demonstrate improved health outcomes in the individual models have a poor level of evidence around their clinical and economic impact.

5.3 Strengths and limitations

This economic analysis has included the costs or outcomes that may have been incurred as a result of the errors addressed by the intervention, giving an estimate of the clinical and economic impact of the intervention. Given the current state of evidence around the economic impact of error-reducing interventions, this is a development.

The key limitation of this analysis is the paucity of data upon which to base the estimates of economic impact of the individual errors. Further work is needed to quantify the actual clinical and economic effect of prescribing and monitoring errors, to provide better data to populate the models. Analysis of clinical databases might help us estimate more accurately the costs and benefits of different patterns of care.

As in the within-trial analysis, the costs of the simple feedback and pharmacist intervention arms were assumed to reflect how the interventions would be implemented in practice. There are also many models of this type of service provision, which may affect costs. This economic analysis did not include any intervention costs other than those incurred as a direct result of the intervention. These costs assume no time spent by the practice dealing with errors in both the simple feedback and pharmacist intervention arm. It is not clear which arm this would favour. However, this means that the costs presented are an underestimate of the real cost to the practice.

5.4 Using economic evaluation to evaluate safety in health care

There is an increasing need to assess the value of safety improvements to society. Somewhat lagging behind other industries associated with risky outcomes, there is finally emerging an increasingly evident “safety culture” in health care.¹⁰⁴ The perception of safety in health care has moved from a person-centred paradigm where an individual was blamed, to a system centred paradigm where errors are seen to be expected in a complex system, so reducing errors requires a system-wide approach.¹⁰⁵ However, preventing errors and adverse events entirely can be argued to be infeasible due to the prohibitive costs involved,^{10 105 106} such that there are diminishing returns associated with increased effort required (or resources consumed) to prevent harm from adverse events.¹⁰ An example of this given by Brennan is the impracticability of testing all patients for allergies to antibiotics.¹⁰⁷ In fact this suggests that preventability of adverse events is determined to some extent by affordability.¹⁰ Therefore, the cost effectiveness of safety interventions should be integral to their development, implementation and assessment to allow

prioritisation of spending on suggested safety improvements.¹⁰⁵ However, there are very few empirical examples of this.

There may be barriers to the usefulness of cost effectiveness to justify the expense associated with the intervention. For example, in the case of PINCER, the costs are incurred in primary care for the intervention, but the costs saved may occur in secondary care. Also the costs saved and improved outcomes may be downstream from the initial expenditure, a situation not likely to be compatible with return on investment calculations carried out on an annual basis.

Concerns exist, however, as to what extent standard health economic methods are able to appropriately evaluate interventions to improve safety.^{108 109} Cost effectiveness analysis generally includes direct medical costs and some measure of health consequences, such as quality-adjusted life years (QALYs). However, this conventional position that all we want from healthcare resources consumed is to produce health, is being increasingly challenged.¹¹⁰⁻¹¹³ Recent UK research on local and national decision-making shows that considerations of benefits unrelated to health outcome such as enhancing patient experience and empowerment, public trust and confidence and staff morale were used to make decisions about implementing services.¹¹³ This may be due to the cost effectiveness perspective of using QALYs to define health benefits being narrower than the perspectives of policy-makers, who also include non-health benefits in their deliberation.

Specific examples where health benefits may not drive implementation decisions include diagnostic procedures and interventions with wider social implications. The reassurance associated with a diagnostic procedure may outweigh any potential health benefits, such as ultrasound in normal pregnancy,¹¹² or the reassurance associated with prostate cancer screening.¹¹⁴ Another example of a non-health benefit reflecting the importance of the psychological benefits of “peace of mind” to patients is that associated with autologous blood donation, an intervention known to show very small health benefits for a substantial increase in cost.¹¹⁵ Another example of NHS resources being consumed to produce non-health benefits is interventions that have both individual and society-level consequences, such as those targeting drug¹¹⁶ and alcohol abuse.¹¹¹ Individual consequences are improved health and well-being. Society-level consequences are here defined as the consequences that arise from individuals’ drinking or drug-taking behaviour that can be influenced by an intervention, resulting in non-health outcomes valued by society such as crime reduction and increased engagement with housing, education and employment.¹¹⁶ The increasing use of clinical genetic services provide individuals and their families with knowledge about the

diagnosis, prognosis and risk of a disease, supporting future decisions about treatment choices and lifestyle, but the relative importance of health gain is often low.¹¹⁰

Conventional cost effectiveness analysis methods do not typically incorporate the non-health or extra-consequentialist value of these interventions. Similarly, interventions to improve safety by avoiding medication errors have non-health effects.¹⁰⁸ Errors may be associated with a decreased trust of patients and citizens in healthcare systems and providers, leading to reduced service uptake or political support,¹⁰⁸ lost productivity from healthcare professionals blamed for committing an error,¹¹⁷ and litigation and compensation costs.¹¹ From the perspective of society, compensation costs are simply a transfer from one part of society to another, so there is a zero gain or loss. However, from the perspective of the healthcare provider, these compensation costs can be catastrophic. Attributes of different types of errors unrelated to their effect on patient outcome, such as preventability and controllability, have been shown to be important to healthcare providers in one study.¹⁰⁸ This expressed preference suggests that allocating scarce resources to improve healthcare safety in the most cost-effective way must take account of the health and non-health components of safety outcomes. However, another study suggests that preventability is not valued highly by the public when assessing importance of interventions, so there is clearly more work to be done before this aspect of safety interventions is understood.¹¹⁸

5.5 Implications for policy makers and practitioners

This study suggests that this safety-orientated intervention could be cost effective, given current levels of evidence, if the “right” errors are targeted. Furthermore, it may be worth looking at other errors to see whether intervening in some of these might be more cost-effective than some of the measures used in our trial. For example, having a basket of NSAID errors might prove to be very effective, e.g. NSAIDs in older people without PPI, NSAIDs in renal impairment, NSAIDs in heart failure. However, restricting the PINCER intervention to only the potentially small number of errors where it proved to be strongly cost-effective ignores non-health benefits that may be obtained from PINCER. An alternative viewpoint is that the PINCER intervention works well as a package with a mix of outcomes to focus on, some of which appear to be highly cost-effective and others where there is nevertheless a consensus that it would be good practice to correct the potentially hazardous prescribing or failure to monitor. It is likely that practitioners would want to intervene whether or not it appears to be cost effective.

5.6 Priorities for future research

- Better understanding of the real clinical and economic impact of prescribing and monitoring errors to identify the errors where corrective action is likely to have the most chance of being cost-effective;
- Use of large datasets as one way to obtain better data on the clinical and economic impact of prescribing and monitoring errors as large prospective observational studies may be prohibitively expensive;
- Debate around methodological development of the assessment of safety interventions given their extra-consequentialist nature.

5.7 Conclusions

This is one of the first studies to attempt to estimate the real economic impact of a safety-focused intervention in health care. The mean ICER was £891 per QALY gained (SD 159,494; median -£184; 2.5th percentile: -£40,330; 97.5th percentile £35,275). The probability of PINCER being cost effective does not increase beyond 52% irrespective of how high the decision-maker's ceiling willingness to pay becomes. However, correction of some errors has a larger clinical and economic effect, such that this safety-orientated intervention could be cost effective, given current levels of evidence, if the "right" errors are targeted. Further work is required to address the economic impact of including other errors not included in the PINCER intervention. More importantly, the role of cost effectiveness in allocating resources to safety-focused interventions in health care needs to be examined and explored.

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7 Appendix 1: Patients with a past medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI.

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7.1 Introduction

Around 17 million items for non-steroidal anti-inflammatory drugs (NSAIDs) are prescribed annually in England alone.¹¹⁹ The most commonly prescribed non-selective NSAIDs are diclofenac and ibuprofen.¹²⁰ Between 2003 and 2008, prescribing of NSAIDs (excluding topical) decreased by 16% (to 4.3 million items) and costs have fallen 51% (to £27.3 million) in the last five years. Diclofenac is the most commonly prescribed NSAID, 1.9 million items per quarter (a 6% increase) costing £10.9 million (a 29% decrease). It accounts for 44% of all NSAID items and 40% of the cost. Ibuprofen accounts for 25% (1.1 million) of NSAID items (a 9% decrease) and 10% (£2.6 million) of the cost (a 19% decrease). Naproxen items have increased by 54% to 427,000, costing £2.3 million (a 19% increase). Prescribing of selective inhibitors of COX-2 decreased by 77% between 2003 and 2008 (to 248,000 items, £6.4 million). This represents 6% of NSAID items and 23% of the cost. Prescribing of celecoxib has fallen 74% to 127,000 items, costing £3.4 million (a decrease of 67%). Etoricoxib items have increased by 36% (121,000 items, £3 million). Meloxicam items remains stable, increasing by 5% to 261,000 items with a reduction in cost of 64% (£1.3 million). Items for etodolac have increased by 66% (to 80,000) and costs by 47% (£1.3 million).

These drugs are associated with upper gastrointestinal complications.¹²¹ For example, each year in the UK, NSAIDs cause about 3,500 hospitalisations for, and 400 deaths from, ulcer bleeding in people aged 60 years or above.¹²² Aspirin, even in low doses, is also associated with gastrointestinal complications.^{123 124} People are at high risk of serious NSAID-induced GI adverse events if they have one or more of the following risk factors: age 65 years or older (the risk is twice as high in men as in women); history of GI ulcer, bleeding, or perforation; concomitant use of drugs that increase the risk of GI adverse events; serious co-morbidity, such as CV or renal disease; requirement for prolonged NSAID use; and use of the maximum recommended dose of an NSAID.¹²⁵ A systematic review and meta-analysis investigated NSAIDs and serious GI complications.¹²¹ Pooled relative risks were calculated for different risk factors. NSAID users with advanced age or a history of peptic ulcer disease had the highest absolute risks for upper GI tract bleeding or perforation. Compared with patients aged 25 to 49 years, 60 to 69 year olds had 2.4 times the risk of a GI bleed or

perforation, 70 to 80 years had 4.5 times the risk, and patients over 80 years had 9.2 times the risk. Recent trials have influenced the NICE Osteoarthritis guideline development group to recommend that when offering treatment with an oral nonselective NSAID or a selective inhibitor of COX-2, the drug selected should be co-prescribed with a PPI, choosing the one with the lowest acquisition cost.¹²⁵

7.2 Aim of the study

The principal aims and objectives of this analysis are to:

- Identify and value the impact on patients' health status of patients with a past medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI, by identifying possible health states and the probabilities of making a transition from one to another 'health state';
- Identify and value the resource use associated with managing patients who are, and are not, prescribed ns-NSAID with a PPI;
- Assess the relative costs and outcomes for PPI prescribed and not prescribed group.

7.3 Literature search

The search strategy built on the systematic review and economic evaluation by Brown *et al.* (2006).^{126 127} Additionally, an inclusive search strategy was implemented using the key terms "peptic ulcer" and 'bleeding\$' in combination with NSAIDs (Pubmed, Embase range:1996-2010). Opinion papers were excluded. As the patient group under study is defined as "patients with a past medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI", all references were further screened on available evidence for this specific group. Only data on the defined subgroup were evaluated.

References in English and limited to humans were included. All prospective studies, as well as cohort studies and case control designs were included. The review focused on studies of adults (18 years or older). The focus of the analysis is on patients with a past medical history of peptic ulcer (PU). Data were gathered from studies which focused on this specific group of patients or from subgroups that were studies in more general patient groups. Only interventions with non-selective NSAID (nsNSAIDs) were considered. Treatment with PPIs or H₂ receptor antagonists (H₂RA) was not excluded.

7.4 Decision-analytic model for economic analysis

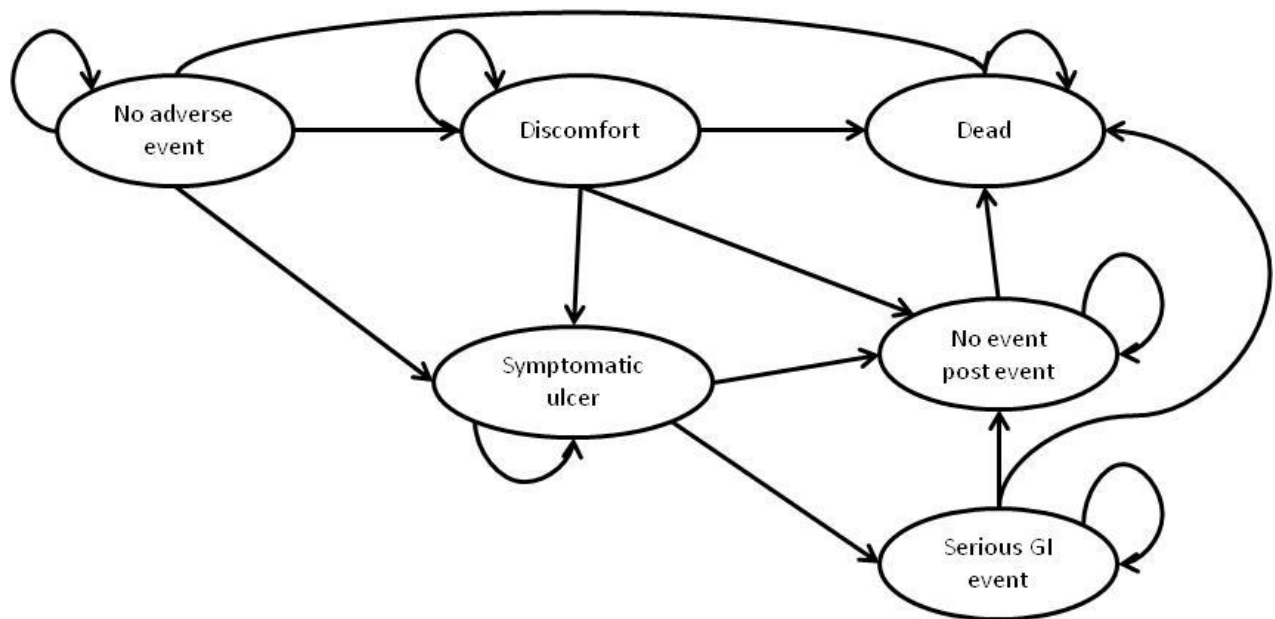
7.4.1 The decision-analytic model

The decision-analytic model describes the possible treatment pathways of patients with a past medical history of peptic ulcer prescribed a non-selective NSAID and no PPI.

There are many published decision-analytic models in relating to the use of NSAIDs (see Brown et al.¹²⁶). These models primarily concentrate on two areas: switching between NSAIDs and the point at which gastro-protective agents or COX-2 inhibitors need to be added to therapy. The ACCES Model (Arthritis Cost Consequence Evaluation System)⁴⁷ is a model that examines the use of different gastro-protective agents and COX-2 inhibitors. In the first model by Brown et al, four pathways were defined in using NSAIDs: no adverse event; discomfort; symptomatic ulcer and serious gastrointestinal events.¹²⁶ These specifications of possible consequences were believed not being different in our specific patient group and the main structure of events was kept in our model.

Figure 23 represents the decision-analytic model that is used in this analysis. The data requirements for population of this model can be divided into probabilistic, health status and resource use data.

Figure 23: Markov model for patients with a past medical history of peptic ulcer who have been prescribed a non-selective NSAID and no PPI



The model has a 3-month cycle.

7.4.2 Probabilities of moving from one state to another

Transitions between states were determined for the error (see Table 19) and non-error (see Table 20) group.

Table 19 Probabilities for the 3 month-cycle Markov model in the error group for NSAIDs

Transition probabilities for patients in the error-group										
	Transition to:									
Transition from:	No adverse event	GI Discomfort	Symptomatic ulcer	Serious GI event	No GI following initial event	further GI	Death			
No GI AE	0.829	0.154	0.0142	0.0002		0.0	0.0003			
Discomfort	0.0	0.188	0.0069	0.00015		0.802	0.0003			
Symptomatic ulcer	0.0	0.148	0.0183	0.00039		0.824	0.001			
Serious GI event	0.0	0.148	0.0183	0.00039		0.725	0.1083			
No GI post GI	0.0	0.0985	0.0001	0.0001		0.894	0.0003			
Death	0.0	0.0	0.0	0.0		0.0	1.00			

Table 20 Probabilities for the 3 month-cycle Markov model in the non-error group for NSAIDs

Transition probabilities for patients in the non-error-group									
	Transition to:								
Transition from:	No adverse event	GI Discomfort	Symptomatic ulcer	Serious GI event	No GI following initial event	further GI	Death		
No GI AE	0.894*	0.099	0.0047	0.0001		0.0	0.0003		
Discomfort	0.0	0.188	0.0069	0.00015		0.802*	0.0003		
Symptomatic	0.0	0.148	0.0183	0.00039		0.824*	0.001		

ulcer							
Serious	GI	0.0	0.148	0.0183	0.00039	0.725*	0.1083
event							
No GI	post	0.0	0.0985	0.0001	0.0001	0.894*	0.0003
GI							
Death		0.0	0.0	0.0	0.0	0.0	1.00

*1-(sum of other probabilities)

7.4.2.1 No GI adverse event -> discomfort

For the error group we assumed that patients were taking NSAIDs. The transition probabilities from no GI adverse event to GI discomfort were derived from Maetzel *et al.*⁴⁶ The data were taken from the MUCOSA study as this provides the largest sample size of patients, the most naturalistic design and the most comprehensive follow-up of treatment pathways.⁴⁶ Transition probabilities were calculated from the event rates, assuming a fixed rate over time.¹²⁸

7.4.2.2 No GI adverse event -> symptomatic ulcer

The transition probability from 'no GI adverse event' to symptomatic ulcer was also based on the data published by Maetzel *et al.*⁴⁶ Transition probabilities were calculated from the event rates, assuming a fixed rate over time.

7.4.2.3 No GI adverse event -> serious GI event

The transition probability from no GI adverse event to serious GI event was derived from the MUCOSA study and recalculated from the event rate.⁴⁶

7.4.2.4 No GI adverse event-> death

No information could be found on the transition probability from No GI adverse event to death for the specific patient group under study. We used the age standardised mortality rate for England and Wales⁴⁸ to calculate the transition probability. The age-standardised mortality rate was estimated at 11752 per 1,000,000 people. Based thereupon, the transition probability was calculated for a 3-month cycle.

7.4.2.5 Transitions after a first event

After the experience of any event (discomfort, symptomatic ulcer, serious GI event), we assumed that patients were treated as in the non-error group. Therefore transition probabilities were set equal in both groups.

7.4.2.6 No GI adverse event to other health states

The calculations of the transition probabilities were based on the risk reduction in the error group. In the study by Pettit et al (2000),⁴⁷ risk reductions were documented for patients who were prescribed an NSAID with PPI compared with patients with NSAIDs alone. These relative risks were multiplied with the estimated transition probability in the error group. The transition probability from 'No GI adverse event' to death was assumed to be same as in the error group, i.e. age standardised mortality.

7.4.2.7 Transition probabilities for a subsequent event after an event had already occurred

The transition probabilities after an event had occurred were calculated using the relative risks for subsequent events, given the initial event, published by Pettit et al (2002).⁴⁷ We assume that risks are fixed, independent how many events patients experienced previously.

7.4.2.8 Discomfort -> death

No information was found on the increased risk for death after an episode of discomfort. We assumed this risk was the same as for patients who have not experienced a GI event.

7.4.2.9 Symptomatic ulcer -> death

Probabilities were taken from Armstrong et al., 1987 in Ohmann *et al.* 2005.⁴⁹

7.4.2.10 Serious GI event-> death

Observational data on death rates were obtained from Blower and colleagues, which provided the most relevant, detailed and up-to-date information on death rates associated with hospitalisation for a gastric bleed in the UK.¹²⁶ These death rates were corresponded with death rates in Danish hospitals,^{129 130} adding external validity in generalisation of probabilities.

7.4.3 Required resource use and unit costs

For each unit cost used in the analyses, a detailed description of the source is provided in Table 21. Deterministic ranges had to be used for most parameters and correspond with 25th and 75th percentile of the unit cost found in the national database. The resource use is described below and is based on the current practice as described by Brown *et al.*¹²⁶

Table 21 Sources of unit costs for NSAIDs

Cost parameter	Units	Data source	Unit Cost (£)
Ibuprofen 400mg	Monthly cost	Drug Tariff ¹³¹	2.37
PPI: Omeprazole 20 mg (ddd)	Monthly cost	Drug Tariff ¹³¹	31.31
Omeprazole 40mg intravenous	Number of courses	BNF ¹³²	88.57
Helicobacter pylori test	Number of tests	BNF ¹³²	20.75
Diagnostic endoscopy	Number of procedures	Department of Health (ranges for 50% NHS trusts) ⁶⁸	418.00 (min: 347.00 max:728.00)
Therapeutic endoscopy	Number of procedures	Department of Health (ranges for 50% NHS trusts) ⁶⁸	462.00 (min: 354.00 max:761.00)
GP consultation costs	Number of consultations	Curtis 2010 ¹³³	34.00
Inpatient care gastroenterology	Number of days	Curtis 2010 ¹³³	256.00
Intensive care unit	Number of days	Department of Health (ranges for 50% NHS trusts) ⁶⁸	1716.00 (min: 1517.00 max: 1931.00)
Surgery for Gastrointestinal Bleed - Very Major Procedures	Number of procedures	Department of Health (ranges for 50% NHS trusts) ⁶⁸	3983.00 (min:1467.00 max:5414.00)
Outpatient stay	Number of	Department of Health (ranges	96.00

	days	for 50% NHS trusts) ⁶⁸	(min:76.00 max:122.00)
Standard set of laboratory tests	Number of procedures	Curtis 2008 ¹³³	27.00
Blood products: Standard red cells	Number of doses	Baseline National Price 2007-2008	131.00

7.4.3.1 No GI adverse event

The baseline cost of treating patients for 3 months included only the acquisition cost of drugs for 3 months. Patients are assumed to be on ibuprofen 400mg, at a maintenance dose of 1.2g daily.¹³² No other resource use was included.

7.4.3.2 Discomfort

These patients are assumed to receive one month of original drug therapy, return to the GP for one extra visit and then be switched to an alternative therapy, with no further investigation. The alternative therapy was assumed to be adding omeprazole 20mg to the initial drug management.

7.4.3.3 Symptomatic ulcer

These patients are assumed to return to the GP for one extra visit and then be switched to an alternative therapy, with further investigation as an outpatient. The alternative therapy was assumed to be adding omeprazole 20mg to the initial drug management.

7.4.3.4 Serious GI event

These patients are assumed to receive 1 month of original therapy, and be admitted to the hospital for further investigation and medical management of their bleeding. Therapy switch was initiated after hospital discharge and was similar to the scenarios above.

7.4.4 Utility weights for health states

We assumed that these were adults without any specific diseases. Therefore the baseline utility weight for patients in the study group who did not experience an adverse event was taken from UK EQ-5D population norms for age 40-49 (merged men and women): 0.85 (SD

0.16).⁵². The utility decrements for the other health states were obtained from a study on cost-utility on nsNSAIDs.⁵³ (Table 22) This study used states that were comparable with the states we defined in our model. The health state 'discomfort' in our model corresponds with the health state 'moderate dyspepsia' in Spiegel's model. The utility weight for 'unresolved dyspepsia' was used for our health state 'symptomatic ulcer'. For the health state 'serious GI', a mean utility was calculated over 3 months based on the temporary utilities in Spiegel *et al* for inpatient treatment for ulcer hemorrhage (0.46 for 10 days) and the dyspepsia after the inpatient stay (0.87 for 80 days). The state 'no GI post GI' was considered as resolved dyspepsia in Spiegel's study and the utility weight was assumed to be same as 'No symptoms'.

Table 22: Health states for Markov model (NSAIDs)

Health state	Utility weight
No GI adverse events	0.85 ⁵²
Discomfort	0.760 ⁵³
Symptomatic ulcer	0.720 ⁵³
Serious GI event	0.674 ⁵³
No further GI event following initial GI event	0.85 ⁵²
Death	0

8 Appendix 2 Patients with a history of asthma who have been prescribed a beta-blocker

Lead author: Koen Putman

8.1 Introduction

Traditionally, β -blockers have not been used in asthmatic patients because of the risk of bronchoconstriction. However, β -blockers are becoming increasingly useful in patients with cardiovascular disease, and the pressure to use them in asthmatics is increasing. Following case reports of bronchoconstriction in asthmatics caused by β -blockers, some resulting in death, the Committee on Safety of Medicines issued the following advice:

*"... β -blockers, even those with apparent cardioselectivity, should not be used in patients with asthma or a history of obstructive airways disease, unless no alternative treatment is available. In such cases the risk of inducing bronchospasm should be appreciated and appropriate precautions taken."*¹³²

Small-scale safety studies confirm that non-cardioselective β -blockers do cause bronchoconstriction, which can be severe in some asthmatics.¹³⁴⁻¹³⁹ A number of studies have shown that topical timolol eye drops cause bronchoconstriction, and reduce the efficacy of bronchodilator therapy.¹⁴⁰⁻¹⁴² Betaxolol eye drops do not appear to have this effect.^{141 142} A Cochrane review of short-term cardioselective β -blocker use in reversible airways disease found a statistically significant 7.5% reduction in FEV1 with single-doses of β -blockers, which was responsive to β_2 -agonist therapy (4.63% increase in FEV1).¹⁴³ This meta-analysis also suggested that cardioselective β -blockers did not produce clinically significant adverse respiratory effects in patients with mild to moderate reactive airways disease.¹⁴³ In fact, β -blocker therapy after AMI may be beneficial for COPD or asthma patients with mild disease.¹⁴⁴

We describe the clinical and economic consequences of prescribing β -blockers in people with a recorded diagnosis of asthma: OM2 'Patients with asthma who had been prescribed a β -blocker'. More specifically this refers to those with a computer-coded diagnosis of asthma, at least six months prior to data collection that had a computer record of one or more prescriptions for a β -blocker (oral preparations or eye drops) in the six months prior to data collection. The denominator for this outcome measure was patients with a computer-coded diagnosis of asthma, at least six months prior to data collection.

8.2 Aim of the study

The principal objectives of this analysis are:

- Identify and value the impact on patients' health status of patients taking β -blockers with a history of asthma, by identifying possible health states and the probabilities of making a transition from one to another 'health state';
- Identify and value the resource use associated with managing patients who are, and are not, prescribed β -blockers;
- Assess the relative costs and outcomes for the error and non-error group.

8.3 Literature search

A literature search was conducted. References in English and limited to humans were included to 2010. This search produced 524 references. First, a selection was made on title and/or abstract. Studies were included if they examined issues on the incidence and/or prevalence of respiratory problems caused by taking β -blockers. After this selection, two references remained eligible for this research question. Finally, reference lists of the retrieved references of the first search were hand-searched. A large majority of the papers assessed short term administration of β -blockers. Studies with a treatment period of less than a week were excluded. Many other studies lacked information on the specific patient subgroups discussed here (patients with asthma prescribed β -blockers). Data from the following studies were included in the final modelling: one clinical trial¹⁴⁵ three observational studies,^{55 56 144} and two economic studies.^{54 146}

8.4 Decision-analytic model for economic analysis

8.4.1 The decision-analytic model

The decision-analytic model describes the possible treatment pathways of patients treated with non-selective β -blockers, with a diagnosis of asthma. Very little clinical or economic data were available to populate this decision model. Therefore the model is quite simple, uses data from a range of sources and oral and ocular exposure to non-selective β -blockers were combined within the model. The defined population of interest was adult patients with asthma prescribed the selective β -blocker atenolol, as the representative agent for the group.

Five acute exacerbation levels in asthma can be identified from the literature and from discussion with clinical experts: brittle asthma, moderate asthma exacerbation, acute severe

asthma and life threatening asthma and near-fatal asthma.¹⁴⁷ Each of these levels can be identified by objective criteria. However, discussion with PINCER clinical colleagues suggests that these various levels are difficult to distinguish in practice, do not reflect decision-making and are therefore less useful in economic modelling. From a pragmatic point of view, the five levels above were collapsed to three levels for this model: no exacerbation; brittle asthma was combined with moderate exacerbation and life-threatening asthma was combined with near-fatal asthma. The model developed by Price *et al*⁵⁴ and also used by Steuten *et al*¹⁴⁶ also collapse acute asthma exacerbations into two levels of severity, with a severe exacerbation being treated in secondary care, and a moderate exacerbation being treated in primary care.

The Price and Steuten model divided asthma further into “well controlled” and “suboptimal control”, the latter indicating mild exacerbations managed by the patient without health service contact. In our model the health state “no symptoms” reflected both these health states, due to the lack of data available regarding the effect of selective or non-selective β -blockers on inducing these mild exacerbations.

In our model, in case of an event (moderate or severe exacerbation), the patient returns to the health state of ‘No adverse event, post event’ except for death (patient exits the model). This health state was required because we assume that once a patient has experienced an event after prescription of a selective β -blocker, they are then transferred to another cardiovascular drug, with associated probabilities, health state and resource consumption.

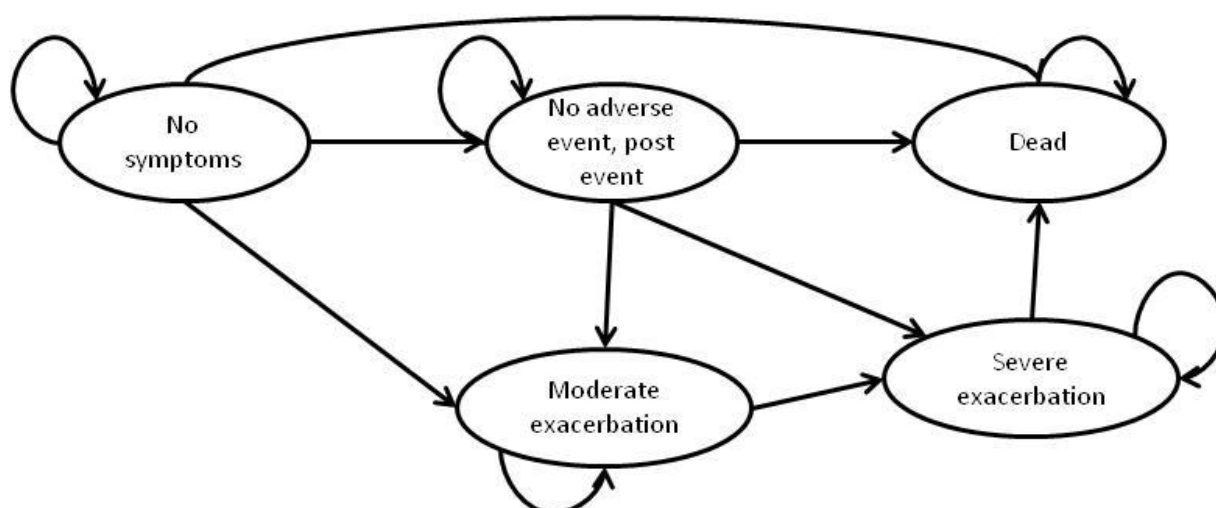
So, five health states were defined (Figure 24):

1. No symptoms (patient either well controlled, or minor exacerbations not requiring health care provider input). Patients can remain in this state for more than one 3 month period.
2. Moderate exacerbation (acute asthma event requiring management in a primary care setting). Patients can remain in this state for more than one 3 month period.
3. Severe exacerbation (severe acute asthma event requiring management in a secondary care setting). Patients cannot remain in this state for more than one 3 month period.
4. No adverse event, post event (patient with no symptoms and with no prescribing error). Patients can remain in this state for more than one 3 month period.

5. Death (absorbing state)

The model has a 3-month cycle. Due to lack of evidence to the contrary, we assumed that the likelihood for exacerbation or death did not change with exposure time to β -blockers. Therefore, no memory was included into the model.

Figure 24 Markov model for patients with asthma and a β -blocker prescription



8.4.2 Probabilities of moving from one state to another

The transitions from one state to another were defined for the error group and non-error group separately. The probabilities required to populate the model are summarised in Table 23 and Table 24.

Table 23 Probabilities for the 3-month cycle Markov model in the error groups for Beta-blockers

Transition probabilities for patients in the error group					
Transition from:	Transition to:				
	No symptoms	No symptoms post-event	Moderate exacerbation	Severe exacerbation	Death
No symptoms	0.955*		0.031 ⁵⁶	0.007 ⁵⁵	0.008
No symptoms post event		0.982*	0.008 ⁵⁴	0.002 ⁵⁴	0.008

Moderate exacerbation	0.664*	0.111 ⁵⁴	0.111 ⁵⁴	0.114
Severe exacerbation	0.797*			0.203
Death				1.000

*1-(sum of other probabilities)

Table 24 Probabilities for the 3-month cycle Markov model in the non-error groups for Beta-blockers

Transition to:					
Transition from:	No symptoms	No symptoms post event	Moderate exacerbation	Severe exacerbation	Death
No symptoms	0.982*		0.008 ⁵⁴	0.002 ⁵⁴	0.008
No symptoms post event		0.982*	0.008 ⁵⁴	0.002 ⁵⁴	0.008
Moderate exacerbation		0.664*	0.111 ⁵⁴	0.111 ⁵⁴	0.114
Severe exacerbation		0.797*			0.203
Death					1.000

*1-(sum of other probabilities)

8.4.2.1 No symptoms → moderate exacerbation

For the error group the transition probability from 'no symptoms' to 'moderate exacerbation' is derived from Chen *et al.*⁵⁶ This study concerned a UK-study on contraindicated drug combinations in four general practices. In total, 547 patients with asthma could be identified as having been prescribed non-selective β -blockers (levobunolol, nadolol, propanolol, sotalol, timolol or timolol and propranolol combined). The reason for the administration of non-selective β -blockers was not presented. In this group, 17 patients experienced breathing problems probably or possibly attributed to the use of non-selective β -blocker prescription (3.1%). The level of exacerbation was not further stipulated. We assumed it concerned moderate exacerbation due to the fact that no hospital admissions were reported.

The transition probabilities for the non-error group are taken from Price *et al.*⁵⁴ The patients in the Price economic model were asthma patients taking regular asthma medication.

8.4.2.2 No symptoms → severe exacerbation

In the error group, the transition probability from ‘no symptoms’ to ‘severe exacerbation’ is derived from Brooks *et al.*⁵⁵ In this American study on 11,592 patients with asthma and/or COPD, 3752 patients were prescribed β -blockers. In this group, 30 hospitalisations were recorded (over a total of 2123 patient years). We assumed that it concerned patients with severe exacerbation due to the fact that they were admitted to hospital. We estimated the 3-months probability for hospitalisation at 0.0071.

The transition probabilities for the non-error group are taken from Price *et al* (2002).⁵⁴

8.4.2.3 No symptoms or no symptoms post-event → death

The transition probability from ‘no symptoms’ to ‘death’ and from ‘no symptoms post-event’ to death are assumed to be non asthma-related and equal the age standardised death rate for UK (Office for National Statistics).

8.4.2.4 Moderate exacerbation → death

The transition probability from ‘moderate exacerbation’ or ‘severe exacerbation’ to death are defined as the sum of probability for death following moderate/severe exacerbation⁵⁴ and the age-standardised death rate for UK.

8.4.2.5 Severe exacerbation → death

The transition probability from ‘moderate exacerbation’ or ‘severe exacerbation’ to death are defined as the sum of probability for death following moderate/severe exacerbation⁵⁴ and the age-standardised death rate for UK.

8.4.2.6 No symptoms post-event → moderate or severe exacerbation

In the non-error group, we assume that after an event patients are well monitored and the prescription of β -blockers is stopped. Transition probabilities from ‘no symptoms, post event’ to ‘moderate exacerbations’ and ‘severe exacerbations’ are assumed to be equal to those from “no symptoms”.

In the error group, we assume that after an event patients the prescription of β -blockers is not modified. Hence, transition probabilities from 'no symptoms, post event' to 'moderate exacerbations' and 'severe exacerbations' are assumed to be equal as from 'no symptoms'.

8.4.3 Utility weights for health states

Utility weights for all health states in the model were obtained from a model developed by Steuten *et al.*¹⁴⁶ The utilities were derived from trial data (658 Dutch adults with asthma managed with daily inhaled corticosteroids), were measured using EQ-5D,⁵² and standard error, alpha and beta parameters were reported and could be used in our model (see Table 25).

Table 25 Health states for Markov model (Beta-blockers)¹⁴⁶

Health state	Utility weight (SE)	Alpha, beta parameters
No symptoms	0.73 (0.03)	159.14, 58.86
No symptoms post event	0.73 (0.03)	159.14, 58.86
Moderate exacerbation	0.67 (0.02)	369.67, 182.08
Severe exacerbation	0.66 (0.04)	91.91, 47.35
Death	0	-

8.4.4 Required resource use and unit costs

No data on resource use associated with management of people with asthma who are prescribed β -blockers could be retrieved so the resource use associated with each health state was obtained from a range of published sources.

8.4.4.1 No symptoms

The baseline cost of treating a patient included only the acquisition cost of drugs for 6 months. One of the most commonly prescribed beta-blockers was considered as standard drug. Maintenance dosages as suggested by the BNF were included. Atenolol at 100mg/day was used. No other resource use was included in this health state.

8.4.4.2 No symptoms post-event

Patients who have experienced an adverse event return to this state.

8.4.4.3 Moderate exacerbation

These patients are assumed to experience mild/moderate exacerbation of their asthma with an acute visit to their GP for treatment, followed by a follow-up visit at the surgery with a therapy switch away from β -blockers.¹⁴⁷ The acute treatment comprises 100 puffs of salbutamol (4-6 puffs /10-20 min, 15 cycles simulated), followed by a prescription of oral prednisolone for 5 days.

8.4.4.4 Severe exacerbation

These patients are assumed to experience severe exacerbation of their asthma with an admission to hospital, followed by a follow-up visit at the surgery with a therapy switch away from β -blockers. The hospital stay was considered to be 3.4 days for the reason of asthma with complications (mean length of stay by Hospital Episodes Statistics 2006-2007). The acute event starts at the beginning of the cycle.

8.4.4.5 Death

Patients can die from non-asthma related causes or following exacerbation. In the first instance age standardised death rates for UK are applied, with no costs. In the latter, patients are assumed to experience exacerbation of their asthma with an admission to the hospital. It is assumed that patients die on the 1st day of admission. The acute event starts at the beginning of the cycle.

For each unit cost used in the analyses, a detailed description of the source is provided in Table 26, with the cost for each health state detailed in Table 27. Deterministic ranges had to be used for most parameters and correspond with 25th and 75th percentile of the unit cost found in the national databases.

Table 26 Sources of unit costs (Beta-blockers)

Cost parameter	Resource use	Source of unit costs	Unit cost	Min	Max
GP visit	Number of visits	Curtis 2008 ¹³³	£34	£34	£34
Inpatient care Asthma	Number of days	DOH reference costs 2008-2009 ⁶⁸	£1218	£819	£1599

Salbutamol 100 µg/puff	Monthly cost	BNF ¹³²	£4.49	£4.49	£4.49
Prednisolone 5mg at 40mg/day*	Daily cost	BNF ¹³²	£2.74	£2.74	£2.74
Atenolol 25-50 mg/day*	Monthly cost	BNF ¹³²	£1.65	£1.65	£3.30

*The dosages are based on the recommendations as outlined in the BNF.¹⁴⁷

Table 27 Cost per patient for each health state (Beta-blockers)

Health state	Resource use	Units	Mean cost	Min	Max
No symptoms	Medication: Atenolol	90	£5	£5	£10
Moderate exacerbation	GP visit	2	£68	£68	£68
	Medication: Salbutamol	100	£1	£1	£1
	Medication: Prednisolone	5d	£3	£3	£3
Severe exacerbation	Hospital admission	3.4	£142	£2786	£5438
Death after hospital admission	Hospital admission	1	£1218	£819	£1599

9 Appendix 3: Patients aged 75 years and older who have been prescribed an Angiotensin-Converting Enzyme Inhibitor (ACEI) long-term who have not had a recorded check of their renal function and electrolytes in the previous 15 months

Lead author: Nick Verhaeghe

9.1 Introduction

Angiotensin-Converting Enzyme Inhibitors (ACEI) have various indications including hypertension, heart failure, left ventricular dysfunction and diabetic nephropathy.¹⁴⁸ However, ACEIs, by altering glomerular perfusion, may result in a decrease in creatinine clearance, and in an increase in serum creatinine and serum potassium.¹⁴⁹ Hence, the risk of developing renal dysfunction is ongoing and regular monitoring helps to reduce the incidence of adverse events such as acute renal failure (ARF) and hyperkalaemia because creatinine and potassium levels along with other electrolytes can be kept within the optimal range, by adjusting doses or stopping the ACEI.

Yet, a retrospective cohort study revealed that the proportion of ambulatory patients with ACEI therapy who received serum creatinine and serum potassium monitoring in a one-year period was only 67.5%. So, nearly one-third of ambulatory patients treated with ACEI did not have at least one serum creatinine or serum potassium level evaluated in the one-year period.¹⁵⁰

We describe the clinical and economic consequences of monitoring, and not monitoring, renal function and potassium levels: OM3 'Patients aged 75 years and older who have been prescribed an ACEI long-term who have not had a recorded check of their renal function and electrolytes in the previous 15 months'.

9.2 Aim of the study

The principal objectives of this analysis are to:

- Identify and value the impact on patients' health status of patients taking ACEI who are and are not monitored for 15 months, by identifying possible health states and the probabilities of making a transition from one to another 'health state';

- Identify and value the resource use associated with managing patients who are, and are not, monitored;
- Assess the relative costs and outcomes for the monitored and unmonitored group.

9.3 Literature search

The review focused on studies of adults (18 years or older). The focus of the analysis is on patients taking ACEIs for hypertension. A literature search was conducted using the search string '(Angiotensin-Converting Enzyme Inhibitors OR ACE Inhibitors OR Enalapril OR Captopril OR Lisinopril OR Perindopril OR Ramipril OR Fosinopril) AND (Hyperkalemia OR Hyperkalaemia OR Kidney Failure, Acute OR Acute Renal Failure). References in English and limited to humans were included to 2010. This search produced 1404 references. After excluding duplicate records, 1218 references remained for further evaluation.

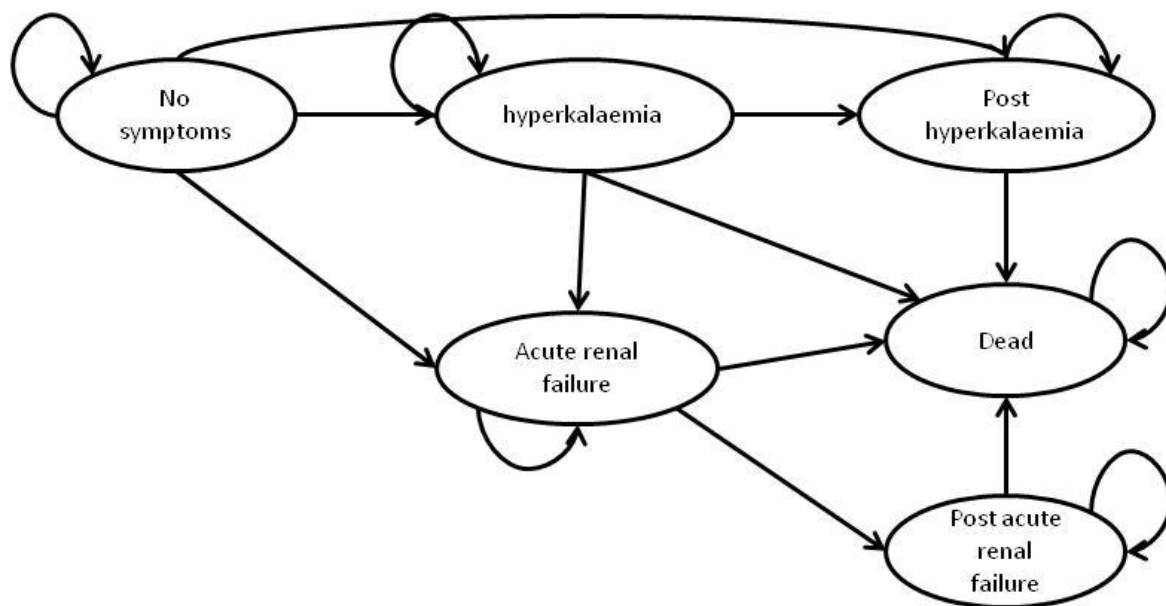
First, a selection was made on title and/or abstract. Studies were included if they examined issues on the incidence and/or prevalence of hyperkalaemia and ARF in patients treated with ACEI. After this selection, 21 references remained. Subsequently, full text of the retrieved references of the previous selection was evaluated. After this selection, two references remained eligible for this research question. Finally, reference lists of the retrieved references of the first search were hand-searched. This search produced another two references.

9.4 Decision-analytic model for economic analysis

9.4.1 The decision-analytic model

The decision-analytic model describes the possible treatment pathways of patients treated with ACEI, who were monitored, or not monitored in the previous 15 months. In the model (Figure 25) six 'health states' are identified: 'no symptoms', 'hyperkalaemia', 'No hyperkalaemia post hyperkalaemia', 'acute renal failure', 'no acute renal failure post acute renal failure' and 'death'. The model has a three-month cycle. The model is defined so that only one transition from one to another state is possible within each three months.

Figure 25 Markov model of patients treated with ACEI, not monitored in the previous 15 months



In this model we assume that, once clinical manifestations of hyperkalaemia appear or the clinical state of ARF is reached, therapy and monitoring of serum potassium and serum creatinine levels will be installed or ACEI therapy will be stopped. This is defined as the states 'post hyperkalaemia' and 'post acute renal failure (no ARF post ARF)'.

9.4.2 Defining 'Hyperkalaemia' and 'Acute Renal Failure' for the model

9.4.2.1 Hyperkalaemia

Normal levels of potassium are kept between the concentrations of 3.5 and 5.0 mmol/l. Hyperkalaemia is defined as a serum potassium level greater than 5.0 mmol/l.¹⁴⁹ In the literature, there is little agreement on what constitutes mild, moderate or severe hyperkalaemia. For example, in a trial by Amir *et al.*¹⁵⁰ mild hyperkalaemia was defined as a serum potassium level 5.1-5.5 mmol/l and severe hyperkalaemia as a serum potassium level >5.5 mmol/l. In another trial, de Denus *et al.*⁵⁷ defined hyperkalaemia as a serum potassium level >5.5 mmol/l.

The initial clinical indicator of hyperkalaemia is the presence of ECG abnormalities which suggests arrhythmias or, in much more serious circumstances, cardiac arrest. It has been estimated that most patients show ECG abnormalities once their serum potassium concentration ≥ 6.5 mmol/l.^{151 152} However, tall tented T waves are possible in serum potassium levels between 5.5 and 6.5 mmol/l.¹⁵³ ACEI therapy should be discontinued if the serum potassium concentration increases to 5.5 mmol/l or higher.^{154 155} For our model, we defined clinical hyperkalaemia as a serum potassium level ≥ 5.5 mmol/l.

9.4.2.2 Acute renal failure

ARF represents an acute condition that may vary in duration of time (hours to months), with the potential for (partial) recovery, non-recovery or death. This potential for multiple disease states of varying lengths adds complexity to the definitions that may be used to define ARF.¹⁵⁶

ARF is defined according to the RIFLE classification.¹⁵⁷ This classification was developed by the US Acute Dialysis Quality Initiative, a group of experts in acute kidney dysfunction, consisting of nephrologists and intensivists. RIFLE stands for the increasing severity classes Risk, Injury and Failure, and the two outcome classes Loss and End-stage kidney disease. According to recommendations, ACEI therapy should be discontinued if serum creatinine concentration increases by more than 88 mmol/l from baseline or the repeat value shows a progressive increase.^{154 155}

9.4.3 Probabilities of moving from one state to another

The probabilities required to populate the model are summarized in Table 8.

Table 28 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups for ACEI

Pathway	Not monitored		Monitored	
	Probability	Source	Probability	Source
No symptoms →No symptoms	0.979 (1-0.016- 0.001-0.004)	Net of other probabilities at this node	0.988 (1- 0.008-0.0005- 0.004)	Net of other probabilities at this node
No symptoms→ Hyperkalaemia	0.016	See explanatory notes below	0.008	De Denus <i>et al.</i> (2006) ⁵⁷
Hyperkalaemia→ Post Hyperkalaemia	0.996 (1-0.004)	See explanatory notes below	0.996 (1- 0.004)	See explanatory notes below
No symptoms→ARF	0.0010	Mittalhenkle <i>et al.</i> (2008); ⁵⁸ Knight <i>et al.</i> (1999) ⁵⁹	0.0005	Baraldi <i>et al.</i> (1998) ⁶⁰

ARF → Post ARF	0.974 (1-0.026)	See explanatory notes below	0.974	See explanatory notes below
No symptoms→Dead	0.004	Hansson <i>et al.</i> (1999); ⁶¹ Garg <i>et al.</i> (1995) ⁶²	0.004	Hansson <i>et al.</i> (1999); ⁶¹ Garg <i>et al.</i> (1995) ⁶²
Hyperkalaemia →Dead	0.004	See explanatory notes below	0.004	See explanatory notes below
Post hyperkalaemia →Dead	0.004	See explanatory notes below	0.004	See explanatory notes below
ARF→Dead	0.026	Wynckel <i>et al.</i> (1998) ⁶³	0.026	Wynckel <i>et al.</i> (1998) ⁶³
Post ARF→Dead	0.026	Wynckel <i>et al.</i> (1998) ⁶³	0.026	Wynckel <i>et al.</i> (1998) ⁶³

9.4.3.1 No symptoms→hyperkalaemia

No evidence in the literature was found for the transition probability from the state 'no symptoms' to the state of 'hyperkalaemia'. For this probability an assumption was made. The assumption is based on the evidence that elevated serum creatinine levels are independently associated with hyperkalaemia.^{158 159} In this sense, the ratio of the incidence rates of ARF in monitored (0.0005) versus not-monitored (0.001) patients was used to calculate the incidence of hyperkalaemia in not monitored patients (Table 29).

Table 29 Derivation of transition probability from no symptoms to hyperkalaemia (ACEI)

	Hyperkalaemia	Acute Renal Failure
Not monitored	0.008 X (0.001/0.0005) = 0.001 0.016	
Monitored	0.008	0.0005

9.4.3.2 Hyperkalaemia→No Hyperkalaemia Post Hyperkalaemia

This transition probability refers to the proportion of patients moving from the state of 'hyperkalaemia' to the 'post hyperkalaemia' state minus the transition probability of the state of 'hyperkalaemia' to the state of 'dead' (0.004).

9.4.3.3 No symptoms→ARF

To calculate the transition probability from the state 'no symptoms' to the state of 'ARF' in patients treated with ACEI, data of two separate studies were combined. Mittalhenkle *et al.* conducted a prospective, population-based, observational cohort study on cardiovascular risk factors and the incidence of ARF in adults ≥ 65 years.⁵⁸ ARF developed in 225 (3.9%) of the 5731 community-dwelling participants during a median follow up period of 10.2 years. Knight *et al.* analysed data from the SOLVD trial.⁵⁹ They found that patients who were treated with enalapril had a relative risk of 1.33 for developing decreased renal function, compared to controls (placebo).

The median follow up period in the study of Mittalhenkle *et al.* was 10.2 years.⁵⁸ For our model, we converted that incidence rate to a three-month incidence. Secondly, this three-month incidence was multiplied by the relative risk of 1.33 found by Knight *et al.*⁵⁹

9.4.3.4 ARF→No ARF Post ARF

This transition probability refers to the proportion of patients moving from the state of 'ARF' to the 'post ARF state' minus the transition probability of the state of 'ARF' to the state of 'dead' (0.026).

9.4.3.5 ARF→Dead / Post ARF→Dead

The transition probability from the state of 'ARF' to the state of 'Dead' was derived from a study by Wynckel *et al.*⁶³ In this retrospective study, 64 patients (mean age 71.2 ± 11.6 years) who had had ARF as a consequence of taking ACEI were prospectively followed for a period of 5 years. During this follow up period 26 (40.625%) of the 64 patients died. Again, the three-month mortality rate was calculated.

9.4.3.6 Hyperkalaemia→Dead / Post Hyperkalaemia→Dead

The probability of the pathway 'no symptoms'→'dead' was appropriate as probability score for the transition from 'hyperkalaemia'→'dead'. In a study of indications for hospitalisation of patients with hyperkalaemia, no fatalities resulting directly from hyperkalaemia were observed.¹⁶⁰

9.4.3.7 No symptoms→Dead

Finally, the transition probability from 'no symptoms' to the state of 'dead' had to be determined. According to statistics of the American Heart Association the proportion of heart failure versus hypertension is 10/90. This is consistent with the study of Sadjadi *et al.*¹⁶¹ In this study, of 1163 patients treated with ACEI, 91.7% of participants had a diagnosis of hypertension, while 9.5% had a diagnosis of heart failure. The ratio 10/90 (heart failure/hypertension) was then used to determine the pathway probability from 'no symptoms' to 'dead'.

To determine this probability, two sources were used. Garg & Yusuf evaluated the effect of ACEI on mortality and morbidity in patients with systematic congestive heart failure.⁶² The authors obtained data for all completed, published or unpublished randomized placebo-controlled trials of patients treated with ACEI for at least eight weeks. The data of trials with a duration of three months provided a mortality rate of 129/3870 (3.33%) patients. Hansson *et al.* studied the effect of ACEI compared with conventional therapy on cardiovascular morbidity and mortality in hypertension.⁶¹ In this prospective, randomized open trial (mean follow up period of 6.1 years) 76 of 5492 (1.38%) patients treated with captopril died of a cardiovascular cause.

People who are in the 'no hyperkalaemia post hyperkalaemia' or 'no ARF post ARF' state stay in this state because we assume that, after the appearance of clinical hyperkalaemia or ARF, these patients will be closely followed and serum creatinine and serum potassium levels will be monitored. These patients go to the monitored group.

9.4.3.8 Derivation of probabilities for the 'monitored' group

In the monitored group, only the transition probabilities from 'no symptoms' to 'hyperkalaemia' and 'no symptoms' to 'acute renal failure' differ in comparison with the probabilities in the not monitored group.

In a retrospective analysis of the results of the Studies of Left Ventricular Dysfunction (SOLVD) the incidence of hyperkalaemia in 3364 patients, treated with enalapril, was evaluated.⁵⁷ Specifically, a retrospective analysis of serum potassium levels, who were reported at baseline and during follow up of the SOLVD trial, were described. In this sense, this retrospective study was appropriate for our study. During the mean follow up time of 2.7 years, hyperkalaemia (serum potassium level ≥ 5.5 mmol/l occurred in 7.8% of patients.

In a retrospective analysis of 1528 hospitalized patients in a 30-month period, the incidence of ARF in patients (mean age 72 years) treated with ACEI was 0.52%.⁶⁰

9.4.4 Utility weights for health states

9.4.4.1 No symptoms

The utility weight for the state of 'no symptoms' was obtained from the data of the 1998 public health survey of a sample of the Stockholm County Population. This study included a subsample of patients with hypertension. Mean (SD) value of health-related quality of life (HRQOL) for this subsample was 0.78 (0.013).⁶⁴

9.4.4.2 Hyperkalaemia

No utility weight for "hyperkalaemia" or "electrolyte disorders in general" was found in the literature. As clinical hyperkalaemia requires urgent medical management,^{151 162} it was decided to derive the utility weight for 'hyperkalaemia' from a study of patients with coronary heart disease and heart failure admitted to hospital. The mean HRQOL of the patients in this study was 0.60 (no range reported).⁶⁷

9.4.4.3 No hyperkalaemia Post hyperkalaemia

As mentioned above no utility weight for "hyperkalaemia" was found in the literature. Again, an assumption was made. The utility weight of this state was obtained from a study of utility loss following cardiovascular events. In that study, HRQOL was measured three months after the first cardiovascular event. This resulted in a mean (SD) value of 0.73 (0.19).⁷⁰

9.4.4.4 Acute Renal Failure

The utility weight for "ARF" was obtained from a study evaluating the HRQOL of patients with renal failure while receiving haemodialysis. The mean (SD) value for this patients was 0.44 (0.32).⁶⁹

9.4.4.5 No ARF Post ARF

This utility weight was obtained from a study of 703 patients receiving renal replacement therapy for ARF during 1998-2002 at Helsinki University Central Hospital. HRQOL was evaluated during a follow up period (median follow up time: 2.4 years) resulting in a mean value of 0.68 (no range reported).⁷¹

9.4.5 Required resource use and unit costs

Table 30 summarises required resource use and unit costs.

Table 30 Summary of resource use and cost in each ACEI health state

State	Cost/3 Months	Source
NO SYMPTOMS		
Regular GP visit	£34	¹⁶³
drug costs	£5.49	⁶⁶
HYPERKALAEMIA		
Hospital-based management	£1480	¹⁶⁴
ACUTE RENAL FAILURE		
Hospital-based management	£3043	¹⁶⁴
NO HYPERKALAEMIA POST HYPERKALAEMIA		
GP visit	£112	
Drug costs	£5.49	⁶⁶
NO ARF POST ARF		
GP visit	£112	¹⁶³
Drug costs	£5.49	⁶⁶

9.4.5.1 Cost No symptoms

The baseline cost of treating a patient for a period of three months included the acquisition of an ACEI for that period and a GP visit for obtaining a prescription for the ACEI. The cost is based on the ACEI captopril at the maintenance dose of 25 mg twice a day. This results in an annual cost of £21.96 (£5.49 per three months). Captopril was assumed as an appropriate choice as this was used in the ELITE trial.¹⁶⁵ The cost of a GP visit was obtained from the annual report “Unit Costs of Health & Social Care”.¹⁶³

9.4.5.2 Cost hyperkalaemia

The cost of hyperkalaemia was obtained from the National Schedule of Reference Costs 2008-09 NHS Trusts.¹⁶⁴ No detailed costs for hyperkalaemia was found, so it was assumed that the unit cost of “electrolyte disorders in general” (£1480) was most suitable.

9.4.5.3 Cost acute renal failure

The unit cost for ARF was also obtained from the National Schedule of Reference Costs 2008-09 NHS Trusts¹⁶⁴ and comprised a unit cost of £3043.

9.4.5.4 Cost No Hyperkalaemia Post Hyperkalaemia

When serum potassium levels are normalized, ACEI therapy can be re-initiated. The serum potassium concentration should be checked within one week after the ACEI has been started. If this concentration is normal, the dose of the drug can be titrated upward. With each increase in the dose, the serum potassium concentration should be measured again one week later.¹⁶⁶ In our model, this results in three GP visits in the three month period. The drug cost is based on captopril at the maintenance dose of 25 mg twice a day.

9.4.5.5 Cost No ARF Post ARF

ACEI therapy should be restarted on a low dose and then gradually titrated up to the targeted or maximum tolerated dose. After one week, serum creatinine and urea levels should be checked. After a further month the patient should be monitored again. This process should continue until the patient is stabilized at the target dose, and the serum potassium and serum creatinine levels are constant.⁶⁶ In our model, this results in three GP visits in the three month period. The drug cost is based on captopril at the maintenance dose of 25 mg twice a day.

10 Appendix 4: Patients receiving methotrexate for at least three months who have not had a recorded full blood count and/or liver function test within the previous three months

Lead author: Nick Verhaeghe

10.1 Introduction

Methotrexate is a commonly used drug in the treatment of psoriasis and rheumatoid arthritis (RA). Oral methotrexate is associated with adverse incidents and deaths in the NHS and worldwide. In this sense, full blood counts (FBC) and liver function tests (LFT) are recommended at 1-3 monthly intervals to clinically evaluate and monitor the patient, and prevent methotrexate toxicity.¹⁶⁷ In a study, members of the British Association of Dermatologists were asked about their methotrexate prescribing and monitoring practices. Three hundred and sixty-five questionnaires were analysed. During commencement or when changing dose, 77% checked FBC 1-2 weekly, and 64% checked LFT 1-2 weekly. Once established at a steady dose, 49% checked FBC and LFT 9-12 weekly, suggesting inadequate monitoring.¹⁶⁸

In a study of an intervention to improve laboratory monitoring at initiation of drug therapy in ambulatory care, for methotrexate, no statistically significant differences between the intervention and the usual care group were found. For methotrexate, in the intervention group 90.7% of dispensing and in the usual care group 88.6% were monitored ($p=0.43$).¹⁶⁹ Little data are available to assess the level of monitoring of long term methotrexate therapy. There are no studies examining the economic impact of monitoring or not monitoring methotrexate.

We describe the clinical and economic consequences of monitoring, and not monitoring, full blood count and/or liver function test in patients receiving methotrexate for at least three months.

10.2 Aim of the study

The principal objectives of this analysis are:

- Identify and value the impact on patients' health status of patients taking methotrexate who have not had a recorded full blood count and/or liver function test

within the previous three months, by identifying possible health states and the probabilities of making a transition from one to another 'health state';

- Identify and value the resource use associated managing patients who are, and are not, monitored;
- Assess the relative costs and outcomes for the monitored and unmonitored group.

10.3 Literature search

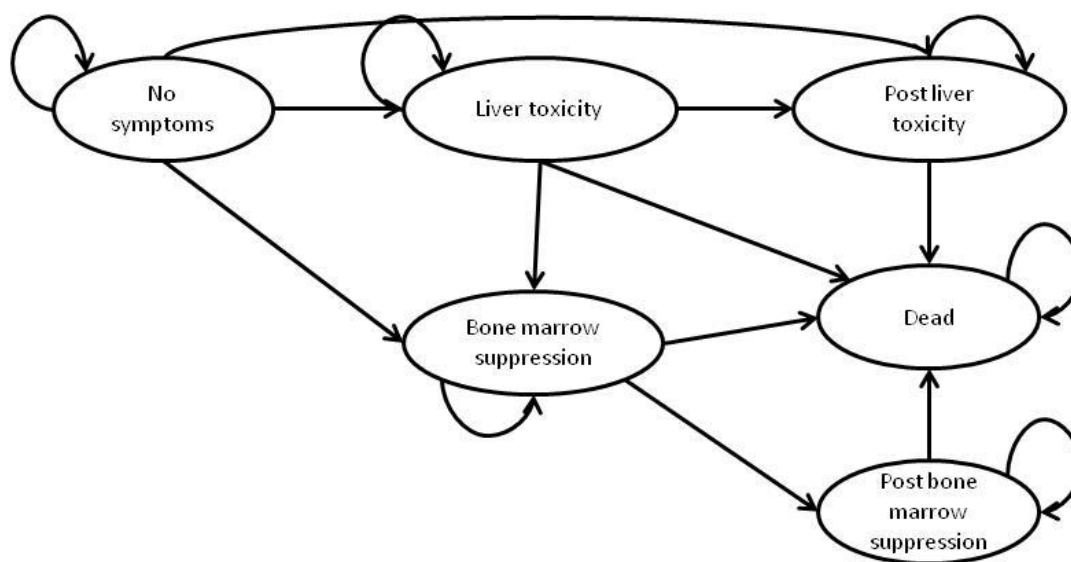
A literature search was conducted through the electronic database Medline using the search string '(Methotrexate) AND (Drug Monitoring OR Blood Cell Count OR Liver Function Tests)'. References in English and limited to humans were included to 2010. This search produced 263 references. First, a selection was made on title and/or abstract. Studies were included if they examined the incidence of abnormal liver function tests and abnormal full blood counts in patients treated with methotrexate. After this selection, 13 references remained. Subsequently, full text of the retrieved references of the previous selection was evaluated. Finally, reference lists of the retrieved references of the electronic search were hand-searched. This search produced another two references.

10.4 The decision-analytic model

The decision-analytic model describes the possible treatment pathways of patients treated with methotrexate, who have not had a recorded full blood count and/or liver function test within the previous three months. In the model (Figure 26), six 'health states' are identified: 'no symptoms', 'liver toxicity', 'bone marrow suppression', 'no liver toxicity post liver toxicity', 'no bone marrow suppression post bone marrow suppression' and 'dead'. The model has a 3-month cycle. The model is defined so that only one transition from one to another state is possible.

In this model we assume that, once clinical manifestations of liver toxicity or bone marrow suppression appear, therapy and monitoring will be installed or methotrexate therapy will be stopped. This is defined as the states 'no liver toxicity post liver toxicity' and 'no bone marrow suppression post bone marrow suppression'.

Figure 26 Markov model of patients treated with methotrexate



10.5 Probabilities of moving from one state to another

The probabilities required to populate the model are summarized in Table 31.

Table 31 Probabilities for the 3-month cycle Markov model in the monitored and not monitored groups (methotrexate)

Pathway	Not monitored			Monitored		
	Probability	Source		Probability	Source	
No symptoms → No symptoms	0.9474-0.0434-0.0038-0.0054)	(1- Net of other probabilities at this node		0.9686-0.0228-0.0032-0.0054)	(1- Net of other probabilities at this node	
No symptoms → Liver Toxicity	0.0434	Malatjalian et al. (1996) ⁷²		0.0228	Haustein & Rytter (2000) ⁷³	
Liver Toxicity → Post Liver Toxicity	0.9946-0.0054)	(1- See explanatory notes below		0.9946-0.0054)	(1- See explanatory notes below	
No symptoms → BMS	0.0038	Bologna et al. (1997) ⁷⁴		0.0032	Haustein & Rytter (2000) ⁷³	
BMS → Post BMS	0.9270-0.0730)	(1- See explanatory		0.9270(1-0.0730)	See explanatory	

		notes below		notes below
No symptoms→Dead	0.0054	Choi et al. (2002) ⁷⁵	0.0054	Choi et al. (2002) ⁷⁵
Liver Toxicity→Dead	0.098	Berman et al. 2011 ⁷⁶	0.098	Berman et al. 2011 ⁷⁶
BMS→Dead	0.0730	Lim et al. (2005) ⁷⁷	0.0730	Lim et al. (2005) ⁷⁷
Post-BMS →Dead	0.0054	Choi et al. (2002) ⁷⁵	0.0054	Choi et al. (2002) ⁷⁵

BMS: bone marrow suppression

10.5.1 Derivation of probabilities for the ‘not monitored’ group

10.5.1.1 No symptoms→Liver Toxicity

The transition probability from the state of ‘No symptoms’ to the state of ‘Liver Toxicity’ was derived from a study of methotrexate hepatotoxicity in psoriatic patients.(4) In this retrospective study, liver biopsies of 104 patients were analyzed. Mean follow up was 3.8 years. 49% of all patients had grade III liver biopsies. For the model, the three-month incidence rate was calculated.

10.5.1.2 Liver Toxicity →No liver Toxicity Post Liver Toxicity

This transition probability refers to the proportion of patients moving from the state of ‘Liver Toxicity’ to the ‘Post Liver Toxicity’ state minus the transition probability of the state of ‘Liver Cirrhosis’ to the state of ‘Dead’.

10.5.1.3 No symptoms→Bone Marrow Suppression

The transition probability from the ‘No symptoms’ state to the state of ‘BMS’ was derived from a long-term follow up study of 453 rheumatoid arthritis patients treated with methotrexate (6). Mean follow up period was 35.2 months. The incidence of haematological side effects was 4.4%. For the model, the three-month incidence rate was calculated.

10.5.1.4 Bone Marrow Suppression →No BMS Post BMS

This transition probability refers to the proportion of patients moving from the state of ‘BMS’ to the ‘Post BMS’ state minus the transition probability of the state of ‘BMS’ to the state of ‘Dead’.

10.5.1.5 No symptoms → Dead

Mortality rate was derived from a cohort study including 1240 patients with rheumatoid arthritis seen in an outpatient rheumatology facility.⁷⁵ The mean length of follow up was 6 years. Of 588 patients treated with methotrexate 72 had died. For the model, the three-month incidence rate was calculated.

10.5.1.6 Post-BMS → Dead

Due to lack of other evidence, we used the same probability as for No symptoms → Dead.

10.5.1.7 Liver Toxicity → Dead

No evidence on transition probability from the state of 'Liver Toxicity' to the state of 'Dead' was found for the group of patients considered in this study. Transition probabilities were derived from an American cohort of 447 patients with advanced liver disease who were admitted to the hospital.⁷⁶ The 90-day mortality of patients with no readmission within 30 days of admission was calculated at 9.8% compared to 26.8% in patients who experienced readmission within 30 days. Taking a conservative approach, we used 0.098 as transition probability from the state of "Liver toxicity" to the state of "Dead".

10.5.1.8 Bone Marrow Suppression → Dead

This transition probability was derived from a study of methotrexate-induced pancytopenia.⁷⁷ Seven of 25 patients died. Median follow up time was 13 months.

10.5.2 Derivation of probabilities for the 'monitored' group

10.5.2.1 No symptoms → Liver Toxicity

The transition probability from the 'No symptoms' state to the state of 'BMS' was derived from a retrospective study of 157 psoriasis patients treated with methotrexate.⁷³ Mean follow up period was 54.7 months. The incidence of liver toxicity was 14.08%. For the model, the three-month incidence rate was calculated.

10.5.2.2 No symptoms → Bone Marrow Suppression

The transition probability from the 'No symptoms' state to the state of 'BMS' was derived from a retrospective study of 157 psoriasis patients treated with methotrexate.⁷³ Mean follow up period was 54.7 months. The incidence of bone marrow suppression was 5.73%. For the model, the three-month incidence rate was calculated.

10.5.3 Required resource use and unit costs (Table 32)

Table 32 Summary of resource use and costs in each health state in methotrexate

STATE	COST/3 MONTHS	SOURCE
NO SYMPTOMS		
Regular GP visit	£34	¹³³
Drug costs	£6.07	Trust Guideline for use of oral methotrexate ⁷⁹
COST LIVER TOXICITY		
Hospital-based management	£2472	⁶⁸
COST BONE MARROW SUPPRESSION		
Hospital-based management	£2776	⁶⁸
COST NO LIVER TOXICITY POST LIVER TOXICITY		
GP visit	£112	Based on three GP visits in a three-month period
Drug costs	£6.07	Trust Guideline for use of oral methotrexate ⁷⁹
COST NO BMS POST BMS		
GP visit	£112	Based on three GP visits in a three-month period
Drug costs	£6.07	Trust Guideline for use of oral methotrexate ⁷⁹

10.5.3.1 Cost No symptoms

The baseline cost of treating a patient for a period of three months included the acquisition of methotrexate for that period and a GP visit to obtain a prescription. The three-month cost of methotrexate is based on a maintenance dose of 10mg/week (annual cost: £24.29).^{79 132} The cost of a GP visit was obtained from the annual report “Unit Costs of Health & Social Care”.¹³³

10.5.3.2 Cost Liver Toxicity

The cost of liver toxicity was obtained from the NHS Schedule of Reference Costs.⁶⁸ No detailed cost for Liver Cirrhosis was found. Therefore, it was assumed that the unit cost of ‘Liver Failure Disorders’ was most suitable.

10.5.3.3 Cost Bone Marrow Suppression

The unit cost of BMS was also obtained from the NHS Schedule of Reference Costs.⁶⁸ Like for Liver Toxicity no specific unit cost was found for BMS. Therefore, it was assumed that the unit cost of 'pyrexia of unknown origin' was most suitable.

10.5.3.4 Cost No Liver Toxicity Post Liver Toxicity

It is recommended that Liver Function Tests (LFTs) should occur until 4 weeks after the last dose increase and then 6 weekly after 2 – 3 months.⁷⁹ In our model, three GP visits are included in the three-month period. The cost of methotrexate is based on a maintenance dose of 10mg/week (annual cost: £24.29).^{79 132}

10.5.3.5 Cost No BMS Post BMS

Monitoring of full blood counts should occur at baseline and until 4 weeks after the last dose increase and then 6 weekly after 2 – 3 months. In our model, three GP visits are included in the three-month period. The three-month cost of methotrexate is based on a maintenance dose of 10mg/week (annual cost: £24.29).^{79 132}

10.5.4 Utility weights for health states

10.5.4.1 No symptoms

The utility weight for the state of 'No symptoms' was derived from a trial of health-related quality of life of patients with psoriasis taking methotrexate.⁷⁸ The mean EQ-5D index score for that population was 0.90.

10.5.4.2 Liver toxicity

The utility weight for liver toxicity was derived from a systematic review of health-state utilities in liver disease.⁸⁰ A utility weight of 0.76 was found for 'decompensated cirrhosis' and appeared to be relevant.

10.5.4.3 No liver toxicity post liver toxicity

The utility weight for this state was derived from the same paper as for the utility weight for liver toxicity. For the post state, a utility weight of 0.84 appeared to be appropriate.

10.5.4.4 Bone Marrow Suppression

No utility weights were found in the literature for patients with bone marrow suppression taking methotrexate. An estimated utility weight of 0.75 was assigned.

10.5.4.5 No Bone Marrow Suppression Post Bone Marrow Suppression

Also for this state, no utility weights were found in the literature. An estimated utility weight of 0.80 was assigned.

11 Appendix 5: Patients receiving lithium for at least three months who have not had a recorded check of their lithium levels within the previous three months

Lead author: Matthew Franklin and Rachel Elliott

11.1 Introduction

Lithium is used in the prophylaxis and treatment of mania, in the prophylaxis of bipolar disorder (manic depressive disorder),¹⁷⁰ and is an effective adjunctive treatment in resistant recurrent depression.¹⁷¹ Bipolar disorder is characterised by recurrent manic or hypomanic and depressive episodes. This leads to patterns of stability and relapse with associated impaired health related quality of life.¹⁷² Suicidal behaviour is common and associated with high mortality.¹⁷³ Lithium has become one of the most common therapies administered to bipolar patients due to its benefits in reducing the frequency of relapse, particularly into manic episodes.¹⁷⁴ It also reduces attempted and completed suicide in this population.^{173 175}

Lithium is a drug associated with many adverse effects and 75-90% patients treated with lithium will show signs or symptoms of toxicity during their treatment.¹⁷⁶ The decision to give prophylactic lithium usually requires specialist advice, based on careful consideration of the likelihood of mania recurrence in the individual patient, and the benefit weighed against the risks. Long term use of lithium therapy has been associated with thyroid disorders, renal impairment and mild cognitive and memory impairment.¹⁷⁰ Long term lithium therapy should therefore only be undertaken with regular monitoring of thyroid function. The need for continued therapy should be assessed regularly and patients should be maintained on lithium after 3-5 years only if benefit persists. Poor patient adherence (18-52%) has been reported due to side effects and perceived lack of efficacy against depressive episodes.⁸² Sub-therapeutic levels, and associated increased risk of relapse, are usually associated with poor adherence.¹⁷⁷ There is a further risk of rebound mania and increased risk of suicide on withdrawal.^{178 179} Other drugs, such as sodium valproate, are used as alternative mood stabilisers, but none have been found to be more effective than lithium.^{180 181}

Lithium has a small therapeutic/toxic ratio¹³² so should not be prescribed unless facilities for monitoring serum lithium concentrations are available. Doses are adjusted to achieve serum lithium concentration of 0.6-1.2 mmol/litre. Serum lithium concentrations below 0.6 mmol/litre have little, or no, effect on reducing risk of relapse. Over-dosage, usually with a serum-lithium concentration of over 1.5 mmol/litre, may be fatal and toxic effects include tremor,

ataxia, dysarthria, nystagmus, renal impairment and convulsions. If these potentially hazardous signs occur treatment should be stopped, lithium concentrations checked and steps taken to reverse lithium toxicity. Serum lithium concentrations above 2 mmol/litre require emergency poisoning treatment. To keep serum lithium concentration levels within a therapeutic range, therapeutic drug monitoring (TDM) should be carried out every three months.¹³²

We describe the clinical and economic consequences of monitoring, and not monitoring, lithium levels: OM7 'Patients receiving lithium for at least three months who have not had a recorded check of their lithium levels within the previous three months'.

11.2 Aim of the study

The principal objectives of this analysis are:

- Identify and value the impact on patients' health status of patients taking lithium who are and are not monitored for 3 months, by identifying possible health states and the probabilities of making a transition from one to another 'health state';
- Identify and value the resource use associated with managing patients who are, and are not, monitored;
- Assess the relative costs and outcomes for the monitored and unmonitored group.

11.3 Literature search

The key words 'bipolar' and 'lithium', 'TDM' OR 'therapeutic drug monitoring' OR 'monitoring' OR 'therapy' OR 'therapeutic' OR 'manic' OR 'depressive' OR 'depressed' OR 'depression' OR 'suicide' OR 'suicide rates' were used to extract potentially relevant papers.

11.4 Decision-analytic model for economic analysis

11.4.1 The decision-analytic model

The decision-analytic model describes the possible treatment pathways of patients treated with lithium that were monitored, or not monitored in the previous 3 months.

During the literature search two systematic reviews^{182 183} of decision analytic models based on bipolar disorder, and a bipolar disorder model designed by NICE⁸³, looked at the effect lithium (among other medications) has on reducing relapse rates as well as suicide in this

population group. NICE guideline number 38⁸³ introduces a model for the medical and pharmacological management of bipolar disorder. This model takes into account the long term treatment of bipolar disorder using a number of pharmacological drugs, including the use of lithium. No published model was found that examined the effect of medication monitoring on the disease.

The efficacy and toxicity profile of lithium is complex. Patients can be in one of three states in terms of serum lithium concentration:

- Therapeutic range (0.6-1.2 mmol/litre)
- Sub-therapeutic range (<0.6 mmol/litre)
- Supra-therapeutic range (>1.2 mmol/litre)¹³²

11.4.2 Modelling efficacy

Lithium has two main benefits when used in patients with bipolar disorder: reducing the probability of relapse into a manic or depressive state and reducing the probability of suicide. In our model, we assume that patients in the therapeutic and supra-therapeutic range have the same relapse and suicide incidence, in the absence of evidence to the contrary. Patients who are sub-therapeutic do not realise these benefits of lithium.

11.4.3 Modelling toxicity

Previously published lithium models do not take into account the effects of lithium intoxication,^{182 183} possibly because the literature around lithium intoxication is complex and sometimes contradictory. Adverse effects occur at therapeutic and sub-therapeutic levels, and relapse or non-response can occur when the drug is within therapeutic range.¹⁷⁷ Waring et al. (2007) describes three patterns of lithium toxicity:¹⁸⁴

1. Acute intoxication: in patients not receiving lithium previously
2. Acute-on-therapeutic intoxication: acute ingestion whilst on lithium therapy
3. Chronic intoxication: arising insidiously over time due to lithium accumulation.

The first two types would present as an acute overdose and incidence would not be affected by the presence or absence of lithium monitoring. Patients in the supra-therapeutic state are generally so because of a long term increase of serum concentration in the body due to long-term lithium therapy, ie chronic intoxication.¹⁸⁴ Waring and other authors suggest there

is a limited link between serum levels of lithium and toxicity severity.^{177 184 185} This means that the adverse effects of being supra-therapeutic are not clearly different from those in therapeutic or sub-therapeutic levels as chronic toxicity occurs due to long term accumulation, so can present at any serum lithium concentration.¹⁷⁷ The use of lithium TDM may reduce the chance of this occurring over the long term, but this is not clear. Monitoring renal and thyroid function may be more useful.

11.4.4 Varying definitions of relapse

Relapse has been defined variably in different studies around the use of lithium in bipolar disorder.^{170 174} Most definitions of mania relapse include some form of clinical action, such as initiation of a new treatment or admission to hospital, or use of a rating scale, whereas depressive episodes are generally defined by pharmacological intervention or study withdrawal.¹⁷⁰ Young *et al* consider that “this discrepancy in defining mood episode type may misinterpret the extent of lithium’s influence on preventing depressive relapse in relation to the extent in prevents manic relapse.”¹⁷⁰ Our model uses relapse rates as defined in the literature.

11.4.5 Model structure: Markov states

The Markov model has five states: ‘stable (supra-therapeutic and therapeutic)’, ‘stable (sub-therapeutic)’, two relapse states (where the patient can only stay for one cycle): ‘manic’ and ‘depressed’, ‘dead/suicide’, this last absorbing state equating to exiting the model. (Figure 27) Patients can move from stable states to relapse states, or commit suicide (combined with dead state). Patients can move from any state to dead from other causes of death.

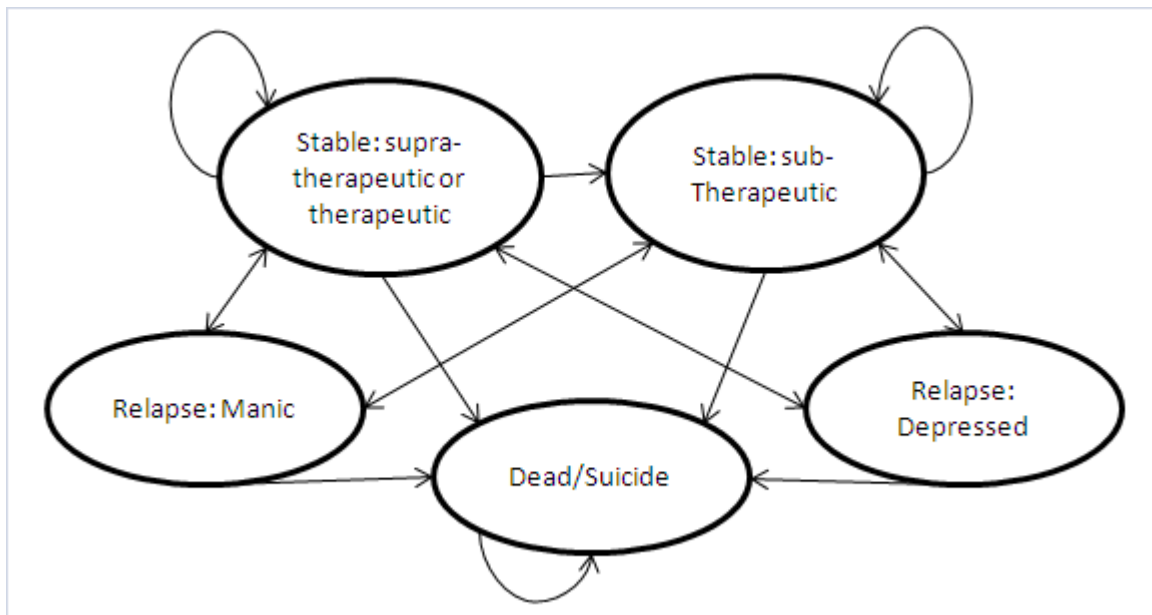


Figure 27: Markov model of adults with bipolar disorder treated with lithium

11.4.6 How does monitoring affect outcome?

11.4.6.1 Modelling the effect of lithium monitoring on toxicity

In section 5.4.3 we outlined the lack of relationship that has been demonstrated between serum levels and chronic toxicity. Acute intoxication (as in an overdose) would present clinically, so TDM has limited value here. As there were no clear primary data to populate the model for lithium acute or chronic toxicity, it was excluded from the model on the assumption that it would probably occur equally between monitored and un-monitored patients.

11.4.6.2 Modelling the effect of lithium monitoring on patient adherence and associated relapse rates

Sharma *et al*¹⁷⁷ reported proportions of 'sub-therapeutic', 'therapeutic' and 'supra-therapeutic' lithium levels in a sample of requests for monitoring. In this study of over 4000 TDM requests, 29.5% of the requisitions for monitoring had lithium levels in the sub-therapeutic range and 7% were above therapeutic range. In the absence of any other evidence, the data from Sharma *et al* were used to approximate to proportions of patients starting in each lithium level category (sub-therapeutic or supra-therapeutic) where patients do not receive routine monitoring.

Scott and Pope¹⁸⁶ and Schumann *et al*⁸¹ report low adherence levels of 53% and 59.2% respectively in general samples of people taking lithium. Poor adherence to lithium is attributed to lack of acceptance of prophylaxis in general, the effectiveness of lithium and the severity of illness.⁸¹ Published evidence suggests that regular monitoring of lithium levels encourages increased patient adherence to the medication.⁸²

In our model, a pooled mean adherence of 56.1% was used for patients that were not receiving TDM, from Scott and Pope¹⁸⁶ and Schumann *et al*⁸¹. Rosa *et al*⁸² reported that attendance at a regular outpatient mood disorder clinic resulted in 85.6% adherence to lithium treatment leading to therapeutic lithium levels. For this model we assumed the Rosa *et al*⁸² sample represented patients that receive TDM every 3 months. This means that patients being appropriately monitored will be more likely to be in the therapeutic range. Patients without monitoring are less likely to be in the therapeutic range and are more likely to present with relapse (or suicide).

Rosa *et al*⁸² reports 14.4% non-adherence in regularly monitored patients, and the pooled results report 43.9% non-adherence in non-monitored patients. If we hold the assumption that patients that are sub-therapeutic are so because of non-adherence (compared to therapeutic and supra-therapeutic patients being adherent to their medication) then appropriate lithium monitoring (i.e. TDM every 3 months) decreases non-adherence by 32.8% [14.4/43.9]; increasing patients within a therapeutic range by 28.2% (from 70.23% within therapeutic range in the non-monitored group to 90.04% in the monitored group).

It is assumed a person will not move from sub-therapeutic to supra/therapeutic until an adverse event occurs and they begin to be monitored again, i.e. a person will only adhere to their medication, after making the decision to be non-adherent, if they are monitored or an adverse event occurs to change their mind.

11.4.6.3 Modelling the effect of a relapse on subsequent monitoring

A key assumption in the model is that HCPs will correct the error once it has come to their attention, via a clinically evident event. The presentation of the patient with relapse which requires the assistance of a HCP will highlight the lack of monitoring, such that the patient's lithium levels are monitored regularly after this event. Once a person in the "error" arm has experienced a relapse (manic or depressed), it is assumed that this will lead the clinician to monitor lithium levels in future in this patient. Therefore, the patient moves into the "no error" section of the model.

11.4.6.4 Modelling the impact of lithium on suicide incidence

Bipolar disorder has been associated with an increased chance of suicidal tendencies. Angst *et al*⁸⁴ examined 406 patients, with and without long term medication, over a period of 40-44 years. Lithium was one of the long term medications examined during this follow up period. During this study period 8.3% patients defined as suffering from bipolar disorder died due to suicide. When these patient were split into patients that had received long term psychopharmacological treatment and those that had not, the results showed that those patients who had not received long-term treatment had a higher rate of death via suicide than those patients taking medication; 7.1% Vs 11.7% respectively ($p=0.04$).⁸⁴ All other causes of death were also incorporated into the model, taking into account the national age-specific death rate.

All probabilities are summarized in Table 33.

Table 33 Probabilities for the 3-month cycle Markov model in the error and non-error groups for lithium

Transition	Probability	Source	Ref #
Subtherapeutic --> subtherapeutic	0.8147	Net of other probabilities: 1- (0.0725+0.0037+0.1091)	N/A
Subtherapeutic --> relapse: manic	0.0725	NICE guideline 38	83
Subtherapeutic --> relapse: depressed	0.1091	NICE guideline 38	83
Subtherapeutic --> suicide/dead	0.0037	ONS+Angst 2005	48 84
Supra or therapeutic --> Supra or therapeutic	No error:0.7113 Error: 0.6089	Net of other probabilities: 1-(0.1440+0.0427+0.0987+0.0034) 1- (0.2463+0.0427+0.0987+0.0034)	N/A
Supra or therapeutic --> Sub-therapeutic	No error: 0.1440 Error: 0.2463	Rosa 2007 Schumann adjusted to 3 month rates	82 81
Supra or therapeutic --> relapse: manic	0.0427	NICE guideline 38	83
Supra or therapeutic -->	0.0987	NICE guideline 38	83

relapse: depressed			
Supra or therapeutic --> suicide/dead	0.0034	ONS+Angst 2005	48 84
Relapse: manic --> sub-therapeutic	0.1440	Assumption that 14.4% patients are non-adherent post relapse due to improved monitoring Rosa 2007	82
Relapse: manic --> supra or therapeutic	0.8530	Assumption that 85.3% patients are adherent post relapse due to improved monitoring Rosa 2007	82
Relapse: manic --> suicide/dead	0.003	ONS	48
Relapse: depressed --> sub-therapeutic	0.1440	Assumption that 14.4% patients are non-adherent post relapse due to improved monitoring Rosa 2007	82
Relapse: depressed --> supra or therapeutic	0.8530	Assumption that 85.3% patients are adherent post relapse due to improved monitoring Rosa 2007	82
Relapse: depressed --> suicide/dead	0.003	ONS	48

11.4.7 Health state weights

One study was identified in the literature search that recorded QALYs for bipolar patients receiving lithium therapy.⁸⁵ Revicki *et al* looks at the utility of patients that are suffering from mania that are treated as outpatient or inpatient, as well as recording the difference between patients that are stable on medication or without medication.⁸⁵ It is assumed in the model that patients that do not receive lithium therapy have a better health state than those patients receiving lithium therapy when in a stable state, due to lack of adverse effects relating to lithium.

No study was found that recorded utility for bipolar patients in a depressive relapse. Therefore, data were taken from a sample of patients with major depression.⁸⁶ Revicki and Wood (1998) examined patients that have depression, on fluoxetine.⁸⁶

During a manic or depressive relapse, a proportion of patients will be treated as inpatients (IP) or outpatients (OP), depending on the severity of the relapse. This generates different health status measures (Table 34), and has different implications for resource use consumption (see next section). Patients who relapse whilst on therapy (i.e. therapeutic) tend to have slightly less severe relapses, this assumption is constant with the literature that lithium therapy will reduce the severity of relapse as well as rate of relapse.¹⁷⁴ In our model the reduced probability is represented in the model by the lower chance of initial relapse – patients who are sub-therapeutic or therapeutic still have the same chance of this relapse resulting in an inpatient stay or outpatient treatment.

Table 34 Health status weights for lithium model

Health status: utility weights

Severity	Health State	Mean	SD	Source
Mild relapse (from stable supra/therapeutic) Moderate relapse (from stable sub-therapeutic)	Stable (sub-therapeutic)	0.74	0.23	⁸⁵
	Stable (supra/therapeutic)	0.71	0.22	
	Mania (OP)	0.56	0.27	
	Mania (IP)	0.26	0.29	
	Mania (OP)	0.54	0.26	
	Mania (IP)	0.23	0.29	
Mild	Depressed (OP)	0.7	0.2	⁸⁶
	Depressed (IP)	0.33	0.36	
Moderate	Depressed (OP)	0.63	0.19	⁸⁶
	Depressed (IP)	0.27	0.34	
Dead		0	0	

IP=Inpatient OP= Outpatient

11.4.8 Resource use associated with each Markov state

The method of monitoring and treatment used in the costing methods is similar to that highlighted in the NICE Guideline for bipolar disorder.^{83 132}

11.4.8.1 Drug monitoring and general healthcare

Lithium therapy is self-medicated and the amount of medication needed to reach a therapeutic level is dependent on the individual; a dose of 1000mg lithium daily is used as highlighted in the NICE Guidelines.⁸³ In monitored patients, a number of tests are conducted

at different time periods in order to check the patient's lithium serum levels, among other functioning tests. These costs are weighted to a 3 month period (Table 35).

Table 35 Costs of TDM carried out for regularly monitored lithium patients (lithium)

Patients monitoring (months):	Unit price/£	Total 3 months/£	Source
Practice nurse per hour of patient contact	44.00	3.67	Healthcare professional and time assumption, ⁸³ unit cost ¹³³
Serum lithium concentration (3)	2.95	2.95	
Blood urea (6)	0.76	0.38	
Electrolytes (6)	1.53	0.76	
Thyroid function (6)	16.86	8.43	
Glucose test (12)	0.76	0.19	
Total cost		16.38	

The difference in cost between treatment arms is that sub-therapeutic patients are presumed to be taking little or no medication as well as not receiving health professional care (until they have an adverse event) – patients who are not taking their medication will also be adverse to receiving healthcare for the same reason they are adverse to taking their medication (Table 36). Therefore, a cost for lithium therapy and healthcare professional time is not applied to these patients.

Table 36 Healthcare professional resource use in a cycle without an adverse event (lithium)

	Units (per hour)	price Total 3 months/£	Source
Psychiatry outpatient visit: – one 20 minute appointment every three months	42.48	14.16	Healthcare professional and time assumption ⁸³
GP – two 10 minute appointment every three months	183.00	61.00	unit cost ¹³³
CMHT home visit including travel costs – one 30 minute visit every month	70.00	109.20	
Total		184.36	

(CMHT: community mental health team)

11.4.8.2 Resource use for “Stable (therapeutic/supra-therapeutic)”

Therapeutic patients are assumed to be taking medication and receiving healthcare. In the monitored patients, a number of tests are conducted at different time periods in order to check the patient’s lithium serum levels, among other functioning tests. (Table 37)

Table 37 Resource use and unit costs for stable (supra-therapeutic/therapeutic) state for lithium

	Units	Total cost/£
Healthcare provider time	3 months	184.36
Lithium 1000mg od	3 months	7.64
Lithium monitoring (non-error arm)	3 months	16.38
Total cost non error arm (error arm)	3 months	208.38 (192.00)

11.4.8.3 Resource use for “Stable (sub-therapeutic)”

Sub-therapeutic patients are presumed to not be taking medication, therefore a cost for lithium therapy is not applied to these patients. In the monitored patients, a number of tests are conducted at different time periods in order to check the patient’s lithium serum levels, among other functioning tests. (Table 38)

Table 38 Resource use and unit costs for stable (sub-therapeutic) state for lithium

	Units	Total cost/£
Healthcare provider time	0	0
lithium monitoring (non-error arm)	3 months	16.38
Lithium 1000mg od	0	0
Total cost non error arm (error arm)	3 months	16.38 (0)

11.4.8.4 Health care after manic or depressive relapse

In the case of relapse into a manic state a patient can either be treated as an inpatient (80%) or outpatient (20%). Inpatient stays have a mean length of 28 days, outpatient care is also received for a 28 day period.⁸³ Outpatient care is co-ordinated by the specially trained Crisis Resolution Home Treatment Team (CRHTT)), the cost of each are highlighted in Table 39.⁸³

Table 39 Healthcare provider resource use in a cycle with a manic relapse (lithium)

Management of acute manic episode:	Unit price (per day)/£	Total 3 months/£	Source – comments
Inpatient stay per bed day (Inpatient) – 28 days [after an episode]	268	7504	Healthcare professional and time assumption ⁸³ , unit costs ¹³³
CRHTT per contact (Outpatient) – 28 days [after an episode]	198	5544	
CRHTT: Crisis Resolution Home Treatment Team			

If a patient relapses into a depressed state the patient can be treated as an inpatient (10%), or outpatient (20%) or through the use of “Enhanced Outpatient Care”⁸³ (70%); inpatient care and outpatient care is received for a 35 day period, the cost of which are highlighted in Table 40.

Table 40 Healthcare provider resource use in a cycle with a depressive relapse (OM7)

Management of acute depressive episode:	Unit price (per day)/£	Total 3 months/£	Source – comments
Inpatient stay per bed day (Inpatient) – 35 days [after an episode]	268	9380	Healthcare professional and time assumption ⁸³ , unit costs ¹³³
CRHTT per contact (Outpatient) – 35 days [after an episode]	198	6930	
CRHTT: Crisis Resolution Home Treatment Team			

Enhanced Outpatient Care (EOC) is referred to by the service providers as ‘enhanced’ although it does not use care from a team such as CRHTT. Instead, it provides community based health care similarly to that received by patients in a stable state, but at more frequent intervals following an acute depressive episode (Table 41).

Table 41 Enhanced Outpatient Care (EOC) resource use in lithium model

Units price (per hour)	Total 3 months/£	Source
------------------------	------------------	--------

Psychiatry outpatient visit: – five 20 minute appointments	42.48	70.80	Healthcare professional and time assumption ⁸³ , unit cost ¹³³
GP – four 10 minute appointments	183.00	122.00	
CMHT home visit including travel costs – five 30 minute visits	70.00 (+1.40 TC)	182.00	
Total		374.80	
CMHT: Community Mental Health Team; TC: Travel Costs			

After a depressive or manic relapse that results in an inpatient or outpatient care episode, bipolar patients use the same health professionals as they would in a stable state, but in a higher proportion in the three month cycle without a relapse (Table 42). These health professionals do not however visit during the period a patient is being treated as an inpatient or outpatient.

Table 42 Healthcare professional resource use in a cycle with an adverse event (excluding an event that uses EOP) in lithium model

	Unit price (per hour)/£	Total cost/3 months/£	Source
Psychiatry outpatient visit: — four 20 minute appointments	42.48	56.64	Healthcare professional and time assumption ⁸³ , unit cost ¹³³
GP – three 10 minute appointments	183	91.50	
CMHT home visit including travel costs – four 30 minute visits	70.00 (+1.40 TC)	145.60	
Total		293.74	
CMHT; Community Mental Health Team; TC; Travel Costs			

The resource use associated with manic relapse and depressive relapse are summarised in Table 43 and Table 44.

Table 43 Resource use and unit costs for relapse in lithium model: manic state

	Units	Unit Cost/£	Total cost/£
Lithium monitoring (non-error arm)	3 months	(Table 35 Costs of TDM)	16.38

		carried out for regularly monitored lithium patients (lithium))	
Lithium 1000mg od	3 months	0.08	7.64
Inpatient stay (80% patients)	28 days	268.00	7504.00
CRHTT per contact (20% patients)	28 days	198.00	5544.00
Post acute crisis care	2 months	(Table 42)	293.74

Table 44 Resource use and unit costs for relapse in lithium model: depressive state

	Units	Unit Cost/£	Total cost/£
Lithium monitoring (non-error arm)	3 months	(Table 35)	16.38
Lithium 1000mg od	3 months	0.08	7.64
Inpatient stay (10% patients)	35 days	268.00	9380.00
Outpatient care (20% patients)	35 days	198.00	6930.00
Enhanced outpatient care (70% patients)	3 months	(Table 41)	374.80
Post acute crisis care	2 months	(Table 42)	293.74

11.4.8.5 Transition costs

The cost of relapsing into a bipolar state or suicide is the same in both error and non error model and in both treatment arms. Table 45 summarises the cost for each transition state.

Table 45 Transition costs in lithium model

3 Month Transition Cycle Cost	Cost Estimate 3months
Patients remaining stable over 3 month:	
No TDM (Sub therapeutic)	£0.00
TDM (Sub therapeutic)	£16.38
No TDM (Therapeutic)	£192.00
TDM (Therapeutic)	£208.38
Patients experiencing a manic episode over 3 months:	
Inpatient	£7,821.77
Outpatient	£5,861.77
Patients experiencing a depressive episode over 3 months:	

Inpatient	£9,697.77
Outpatient	£7,247.77
EOC	£398.83
Death Or suicide (one inpatient day)	£268.00

12 Appendix 6 - Patients receiving amiodarone for at least six months who have not had a thyroid function test within the previous six months

Lead authors Rachel Elliott and Jasdeep Hayre

12.1 Introduction

Amiodarone is a highly effective anti-arrhythmic drug, with efficacy in suppressing recurrent atrial fibrillation in both congenital and acquired heart disease.^{88 187} However, it is rich in iodine (37%), has a very long half-life (55 days) and it is associated with serious thyroid dysfunction.¹⁸⁸ The clinical significance of this adverse effect is magnified by the target population being either elderly, in whom thyroid dysfunction is already prevalent,¹⁸⁸ or unable to tolerate thyrotoxicosis due to compromised cardiac function.⁸⁸ A recently developed less toxic agent, dronedarone, has not replaced amiodarone, as was originally hoped, due to its lesser efficacy in cardioversion.¹⁸⁹

Amiodarone usage can result in two conditions: amiodarone-induced thyrotoxicosis (AIT), and amiodarone-induced-hypothyroidism (AIH).^{188 190-193} Two types of AIT exist, Type I: the patient has a latent or pre-existing thyroid issue when diagnosed with thyrotoxicosis and Type II: the patient has no pre-existing thyroid issue when diagnosed with thyrotoxicosis. Sometimes patients can present with a mixed picture of Type I and Type II AIT.

AIT and AIH can cause significant patient morbidity^{193 194}. The clinical presentation of AIH is more subtle than that of AIT, which can be more dramatic with life-threatening cardiac instability. Patients may lack cardiac manifestations of AIT due to the intrinsic cardiac inhibitory effects of amiodarone. Patients may instead present with some of the other symptoms of AIT: weight loss, heat intolerance, enlargement of the thyroid glands and emotional instability. AIH can result in a slower heart rate, weight gain, shivers, and emotional instability.¹⁹⁵ Diagnosis and management of AIT in particular is quite complex, with wide variation in practice, including variation in opinions about whether patients should remain on amiodarone in the presence of AIT.⁸⁷

The clinical importance of this morbidity and its economic consequences leads to the need for effective monitoring of patients on amiodarone. We describe the clinical and economic

consequences of monitoring, and not monitoring, thyroid function: OM8 – “Patients receiving amiodarone for at least six months who have not had a thyroid function test within the previous six months”.

12.2 Aim of the study

The principal objectives of this analysis are:

- Identify and value the impact on patients’ health status of patients taking amiodarone who are and are not monitored for 6 months, by identifying possible health states and the probabilities of making a transition from one to another ‘health state’;
- Identify and value the resource use associated with managing patients who are, and are not, monitored;
- Assess the relative costs and outcomes for the monitored and unmonitored group.

12.3 Literature search

The literature search was limited to English and Humans, Adults (18+), up to 2010. The following search was used with the key words ‘amiodarone’ AND ‘induced’ AND ‘thyroid’ were used which resulted in 373 articles, of which 25 were used to inform model design.

12.4 Decision-analytic model for economic analysis

12.4.1 Model population

The prevalence and type of thyroid toxicities is suggested to depend upon the dietary intake of the region or country.^{195 196} Countries with lower dietary intake of iodine are associated with higher rates of hyperthyroidism, whereas countries with higher rates of dietary iodine intake are associated with higher rates of hypothyroidism.^{190 197 198} The incidence of AIT is probably affected by environmental iodine intake, as iodide-induced thyrotoxicosis is commoner in areas of iodine deficiency.¹⁹⁰ The UK and the USA have a high dietary intake of iodine, such that AIT is less common than AIH.^{195 199} Due to this, UK-based evidence was used wherever possible to avoid invalid incidence rates resulting from differences in the baseline prevalence rate of hypothyroidism or hyperthyroidism in the world. Studies conducted in countries with similar iodine dietary intake such as the USA^{188 200} were also considered as valid data to populate the model. If the data were unrelated to the baseline incidence rates of thyroid toxicity and data were unavailable for the UK and USA, data from other countries were used.

Amiodarone is used to treat arrhythmias in both acquired and congenital heart disease.^{88 201} The relationship between age and thyroid disorders, in general, is not clear.²⁰² As a result, an adult population (18+) was assumed to be representative of users of amiodarone.

Literature exploring the relationship between incidence rates of hyperthyroidism²⁰³ and gender¹⁸⁸ for amiodarone users is limited. Bouvy *et al*²⁰² identified an underlying difference in incidence rates for amiodarone-induced-thyroid issues in a cohort of 5522 patients on amiodarone. This suggested that women might be more sensitive to amiodarone than to men, or the dosage between the two sexes differed, but offered no clear explanation. Literature using both men and women was used to populate this model.

12.4.2 Defining AIT and AIH

The Association for Clinical Biochemistry (ACB) and the British Thyroid Association (BTA) have offered UK guidelines for the usage of thyroid function tests (TFTs) in relation to amiodarone usage²⁰⁴ which are used to define thyroid toxicity effects (Table 46).

Table 46 Summary of biochemistry and treatment for amiodarone-induced hyper- and hypothyroidism (amiodarone)²⁰⁴

Thyroid pathology	TSH levels	FT3 and FT4 levels	Treatment	Stop amiodarone?
Subclinical hypothyroidism (AIH)	↑TSH (>10 mU/L)	Normal FT3*, FT4**	Thyroxine	No
Overt hypothyroidism (AIH)	↑TSH (>10 mU/L)	Normal ↓FT4	FT3, Thyroxine	
Subclinical hyperthyroidism (AIT)	↓TSH (approx. <0.2 mU/L)	Normal (↓) ↑FT3, ↑FT4	No consensus on treatment, generally don't treat#	No consensus, stopping is recommended by BNF, ¹³² 90% will
Overt hyperthyroidism (AIT)	↓TSH (<0.1 m U/L)	Normal (↓) ↑FT3, ↑FT4	Type I and Type II treated differently, see text below	stop ²⁰⁵ despite no evidence of benefit ²⁰⁶

TSH: thyroid stimulating hormone; FT3: free tri-iodothyronine; FT4: free thyroxine

*FT3: normal range 3.5 -7.8 nmol/L

**FT4: normal range FT4 (9.0 – 25 pmol/L)

#personal communication (Prof Jayne Franklyn, University of Birmingham)

12.4.3 Incidence of AIH and AIT

Generally, in the Western world the prevalence of AIH ranges from 5 to 22% whilst that of AIT is somewhat lower, affecting 2.0–21% patients.^{88 207 208} Only one study was found reporting incidence rates in the UK.⁸⁸ The longer the patient is on amiodarone, the more probable it is that they will develop AIH/AIT, the mean time from amiodarone treatment initiation to AIH/AIT being 3 years.²⁰¹

12.4.4 Treatment of AIH

Treatment consists of thyroid replacement therapy for both subclinical and overt AIH. Amiodarone is continued.²⁰⁴

12.4.5 Treatment of Type I and Type II AIT

AIT can be associated with increased mortality, especially in patients with impaired left ventricular function.²⁰⁹ Type I AIT is a form of iodine-induced hyperthyroidism, and Type II, a drug-induced destructive thyroiditis. However, mixed/indefinite forms exist that may be caused by both pathogenic mechanisms. Type I AIT usually occurs in abnormal thyroid glands, whereas Type II AIT develops in apparently normal thyroid glands (or small goitres).²⁰⁶ The epidemiology of AIT has changed, as the prevalence of Type II AIT has progressively increased and that of Type 1 has remained constant. Thus, 89% cases are now AIT Type II.²¹⁰

There are no UK guidelines for treatment of AIT and thus, there is some lack of consensus around approach to management.²⁰⁵ Treatment can be divided into the following sections: initial drug treatment, whether to stop amiodarone, follow up thyroid ablation.

12.4.5.1 Initial drug treatment

Thionamides, such as carbimazole, are first-line treatment for Type I AIT; potassium perchlorate may increase the response to thionamides, but is not available, or widely used, in the UK (personal communication, Catherine Stephenson, Medicines Information Manager, Pharmacy, Nottingham University Hospitals NHS Trust). Type II AIT is best treated by oral glucocorticoids.²¹¹ The response very much depends on the thyroid volume and the severity

of thyrotoxicosis. Mixed/indefinite forms may require a combination of thionamides and steroids. Type I patients who do not respond to thionamides within 30 days may have a mixed AIT and are usually coprescribed glucocorticoids at this point.²⁰⁵ AIT of both types can be controlled within about 40 days in most cases.^{206 211} Other treatments such as lithium, iopanoic acid and plasmapheresis are not routinely used.²⁰⁶

12.4.5.2 Whether to stop amiodarone

The decision regarding the continuation or otherwise of amiodarone is a complex one with no absolute answer. Amiodarone may be the only option to manage life-threatening arrhythmias in a patient, so there are cardiovascular risks associated with stopping it. It has such a long half life (and hence no immediate benefit on thyroid status if stopped) and reduces FT4 to FT3 conversion so an initial exacerbation of thyroid symptoms may occur on its cessation.²⁰⁶ If amiodarone is stopped in patients with Type II AIT the majority will become and remain euthyroid within 3–5 months of amiodarone withdrawal.^{87 201} However, patients who stay on amiodarone may also become euthyroid, and there is no compelling evidence of benefit of stopping.^{193 206} (personal communication, Prof Jayne Franklyn, University of Birmingham)

Stopping amiodarone is recommended by the BNF¹³² and experts in the field,²⁰⁶ and 90% of European clinicians report that they will stop amiodarone,²⁰⁵ so, in our model, we assume amiodarone is stopped temporarily once AIT is diagnosed.

12.4.5.3 Follow up thyroid ablation

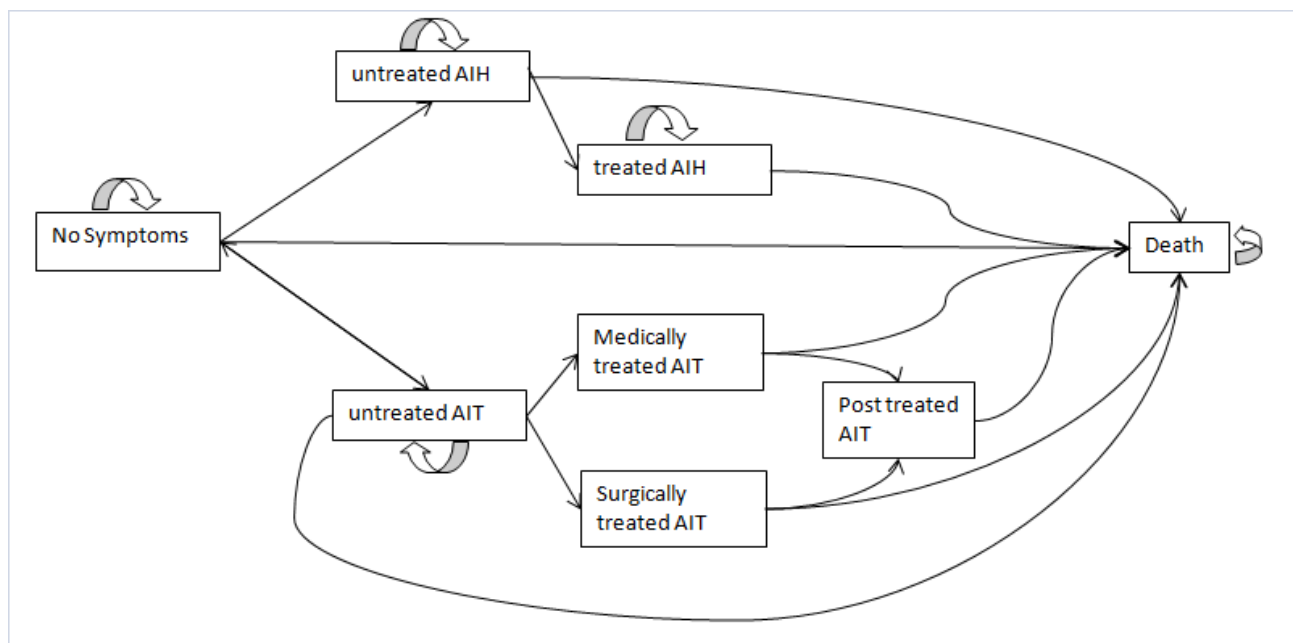
Patients taking amiodarone often need to carry on taking the drug once AIT has been controlled, and then thyroid ablation is often utilised. The indications for reintroduction of amiodarone are uncontrolled recurrent symptomatic paroxysmal atrial fibrillation or recurrent ventricular tachycardia.²¹² In our model, we assume all patients will need to restart amiodarone, although we recognise in practice, some patients may be given alternative treatment. Reports on whether, or how, thyroid ablation occurs, varies between studies. Radioactive iodine treatment,²⁰³ which was significantly used prior to the 1990's but appears to be no longer recommended treatment for AIT in the United Kingdom^{200 213 214} and hence not part of this treatment. In Type I AIT, thyroid ablation via radioactive iodine (RAI) (48%) or thyroidectomy (28%) is the standard approach after euthyroidism has been restored.²⁰⁵ Thyroid radioactive iodine (RAI) uptake values are usually very low or suppressed in Type II AIT, but can range from low to high in Type I AIT despite the iodine load, so can be used in a proportion of patients (personal communication, Prof Jayne Franklyn, University of Birmingham), and has been shown to be effective, allowing reintroduction of amiodarone.²¹²

In Type II AIT, once euthyroidism has been restored, patients are much less likely to need thyroid ablation, and are prone to developing hypothyroidism, so a “wait and see” approach tends to be used in 61% cases,²⁰⁵ with 6% undergoing thyroidectomy.⁸⁷

12.5 The decision-analytic model

The decision-analytic model describes the possible treatment pathways of patients treated with amiodarone, who were monitored, or not monitored in the previous six months. Figure 28 illustrates the Markov model of a cohort of patients with an arrhythmia taking amiodarone for longer than six months.

Figure 28 Markov Model for patients with an arrhythmia and taking amiodarone in the previous 3 months (amiodarone)



12.5.1 Markov states

The Markov model has eight states; ‘No Symptoms’, ‘Untreated AIH’, ‘Treated AIH’, ‘Untreated AIT’, ‘Medically treated AIT’, ‘surgically treated AIT’, ‘Post-treated AIT’ and ‘Death’. The structure of this model relies heavily on two surveys on the diagnosis and management of AIT in Europe.^{87 205}

12.5.2 Effects of amiodarone not included in the model

Users of amiodarone have well documented side effects other than that of the thyroid, such as lung, central nervous system, and skin-related side effects. These other side effects are considered equal in both the intervention and the treatment arm.

12.5.3 How does monitoring affect outcome?

No studies were found that reported the effects of monitoring thyroid function on patient outcomes. In this model, AIT and AIH are assumed to occur whether monitoring takes place or not. If patients are monitored, it is assumed that they will have a lower probability of staying in the state 'Untreated AIH' or 'Untreated AIT', with the associated increased risk of morbidity and mortality.

12.6 Transition probabilities for the model

Table 47 summarises transition probabilities for patients not undergoing regular monitoring (error group)

Table 47 Transition probabilities for the 'error' group (amiodarone)

Transition	Probability (range)	Source	Ref #
No Symptoms --> No Symptoms	0.9622	1-(sum of other probabilities)	n/a
No Symptoms --> AIT untreated	0.0233 (0.0049 - 0.0961)	Thorne et al	88
No Symptoms --> AIH untreated	0.0115 (0.0029 - 0.0201)	Thorne et al	88
No Symptoms --> Death	0.0035	Age-Related Mortality (ONS) + Osman et al	89 215
AIT untreated --> AIT surgical management	0.0081 (0-0.0081)	assumption	See text below
AIT untreated --> AIT medical management	0.0988	Assumption	See text below
AIT untreated --> Death	0.040	Age-Related Mortality (ONS) + Osman et al + Yiu et al	89 90 215
AIT untreated --> AIT untreated	0.8964	1-(sum of other probabilities)	n/a

AIT surgical management --> Post treated AIT	0.9083	1-(sum of other probabilities)	n/a
AIT surgical management --> Death	0.0917	Houghton et al	89 91 215
AIT medical management --> Post treated AIT	0.9965	1-(sum of other probabilities)	n/a
AIT medical management --> Death	0.0035	Age-Related Mortality (ONS) + Osman et al	89 215
Post treated AIT --> Post treated AIT	0.9965	1-(sum of other probabilities)	n/a
Post treated AIT --> Death	0.0035	Age-Related Mortality (ONS) + Osman et al	89 215
AIH untreated --> AIH medical management	0.0995	Assumption	n/a
AIH untreated --> Death	0.0055	Age-Related Mortality (ONS) + Osman et al + Rodondi et al	89 92 215
AIH untreated --> AIH untreated	0.8950	1-(sum of other probabilities)	n/a
AIH treated --> AIH treated	0.9965	1-(sum of other probabilities)	n/a
AIH treated --> Death	0.0035	Age-Related Mortality (ONS) + Osman et al	89 215

Table 48 summarises those transition probabilities that differ for patients undergoing regular monitoring (non-error group)

Table 48 Transition Probabilities that differ for the ‘non-error’ group (amiodarone)

Transition	Probability (range)	Source	Ref #
AIT untreated --> AIT surgical management	0.081 (0-0.081)	Bartalena <i>et al</i>	87
AIT untreated --> AIT medical management	0.9879	1-(sum of other probabilities)	n/a

AIT untreated --> AIT untreated	0	Assumption	n/a
AIH untreated --> AIH medical management	0.9945	1-(sum of other probabilities)	n/a
AIH untreated --> AIH untreated	0.0000	Assumption	n/a

12.6.1 No Symptoms --> AIH or AIT (same value for error and non-error model)

Incidence rates of 'AIH' and 'AIT' can vary greatly.²¹⁴ This review of 20 English-speaking papers (1975-1995) showed an AIT range of 1-23% and AIH within 1-32%, and many reported prevalence rather than incidence. A study based on a cohort of 92 adults (18 to 60 years old) with congenital heart disease with no pre-existing cases of thyroid disorders in London⁸⁸ was used to find the UK incidence of 'AIT' and 'AIH'. Thorne *et al* found that in 1999, 15% of people were diagnosed with AIH in the UK with the mean amiodarone therapy duration of 3.5 years (Range 2.0 – 14.0), adjusted for 3-months the probability is 0.0115 (range 0.0029- 0.0201). Twenty-one percent of patients were diagnosed with AIT with the therapy duration with a mean of 2.5 (Range 0.7 – 12.0) years, adjusted for a period of 3-months the probability was 0.0233 (0.0049 - 0.0807). The incidence of AIT is probably slightly higher than we would expect, however, these were the only data available that allowed us to calculate incidence of AIT, rather than prevalence.

12.6.2 No Symptoms --> Death (same value for error and non-error model)

All-cause Age-Standardised Rates per million based on the European Standard Population released by the Office of National Statistics²¹⁵ were adjusted to the age-standardised 3-monthly death rate. Death rates for asymptomatic patients taking amiodarone were obtained by combining SMR with the increased relative risk of death (RR: 1.2) associated with cardiac arrhythmic conditions.⁸⁹ The probability of death was 0.35% for every 3 months.

12.6.3 AIT untreated --> AIT surgical management (different values for error and non-error model)

If a patient is being monitored regularly, AIT will be picked up and treated within one cycle. The probability of surgical management via thyroidectomy in AIT is 0.081.⁸⁷

There were no studies reporting probability of surgical management of AIT if patients are un-monitored. It was assumed that the probability will be higher than zero as patients may be picked up by chance, at a rate of 10% of the rate in the monitored group.

12.6.4 AIT untreated --> AIT medical management (different values for error and non-error model)

If a patient is being monitored regularly, AIT will be picked up and treated within one cycle. The probability of medical management is assumed to be the net value of 1-(death+surgical management).

There were no studies reporting probability of medical management of AIT if patients are un-monitored. It was assumed that the probability will be higher than zero as patients may be picked up by chance, at a rate of 10% of the rate in the monitored group.

12.6.5 AIT untreated --> Death (same value for error and non-error model)

There is no consensus on death rates for patients taking amiodarone who are also thyrotoxic, as it is not always clear whether patients die due to arrhythmia or thyrotoxicosis. In a recent study of 354 patients with AIT, cardiac death rates doubled from euthyroid to AIT patients ($p=0.08$).⁹⁰ The increased relative risk of death (RR: 1.2) associated with cardiac arrhythmic conditions⁸⁹ was therefore assumed to double to 1.4 in this health state.

12.6.6 AIT surgical management --> Post treated AIT (same value for error and non-error model)

Patients who survive surgical management are assumed to transition to this state.

12.6.7 AIT surgical management --> Death (same value for error and non-error model)

Clinical evidence based from Minnesota, USA ⁹¹ studied all patients who had AIT (N=34) from April 1985 through to November 2002. The death rate due to complication of surgery was 0.088235 for a treatment time of amiodarone adjusted to 3 months. This was then added to the background death rate for this group, of 0.0035.

12.6.8 AIT medical management --> Post treated AIT (same value for error and non-error model)

Patients who survive medical management are assumed to transition to this state.

12.6.9 AIT medical management --> Death (same value for error and non-error model)

In the absence of evidence to the contrary, the probability of death was assumed to be the same as for moving from 'No symptoms' to 'death'.

12.6.10 Post treated AIT --> Post treated AIT (same value for error and non-error model)

Patients who survive post-treated AIT are assumed to remain in this state.

12.6.11 Post treated AIT --> Death (same value for error and non-error model)

In the absence of evidence to the contrary, the probability of death was assumed to be the same as for moving from 'No symptoms' to 'death'.

12.6.12 AIH untreated --> AIH medical management (different values for error and non-error model)

If a patient is being monitored regularly, AIH will be picked up and treated within one cycle. The probability of medical management is assumed to be the net value of 1-(probability of death).

There were no studies reporting probability of medical management of AIH if patients are un-monitored. It was assumed that the probability will be higher than zero as patients may be picked up by chance, at a rate of 10% of the rate in the monitored group.

12.6.13 AIH untreated --> Death (same value for error and non-error model)

There is no consensus on death rates for patients taking amiodarone who have overt hypothyroidism. Evidence suggests that a TSH > 10 mIU increases risk of cardiovascular death by 1.58.⁹²

12.6.14 AIH untreated --> AIH untreated (different values for error and non-error model)

There were no studies reporting probability of medical management of AIH if patients are un-monitored. It was assumed that the probability will be higher than zero as patients may be picked up by chance, at a rate of 10% of the rate in the monitored group.

12.6.15 Post treated AIH --> Death (same value for error and non-error model)

In the absence of evidence to the contrary, the probability of death was assumed to be the same as for moving from 'No symptoms' to 'death'.

12.6.16 AIH treated --> AIH treated (same value for error and non-error model)

Patients who survive post-treated AIH are assumed to remain in this state.

12.7 Health status valuations

Health status valuations were difficult to obtain for users of amiodarone with thyroid complications. Patient-level data from a UK population with atrial fibrillation were used to provide a baseline utility for people with no symptoms (EQ-5D: 0.78 (SD: 0.21)).⁹³ Utility decrements for thyroid toxic events was based on the Quality of Wellbeing scale from Nolan *et al*⁹⁵ which was the only paper that measured the quality of life on a scale of 0 to 1 post-treatment for AIT and AIH. For utility decrement following a thyroidectomy, a clinical expert value was taken from Esnaola *et al*.⁹⁶ Death is assumed to have a health status valuation of 0. (Table 49)

Table 49 Health status valuations for each Markov state (amiodarone)

State	QOL	Source	Ref #
No Symptoms	0.78 (0.21)	Buxton et al 2006	93
Untreated AIH	0.60 (0.21)	Sullivan <i>et al</i>	94
Treated AIH	0.65 (0.21)	Nolan JP, Hypothyroidism, Treatment, First Year	95
Untreated AIT	0.58 (0.21)	Sullivan <i>et al</i>	94
Medically treated AIT	0.76 (0.21)	Nolan JP, Hyperthyroidism, Treatment, First Year	95
Surgically treated AIT	0.73 (0.21)	Esnaola <i>et al</i>	96
Post-treated AIT	0.76 (0.21)	Nolan JP, Hyperthyroidism, Treatment, First Year	95

Death	0	Assumption
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12.8 Resource use associated with each Markov state

Differences in clinical practice between OECD nations were identified, however differences between the USA, UK, Italy and Switzerland appeared to be minimal.^{216 217} There were few differences in the published literature in regards to best practice between the UK²⁰³ and the USA^{195 217} in terms of medication used and procedures made.

12.8.1 No Symptoms

The 'No Symptoms' state is a stable state in which a patient with an arrhythmia has been on a stable and regular dosage of amiodarone for at least six months prior to entering this model. The patient is in a euthyroid state and sees their GP once during this cycle. BTA guidelines indicate that users of amiodarone should have regular consultations with their cardiologist, and we assume 0.5 visits per cycle.²⁰⁴ Patients have a loading dose of amiodarone as an intravenous solution, and is then followed by oral maintenance management.¹³² In this model we assume that the patient is on a maintenance dosage of 200mg daily (Table 50). Patients who are monitored also have 0.5 thyroid tests per 3 month cycle.

Table 50 Resource use and unit costs for “No Symptoms” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
GP visits	1	34.00	34.00	34.00	34.00
Thyroid monitoring (non-error arm)	0.5	4.45	2.23	2.23	2.23
Cardiology consultation	0.5	105.00	52.50	37.50	61.00
Amiodarone 200mg od	90	0.047	4.23	4.23	4.23
Total cost (£) non error (total error arm)			92.96 (90.73)	77.96 (75.73)	101.46 (99.23)

12.8.2 Untreated AIH

In this health state, patients have overt hypothyroidism that is not being managed medically. We assume that the patient is on a maintenance dosage of 200mg daily and sees their GP

once during the cycle, and their cardiologist 0.5 times (Table 51). Patients who are monitored also have 0.5 thyroid tests per 3 month cycle.

Table 51 Resource use and unit costs for “AIH-untreated” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
GP visits	1	34.00	34.00	34.00	34.00
Thyroid monitoring (non-error arm)	0.5	4.45	2.23	2.23	2.23
Cardiology consultation	0.5	105.00	52.50	37.50	61.00
Amiodarone 200mg od	90	0.047	4.23	4.23	4.23
Total cost (£) non error (total error arm)			92.96 (90.73)	77.96 (75.73)	101.46 (99.23)

12.8.3 Treated AIH

In the ‘Treated AIH’ state patients should remain on amiodarone and also receive levothyroxine for hypothyroidism.²⁰⁴ We assume that patients need to see their GP more often, 3 times in the 3-monthly cycle.(Table 52)

Table 52 Resource use and unit costs for “treated AIH” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
GP visits	3	34.00	102.00	102.00	102.00
Thyroid monitoring	1	4.45	14.35	14.35	14.35
Cardiology consultation	0.5	105.00	52.50	37.50	61.00
Amiodarone 200mg od	90	0.047	4.23	4.23	4.23
levothyroxine sodium 100mcg od	90	0.038	3.48	3.48	3.48
Total			151.16	118.84	165.84

12.8.4 Untreated AIT

In this health state, patients have overt hyperthyroidism that is not being managed medically. We assume that the patient is on a maintenance dosage of 200mg daily and sees their GP once during the cycle, and their cardiologist 1 time due to cardiac symptoms (Table 53). Patients who are monitored also have 0.5 thyroid tests per 3 month cycle.

Table 53 Resource use and unit costs for “AIT-untreated” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
GP visits	1	34.00	34.00	34.00	34.00
Thyroid monitoring (non-error arm)	0.5	4.45	2.23	2.23	2.23
Cardiology consultation	1	105.00	105.00	75.00	122.00
Amiodarone 200mg od	90	0.047	4.23	4.23	4.23
Total cost (£) non error (total error arm)			145.46 (143.23)	115.46 (113.23)	162.46 (160.23)

12.8.5 AIT Medical Management

When suffering overt AIT Type I (11% patients with AIT),²¹⁰ a thionamide (carbimazole) is given,²⁰⁴ and a tapering regimen of glucocorticoid (prednisolone) is given in overt AIT Type II¹⁹⁵ (89% patients with AIT).²¹⁰ To simplify the model, no patient is assumed to have mixed AIT due to limited literature on incidence. We assume that patients need to see their GP more often, 3 times in the 3-monthly cycle. We assume patients see an endocrinologist. (Table 54) Patients only stay in this state for one cycle.

Table 54 Resource use and unit costs for “AIT medical management” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
Endocrinology consultation	1	118.00	118.00	84.00	145.00
GP visits	3	34.00	102.00	102.00	102.00
Thyroid monitoring	3	4.45	13.35	13.35	13.35
Cardiology consultation	1	105.00	105.00	75.00	122.00
Carbimazole 10mg od 11% patients	180	0.0551	1.09	0.55	2.19
Prednisolone (30mg od 1 month, 20mg od 1 month, 10mg od 1 month) in 89% patients	360	0.16	51.26	51.26	51.26
Total cost (£)			339.44	274.89 278.94 9	384.53 384.53 2

12.8.6 AIT Surgical Management

Patients suffering AIT Type I have a thyroidectomy if carbimazole is not effective.²¹⁰ We assume that patients need to see their GP more often, 3 times in the 3-monthly cycle. We assume patients see an endocrinologist twice.(Table 55) Patients only stay in this state for one cycle.

Table 55 Resource use and unit costs for “AIT surgical management” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
Endocrinology consultation	2	118.00	236.00	168.00	290.00
GP visits	3	34.00	102.00	102.00	102.00
Thyroid monitoring	3	4.45	13.35	13.35	13.35
Cardiology consultation	1	105.00	105.00	75.00	122.00
Carbimazole 10mg od	180	0.0551	9.92	4.96	19.84
Major Thyroid Procedures without CC	1	2562.00	2561.59	1853.79	3078.14
Total cost (£)			3027.86	2217.10	3625.33

12.8.7 AIT Post-treated

AIT Post-treated is a stable state where a patient has been treated for thyrotoxicosis and is now back on amiodarone, and being monitored closely for thyroid dysfunction. (Table 56)

Table 56 Resource use and unit costs for “AIT Post-treated” (amiodarone)

	Units	Unit Cost	Mean	Min	Max
GP visits	1	34.00	34.00	34.00	34.00
Thyroid monitoring	1	4.45	4.45	4.45	4.45
Cardiology consultation	0.5	105.00	52.50	37.50	61.00
Amiodarone 200mg od	90	0.047	4.23	4.23	4.23
Total			95.18	80.18	103.68

12.8.8 Death

Death is an absorbing state. Death is a result of thyrotoxic crisis, surgical complications²⁰¹ or death from natural causes or the pre-existing arrhythmia condition. The patient will exit the model immediately.